Emma's Atlas of Numbers and Shapes

- A not-so-brief guide to all the math I know -

Emma Bach

2026-09-05

Contents

Part I

Foundations

Chapter 1

An Introduction to Mathematical Reasoning

Chapter 2

Sets

Almost all of modern mathematics can be put in the context of the study of ***sets*** and their ***elements***. Generally, given a collection of objects $a, b, c, d, \ldots$, they can be taken together as a set, generally written in curly brackets and represented by a capital letter:

$$M := \{a, b, c, d, e, \ldots\}$$

However, an approach where any collection of mathematical objects forms a set quickly runs into trouble - famously, ***Russel's paradox*** asks:

> *"Let R be the set of all sets that don't contain themselves. Does R contain itself?"*

If R doesn't contain itself, then by definition it must contain itself, but if it contains itself, then it cannot contain itself! If $R \in R$ is true, then it must be false! We arrive at a classic paradox. The conclusion, then, is that the set of all sets that don't contain themselves cannot exist - we cannot define sets to be any collection of things defined by any property, and must instead establish ground rules about which sets exist and exactly how we can use sets we already know to construct new ones.

These ground rules are given in the form of ***axioms***, which are statements that mathematicians take for granted, which can then be used as a starting point to then derive "the rest of mathematics" from. The following chapters contain a brief and hopefully easily understandable introduction to the ZFC axiom system, which was developed in the early 20th century by Ernst Zermelo and Abraham Fraenkel and is now the most widely accepted axiomatization of mathematics.

2.1 Infinity: $\emptyset$ and $\mathbb{N}$

2.2 Extensionality

To say "b is an element of M", or equivalently "b contains M", we simply write:

$$b \in M$$

One of the most basic axioms of set theory is the ***Axiom of Extensionality***, which states:

Axiom 2.1: Axiom of Extensionality

Two sets A and B are equal if and only if they contain the same elements.

The Axiom of Extensionality establishes that sets are uniquely defined by the objects they contain and have no additional structure. In particular:

1. The order of elements in a set doesn't matter. $\{1, 2\}$ and $\{2, 1\}$ are the same set.

2. The number of times an element appears within the same set doesn't matter. $\{1\}, \{1, 1\}$, and $\{1, 1, 1\}$ are the same set.

Definition 2.2: Subsets

We call a set A a ***subset*** of a set B if and only if every element of A is contained in B. We write "A is a subset of B" as $A \subseteq B$.

The demand that "every element of A is contained in B" can be stated more formally as "$x \in A$ implies $x \in B$". Importantly, this means that every set is a subset of itself - we always have $A \subseteq A$.

This definition gives us another way of phrasing the Axiom of Extensionality - two sets are equal if and only if each is a subset of the other:

$$A = B \Leftrightarrow A \subset B \text{ and } B \subset A$$

This idea is incredibly important for writing out proofs in practice - if a problem asks you to establish that two sets A and B are the same, then the easiest approach is almost always to show that any element of A must be contained in B, and to then show that any arbitrary element of B must be contained in A.

The subset relation is frequently written as $A \subset B$. However, other authors reserve $A \subset B$ to mean "A is a subset of B that isn't equal to B" - we call such a set a ***proper subset of B***. I will therefore attempt to only use the symbols $\subseteq$ and $\subsetneq$, which clear up this ambiguity.

2.3 Specification

2.4 Pairing, Union, and Power Sets

2.5 Relations

2.6 Functions

2.7 Cartesian Products

Definition 2.3: General Products in Category Theory

Let C be a category. Let X_1 and X_2 be objects of C. Then the **product of X_1 and X_2** is an object $X_1 \times X_2$ equipped with two morphisms $\pi_1 : X \to X_1$, $\pi_2 : X \to X_2$ with the property that for every object Y and every pair of morphisms $f_1 : Y \to X_1$, $f_2 : Y \to X_2$, there exists a unique morphism $f : Y \to X_1 \times X_2$ such that:

1. $f_1 = \pi_1 \circ f$
2. $f_2 = \pi_2 \circ f$

We call this property the **universal property of product objects.**

In the case of sets, this definition leads us to the familiar cartesian products, where π_1 and π_2 are the projections to the first and second component of the product respectively. We can generalize the cartesian product to uncountably infinite products as follows:

Definition 2.4: Cartesian Product of sets

Let I be any set. Let $(X_i)_{i\in I}$ be a Family of nonempty sets indexed by I, i.e. there exists a function assigning to each element $i \in I$ an element of $(X_i)_{i\in I}$. Then the cartesian product of $(X_i)_{i\in I}$ is the unique set:

$$\prod_{i\in I} X_i := \left\{ x : I \to \bigcup_{i\in I} X_i \;\middle|\; x_i := x(i) \in X_i \text{ for all } i \in I \right\}$$

Written more simply: Each element of $\prod_{i\in I} X_i$ is a function assigning to each $i \in I$ an element $x(i) \in X_i$, where we generally write the argument in the index ($x_i := x(i)$), to stay consistent with the established notation in the countable case.

This definition is simply a reinterpretation of the definition of finite cartesian products: It amounts to viewing, for example, the tuple $T := (x, y, z)$ as a map:

$$T : \{1, 2, 3\} \to \{x, y, z\}$$

such that $T(1) = x$, $T(2) = y$, and $T(3) = z$.

Definition 2.5: Projections

Let $J \subset I$. Then we define the projection of I to J as the map

$$\begin{aligned}\pi_J := \pi^I_J : \prod_{i \in I} X_i &\to \prod_{j \in J} X_j \\ x &\mapsto \left(x|_J : J \to \bigcup_{j \in J} X_j \right)\end{aligned}$$

which restricts the domain of a given element x of the product to the subset J.

In particular, if J is a set consisting of a single element j, we have:

$$\begin{aligned}\pi_j := \pi^I_{\{j\}} : \prod_{i \in I} X_i &\to X_j \\ x &\mapsto x_j\end{aligned}$$

2.8 Replacement

2.9 Regularity and Well-Foundedness

2.10 Choice

2.11 Cardinality

Part II

Foundations of Analysis

Chapter 1

The Real Numbers $\mathbb{R}$

Chapter 2

The Concept of Distance

2.1 Metric Spaces

A metric space is a set, alongside with a ***distance function***, more commonly known as a ***metric***, d which takes two points and outputs a distance between the points. Naturally, we want this "distance" to actually give us some usable structure on the set.

Now, what pieces of geometric intuition should we bake into our notion of distance? I would argue that the most fundamenal use of "distance" is to be able to ***compare distances between any two pairs of points***. Therefore, our metric should have as its output a ***totally ordered set***.

We also want to be able to add distances together as a way of comparing them - logically, the distance between two points x and y plus the distance between y and z gives us an upper bound to the distance between x and z, since we can always take this path to get from x to z, but there might be a shorter path we haven't found yet. As a formula, this is the ***triangle inequality***

$$d(x,z) \leq d(x,y) + d(y,z)$$

Finally, if we have an ordered group structure, we would really like for our order to be archimedean - the distance between two distinct points should be greater than zero, and it shouldn't be infinitesimal.

This leaves $\mathbb{R}_{\geq 0}$ as the obvious candidate for the domain of our distance function, since any archimedean ordered group is a subgroup of $\mathbb{R}_{\geq 0}$ anways. This shouldn't be too much of a surprise, even without this logical background - I assume that when most people read "distance between two points", they intuitively already assume that these distances will be given by positive real numbers.

Definition 2.1: Metric Spaces

A ***metric space*** is a set M together with a function $d : M \times M \to \mathbb{R}_{\geq 0}$, such that:

1. d is ***positive definite***, i.e. $d(x, y) = 0$ if and only if $x = y$.
2. d is ***symmetric***, i.e. $d(x, y) = d(y, x)$
3. d fulfills the ***triangle inequality***

$$d(x, z) \leq d(x, y) + d(y, z)$$

Theorem 2.2. Let $M^n := \prod_{i=1}^{n} M$ be a finite cartesian product of metric spaces. Then the ***Euclidean distance function***

$$d_\infty(x, y) := \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2}$$

gives a metric on M^n.

Theorem 2.3. Let $M^n := \prod_{i=1}^{n} M$ be a finite cartesian product of metric spaces. Then the ***Chebyshev distance function***

$$d_\infty(x, y) := \max_{i \in [1..n]} d(x_i - y_i)$$

gives a metric on M^n.

Definition 2.4: Balls

Let (M, d) be a metric space. Then the ***open ball of radius r around a point x*** is the set

$$B_r(x) := \{y \in M : d(x, y) < r\},$$

and the ***closed ball of radius r around x*** is the set

$$\overline{B}_r(x) := \{y \in M : d(x, y) < r\},$$

Definition 2.5: Open and Closed Sets

Let (M, d) be a metric space. Then we call a subset $U \subseteq M$ ***open*** (in M) if it contains an open ball (of any radius) around all of its points, i.e.

$$\forall x \in U :$$
$$\exists r \in \mathbb{R}_{\geq 0} :$$
$$B_r(x) \subseteq U.$$

We call U closed if its complement $M \setminus U$ is open.

Intuitively, a set is open if it contains "a little bit of wiggle room" around any point contained in it. This means that sets with a "clear border", such as our closed balls, aren't open. For example, the interval $[0, 1] \subset \mathbb{R}$ doesn't leave any wiggle room around 0 and 1.

Importantly, while it may seem unintuitive, there are in fact sets that are both closed and open at the same time. We call such sets ***clopen***.

Exercise 2.6. *There are exactly two clopen subsets of* $\mathbb{R}$*. What are they?*

Definition 2.7. Let (M, d) be a metric space. We call a subset $N \subseteq M$ ***dense in*** $\boldsymbol{M}$ if every nonempty open subset of M contains an element of N.

Lemma 2.8. Every nonempty open subset of $\mathbb{R}$ is uncountable.

Proof.

1. Since every nonempty open subset of $\mathbb{R}$ contains an open interval (a, b), it suffices to show that open intervals are uncountable.

2. The map

$$\varphi_1 : (a, b) \to \left(-\frac{b-a}{2}, \frac{b-a}{2}\right)$$
$$x \mapsto x - \frac{b+a}{2}$$

is a bijection.

3. The map

$$\varphi_2 : \left(-\frac{b-a}{2}, \frac{b-a}{2}\right) \to (-1, 1)$$
$$x \mapsto \frac{2x}{b-a}$$

is also a bijection.

4. Finally, the map

$$\varphi_3 : (-1, 1) \to \mathbb{R}$$
$$x \mapsto \frac{x}{1 - |x|}$$

is a bijection (the inverse is the map we get by replacing the $-$ with a $+$).

5. Therefore, $\varphi_3 \circ \varphi_2 \circ \varphi_1$ is a bijection $(a, b) \cong \mathbb{R}$, so any open interval is uncountable, so any open subset of $\mathbb{R}$ is uncountable.

□

Theorem 2.9

The set of rational numbers $\mathbb{Q}$ is dense in $\mathbb{R}$. The set of irrational numbers $\mathbb{R} \setminus \mathbb{Q}$ is also dense in $\mathbb{R}$.

Proof.

1. Since any nonempty open subset $U \subseteq \mathbb{R}$ contains an open interval (a, b), it suffices to show that any open interval contains a rational number. We assume without loss of generality that $0 \leq$

$a < b$, since $a < 0 < b$ already implies $0 \in (a, b)$, while $a < b \leq 0$ implies that (a, b) contains a rational number if and only if $(-b, -a)$ contains a rational number.

Now, since the ordering on the reals is archimedean, we can find a natural number $n \in \mathbb{N}$ such that $\frac{1}{n} < b - a$. Now, the set

$$L = \left\{ k \in \mathbb{N} \,\middle|\, \frac{k}{n} > a \right\}$$

is nonempty - if $a = 0$, then $1 \in M$, and if $a \neq 0$, then the archimedean property tells us that there exists a $k \in \mathbb{N}$ such that

$$\frac{1}{k} < \frac{1}{na},$$

which implies $\frac{k}{n} > a$. Since M is a nonempty subset of $\mathbb{N}$, there must be a smallest element m. Since m is minimal, we have $m - 1 \neq M$, which implies

$$\frac{m-1}{n} \leq a,$$

which in turn implies

$$\frac{m}{n} = \frac{m-1}{n} + \frac{1}{n} < a + (b - a) = b$$

Therefore, $\frac{m}{n}$ is a rational number contained in (a, b), so $\mathbb{Q}$ is dense in $\mathbb{R}$.

2. Since any open subset $U \subseteq \mathbb{R}$ is uncountable, while the rationals are countable, U must contain an uncountable number of irrationals. Therefore, $\mathbb{R} \setminus \mathbb{Q}$ is dense in $\mathbb{R}$.

□

Corollary 2.10

The set of rational numbers $\mathbb{Q}$ is neither closed nor open in $\mathbb{R}$.

Proof. Since both sets are dense in $\mathbb{R}$, every open ball containing a rational number must also contain an irrational number, and vice versa. □

2.2 Topologies

Topological spaces generalize metric spaces. By inheriting the concept of open and closed sets from metric spaces, they are able to still capture notions of closeness without putting a numerical value on exactly how close any given pair of points are. The notion of a topology is based on the following insight:

Exercise 2.11. *Let (M, d) be a metric space. Then:*

1. *The empty set is open in M.*

2. *Any union of sets that are open in M is open in M.*

3. *Any finite intersection of sets that are open in M is open in M.*

4. *M is open in M.*

We take this idea and make it a definition:

Definition 2.12

Let X be a set. Then a ***topology on $\boldsymbol{X}$*** is a set $\mathcal{O}$ of subsets of X, called ***open sets***, such that:

1. The empty set is open.

2. Any union of open sets is open.

3. Any finite intersection of open sets is open.

4. X is open.

Most of the topological spaces we care about in practice are already metric spaces, but many proofs flow more elegantly when phrased in the generalized language of topological spaces, and there *are* topological spaces that aren't metric spaces.

Theorem 2.13: Extremal Topologies

Let X be an arbitrary set. Then the following two sets of subsets are always topologies on X:

1. The ***indiscrete topology***, where the only open sets are X and the empty set.

2. The ***discrete topology***, where every subset is open.

> *Proof Sketch.* Both topologies fulfill all requirements trivially. :) "□"

Definition 2.14. We say that a metric d on a space M ***generates*** a given topology $\mathcal{O}$ on M if they give the same open sets, i.e. the open sets in $\mathcal{O}$ are exactly the open sets in (M, d).

Exercise 2.15. 1. *Show that the discrete metric generates the discrete topology.*

2. *Show that there is no metric generating the indiscrete topology.*

Two different metrics can generate the same topology:

Exercise 2.16. *Show that the euclidean metric also generates the discrete topology on $\mathbb{N}$.*

Exercise 2.17. *Let d be a metric on a set M. Let $c \in \mathbb{R}_{>0}$.*

1. *Show that*

$$d_c(x, y) = c \cdot d(x, y)$$

is a metric on M.

2. *Show that d and d_c generate the same topologies on M.*

This fact can be seen as a formalization of the intuition that changing the specific unit we choose to measure space with doesn't change its underlying structure.

Definition 2.18: Interior, Exterior and Boundary

Let X be a topological space. Let $A \subseteq X$. Then:

1. The ***interior*** $\operatorname{int} A$ of A is the set of all points that have an open neighbourhood entirely contained in A.
2. The ***exterior*** $\operatorname{ext} A$ of A is the interior of the complement $X \setminus A$, i.e. the set of all points that have an open neighbourhood entirely contained in $X \setminus A$.
3. The ***boundary*** ∂A of A is the set of all points $x \in X$ lying neither in $\operatorname{int} A$, nor in $\operatorname{ext} A$.

The interior of A is occasionally written as $\mathring{A}$. Intuitively, the interior is the largest open subset of A, and the exterior is the largest open set not containing any point of A.

Exercise 2.19. *Show that:*

1. *The interior of any set is open.*
2. *The exterior of any set is open.*
3. *The interior of A is the union of all open subsets of A.*
4. *The exterior of A is the union of all open subsets of $X \setminus A$.*
5. *The boundary of A is the set of all points $x \in X$ such that every neighbourhood of x contains both points in A and points in $X \setminus A$.*

Definition 2.20: Closure

The closure of A is the set $\overline{A} := A \cup \partial A$ containing A and all of its boundary points.

Exercise 2.21.

1. *Show that $\overline{A}$ is the set of all points $x \in X$ such that every neighbourhood of x contains points in A.*
2. *Show that $\overline{A}$ is closed.*
3. *Show $\overline{A}$ is the intersection of all closed sets containing A.*

Theorem 2.22. Let $B \subseteq \mathbb{R}$ be a bounded subset of $\mathbb{R}$. Then $\inf B$ and $\sup B$ are contained in $\overline{B}$.

Proof. Since B is bounded, completeness of the real numbers tells us that $\inf B$ and $\sup B$ are finite real numbers. If $\inf B \notin B$, then there would exists an open neighbourhood $U_{\inf}$ of $\inf B$ not containing any points of B. Since any neighbourhood of a real number x must contain numbers greater than x, this would give us a lower bound of B greater than $\inf B$, which contradicts the definition of $\inf B$. Similarly, if $\sup B \notin \overline{B}$, we could get a lesser upper bound of B, which is again a contradition. □

2.3 Convergence

Theorem 2.23: Detecting openness and closedness using sequences

Let M be a metric space. Let $A \subseteq M$. Let $x \in M$. Then:

1. $x \in \operatorname{int} A$ if and only if every sequence in M converging to x is eventually in A.
2. $x \in \overline{A}$ if and only if there exists a sequence in A converging to x.

Proof. 1. If $x \in \operatorname{int} A$, then there exists an $\varepsilon > 0$ such that $B_\varepsilon(x) \subseteq A$. If a sequence converges to x, it must thus be eventually in $B_\varepsilon(x)$, and thus eventually in A.

Vice versa, if $x \notin \operatorname{int} A$, then every open ball $B_n := B_{\frac{1}{n}}(x)$ around A contains at least one point not in A, and so we can construct a sequence converging to x that isn't eventually in A by picking these points.

2. If there exists sequence of points in A converging to x, then every neighbourhood of x must contain at least one point in A, so $x \in \overline{A}$.

Vice versa, if $x \in \overline{A}$, then every open ball around x must contain a point from A, so we can construct a sequence in A converging to x by taking the sequence $B_n := B_{\frac{1}{n}}(x)$ of open balls and picking a point $x_n \in A \cap B_n$ for each $n \in \mathbb{N}$.

□

Corollary 2.24.

1. A is closed in X if and only if it contains every limit of every convergent sequence of points in A.
2. A is open in X if and only if every sequence in X converging to a point of A is eventually in A.

2.4 Continuity

Definition 2.25. Let $f : M \to N$ be a function between metric spaces. Then we call f ***uniformly continuous*** if it is continuous and fulfills the additional requirement that, given an ε, we can pick the same δ for all points $x \in M$, i.e:

$$\begin{aligned}
&\forall \varepsilon \in \mathbb{R}_{>0} : \\
&\exists \delta \in \mathbb{R}_{>0} : \\
&\forall x, y \in M : \\
&\quad d(x, y) < \delta \implies d(f(x), f(y)) < \varepsilon
\end{aligned}$$

2.5 Compactness

Exercise 2.26. *Any closed subset of a compact set is compact.*

Exercise 2.27. *A compact subset of a metric space is always closed.*

Theorem 2.28: Compactness in Metric Spaces

1. A subset of a metric space is compact if and only if it is sequentially compact.
2. A compact subset of a metric space is closed and bounded.

Proof. Let $K \subseteq M$ be compact. Since M is a metric space, K is closed. If we cover K using open balls, that cover must have a finite subcover. Thus, K must be contained in a finite union of open balls, which means K must be bounded. $\square$

Theorem 2.29

Let $f : M \to N$ be a continuous function between metric spaces. Let $K \subseteq M$ be compact. Then $f(K)$ is compact.

Proof. Let $K \subseteq M$ be compact. Let $\mathcal{U}$ be an open cover of $f(K)$. Since f is continuous, $f^{-1}[\mathcal{U}] := \{f^{-1}[U] \mid U \in \mathcal{U}\}$ is an open cover of N, which must have a finite subcover $\mathcal{V} \subseteq f^{-1}[\mathcal{U}]$. Since $f : K \to f(K)$ is surjective by definition, $f[\mathcal{V}]$ must be a finite cover of $f(K)$. $\square$

Corollary 2.30: Maximum-Minimum-Theorem

Let $f : K \to \mathbb{R}$ be a function from a compact subset of a metric space into the real numbers. Then f attains a maximum and a minimum.

Proof. By the previous theorem, $f(K)$ is a compact subset of $\mathbb{R}$, i.e. closed and bounded by Theorem 2.28. But then Theorem 2.22 tells us that $\inf f(K)$ and $\sup f(K)$ are already contained in $\overline{f(K)} = f(K)$. $\square$

Lemma 2.31. Let M, N be metric spaces. Let $K \subseteq M$ be compact. Let $f : K \to N$ be continuous. Then the preimage of any compact subset of M is a compact subset of K.

Proof. Since N is a metric space, compact subsets are closed. Since f is continuous, the preimage of a closed subset is closed. Since K is compact, any closed subset of K is compact. □

Theorem 2.32: Countable Metric Tychonoffs Theorem

Let M_1, M_2 be metric spaces. Let $K_1 \subseteq M_1, K_2 \subseteq M_2$ be compact. Then $K_1 \times K_2 \subseteq M_1 \times M_2$ is compact.

Proof. Using whichever bijection $\mathbb{N}^2 \cong \mathbb{N}$ the reader prefers, we can unroll a sequence s_n in $K_1 \times K_2$ into a sequence x_n in K_1 and a sequence y_n in K_2, such that the elements of s_n are in one-to-one correspondence with pairs (x_i, y_j) of elements of x_n and y_n. Since K_1 and K_2 are compact, x_n and y_n have convergent subsequences. Using our bijection, these can be reassembled into a subsequence of s_n, and since both components of this subsequence converge, the subsequence itself converges. Thus, every sequence in $K_1 \times K_2$ has a convergent subsequence. Since sequential compactness and compactness are equivalent in metric spaces by Theorem 2.28, $K_1 \times K_2$ is compact. □

Theorem 2.33

Let M, N be metric spaces. Let $K \subseteq M$ be compact. Then every continuous function $f : K \to N$ is uniformly continuous.

Proof. Assume that $f : K \to N$ isn't uniformly continuous. Then there exists an $\varepsilon \in \mathbb{R}_{>0}$ such that for every $n \in \mathbb{N}$, we can find points $x_n, y_n \in K$ such that $d(x_n, y_n) < 1/n$, but $d(f(x_n) - f(y_n)) \geq \varepsilon$.
Since K is compact, it is sequentially compact by Theorem 2.28, so there exists a subsequence $(x_k) \subset (x_n)$ converging to a point p. Since $d(x_n, y_n) < 1/n$, there also exists a subsequence $x_k \subset y_n$ converging to p. If f was continuous, we would have

$$\lim_{k\to\infty} f(x_k) = \lim_{k\to\infty} f(y_k),$$

contradicting the fact that we have $d(f(x_k) - f(y_k)) \geq \varepsilon$ for all $k \in \mathbb{N}$. □

2.6 Connectedness

2.7 Banach's Fixed Point Theorem

Chapter 3

Infinite Series

3.1 Achilles and the Turtle

3.2 Convergence Tests

3.3 Euler's Number e

3.4 The Exponential Function

Chapter 4

Euclidean Geometry

4.1 The Complex Numbers $\mathbb{C}$

4.2 Circles, Angles and Trigonometry

Chapter 5

Function Spaces

Since we know that a sequence in a finite cartesian product converges if and only if all components converge, it seems natural to generalize this notion to arbitrary, potentially infinite cartesian products, i.e. arbitrary functions:

Definition 5.1: Pointwise Convergence

We say that a sequence of functions $f_n : M \to N$ converges ***pointwise*** to a given function $f : M \to N$ if and only if all components $f_n(x)$ converge in the usual sense, i.e:

$$\begin{aligned} &\forall x \in M : \\ &\forall \varepsilon \in \mathbb{R}_{>0} : \\ &\exists N \in \mathbb{N} : \\ &\forall n \geq N : \\ &\quad d(f_n(x) - f(x)) < \varepsilon \end{aligned}$$

We can't put a metric on N^M in which this form of converges matches regular metric convergence, but the intuition that this is the most natural form of convergence in N^M will be vindicated once we study abstract product spaces in point-set topology, at which point we confirm that in the natural product topology, an infinite cartesian product still converges if and only if every component converges individually.

However, there *is* a natural metric to put on N^M, which leads to a stronger form of convergence:

Definition 5.2. An ***extended metric space*** is a metric space where d can also take on the value ∞.

Theorem 5.3: Supremum Metric

The set N^M of functions from a metric space M to a metric space N can be metrized by the extended metric

$$d_\infty(f,g) := \sup_{x\in M} |f(x) - g(x)|.$$

Additionally, this function is a regular metric on the set $B(M,N) \subset N^M$ of bounded functions $M \to N$.

Note that is simply a direct generalization of the Chebyshev metric 2.3 to infinite cartesian products!

Theorem 5.4: Uniform Convergence

In (N^M, d_∞), a sequence of functions f_n converges to a given function f if and only if

$$\begin{aligned}
&\forall \varepsilon \in \mathbb{R}_{>0} :\\
&\exists N \in \mathbb{N} :\\
&\forall n \geq N :\\
&\forall x \in M :\\
&\quad d(f_n(x) - f(x)) < \varepsilon
\end{aligned}$$

we call this form of convergence of a sequence of functions ***uniform convergence***.

Theorem 5.5. Let $U \subseteq (X,d)$ be open and path-connected. Then there exists a piecewise linear path $\gamma_n : [0,1] \to U$ between any two $p, q \in U$.

Proof Sketch.

1. Since U is path connected, there exists a curve $\gamma : [0,1] \to U$ connecting p and q.
2. We can approximate γ via a piecewise linear curve γ_n by setting
$$\gamma_n\left(\frac{x}{n}\right) := \gamma\left(\frac{x}{n}\right)$$
3. Since γ is continuous and defined on a compact set, it is uniformly continuous by Theorem 2.5. From this, it can be shown that the γ_n converge to γ uniformly.
4. Since $\gamma([0,1]) \subseteq U$ is compact and $X \setminus U$ is closed and disjoint from U, we have $\varepsilon := d(\gamma([0,1]), X \setminus U) > 0$.
5. Therefore, uniform convergence guarantees there is an n such that $d(\gamma_n([0,1]), \gamma([0,1])) < \varepsilon$, which implies $\gamma_n([0,1]) \subseteq U$.

"□"

Part III

Calculus in one Variable

Chapter 1

Derivatives

Chapter 2

Integrals

2.1 The Gamma Function

In this section, we discover some basic properties of the so-called gamma function. We will find that it is the canonical extension of the factorial function to positive real numbers. With some knowledge of complex analyis, it can easily be extended to arbitrary complex numbers, but that is beyond the scope of these notes.

Definition 2.1: Gamma function

For any $x \in \mathbb{R}_{>0}$, the gamma function is defined as the improper integral

$$\Gamma(x) = \int_0^\infty t^{x-1} e^{-t} \, dt$$

Theorem 2.2. The gamma function obeys the recurrence relation

$$\Gamma(x+1) = x \cdot \Gamma(x)$$

Proof. Integration by parts gives:

$$\begin{aligned}
\Gamma(x+1) &= \int_0^\infty t^x e^{-t} \, dt \\
&= \left[-t^x e^{-t}\right]_0^\infty + \int_0^\infty x t^{x-1} e^{-t} \, dt \\
&= 0 - \lim_{n\to\infty} (-t^x e^{-t}) + x \int_0^\infty t^{x-1} e^{-t} \, dt \\
&= x \int_0^\infty t^{x-1} e^{-t} \, dt \\
&= x \cdot \Gamma(x)
\end{aligned}$$

□

Corollary 2.3. For any $n \in \mathbb{N}$, we have

$$\Gamma(n+1) = n!$$

Proof. Our recurrence relation gives

$$\Gamma(n+1) = n! \cdot \Gamma(1)$$

and we have

$$\begin{aligned}
\Gamma(1) &= \int_0^\infty t^{1-1} e^{-t} \, dt \\
&= \int_0^\infty e^{-t} \, dt \\
&= \left[-e^{-t}\right]_0^\infty \\
&= \left(\lim_{n \to \infty} -e^{-n}\right) - (-e^0) \\
&= 0 - (-1) \\
&= 1
\end{aligned}$$

□

Lemma 2.4. We have:

$$\Gamma(x) = 2c^x \int_0^\infty t^{2x-1} e^{-ct^2} \, dt$$

Proof. Substituting $t := cu^2$, we get:

$$\begin{aligned}
\Gamma(x) &= \int_0^\infty t^{x-1} e^{-t} \, dt \\
&= \int_0^\infty (cu^2)^{x-1} e^{-cu^2} \cdot 2cu \, du \\
&= \int_0^\infty c^{x-1} u^{2(x-1)} e^{-cu^2} \cdot 2cu \, du \\
&= \int_0^\infty c^x u^{2x-1} e^{-cu^2} \, du \\
&= c^x \cdot \int_0^\infty u^{2x-1} e^{-cu^2} \, du \\
&= c^x \cdot \int_0^\infty t^{2x-1} e^{-ct^2} \, dt
\end{aligned}$$

□

Proposition 2.5. We have $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$

Proof Sketch. By the last lemma, we have:

$$\begin{aligned}\Gamma\left(\frac{1}{2}\right) &= 2\int_0^\infty t^{2\cdot\frac{1}{2}-1}e^{-t^2}\,dt \\ &= 2\int_0^\infty e^{-t^2}\,dt \\ &= \int_{-\infty}^\infty e^{-t^2}\,dt\end{aligned}$$

The fact that $\int_{-\infty}^\infty e^{-t^2}\,dt = \sqrt{\pi}$ might be familiar if you have a basic knowledge of probability theory, since this is exactly the reason why π appears in the formula for a normal distribution:

$$\mathcal{N}(\mu,\sigma^2)(t) = \frac{1}{\sqrt{2\pi\sigma^2}}e^{\left(-\frac{(t-\mu)^2}{2\sigma^2}\right)}$$

Setting $\mu = 0$ and $\sigma^2 = \frac{1}{2}$, which was Gauss' original definition for the standard normal distribution, we get:

$$\mathcal{N}(0,1)(t) = \frac{1}{\sqrt{\pi}}e^{-t^2},$$

where the factor $\frac{1}{\sqrt{\pi}}$ appears to normalize the integral of the distribution to 1, since it wouldn't be a probability distribution otherwise.

We will formally prove this fact much later, when we introduce the substitution formula for multidimensional Lebesque integration. "□"

Part IV

Multivariable Calculus

Chapter 1

Differentiation in $\mathbb{R}^n$

1.1 Differentials and Partial Derivatives

Definition 1.1: Fréchet Derivative

Let V, W be normed vector spaces. Let U be an open subset of V. Then $f : U \to W$ is called ***Fréchet differentiable*** or just ***differentiable*** at $p \in U$ if there exists a continuous linear map $Df(p) : V \to W$ such that

$$\lim_{h \to 0} \frac{\|f(p+h) - f(p) - Df(p)(h)\|_W}{\|h\|_V} = 0.$$

We call $Df(p)$ the ***differential of f at p***. We call f ***differentiable*** if it is differentiable at every point $p \in U$. We call f ***continuously differentiable*** if the map $Df : p \mapsto Df(p)$ is continuous.

Recall that linear maps are continuous if and only if they are bounded, which is always the case if V is finite-dimensional.

Exercise 1.2. *Verify that if we identify scalars $c \in \mathbb{R}$ with linear maps $x \mapsto c \cdot x$, this definition agrees with the usual definition for $f : \mathbb{R} \to \mathbb{R}$.*

Theorem 1.3. If $f : U \to W$ is differentiable, it is continuous.

Proof. If f is differentiable, we have

$$\lim_{\|h\|_V \to 0} \frac{\|f(p+h) - f(p) - Df(p)(h)\|_W}{\|h\|_V} = 0.$$

This can only be the case if

$$\lim_{\|h\|_V \to 0} \|f(p+h) - f(p) - Df(p)(h)\|_W = 0,$$

i.e. if

$$\lim_{\|h\|_V \to 0} f(p+h) = \lim_{\|h\|_V \to 0} f(p) + Df(p)(h).$$

Since $Df(p)$ is continous with $Df(0) = 0$, this simplifies to

$$\lim_{\|h\|_V \to 0} f(p+h) = f(p),$$

which is the definition of sequential continuity. $\square$

Definition 1.4: Directional and Partial Derivates

Let V and W be normed vector spaces. Let $v \in V$. Let Let $U \subseteq V$ be open and $f : U \to W$. Then we call the limit

$$\frac{\partial}{\partial v} f(p) = \frac{d}{dh} f(p + hv)\bigg|_{h=0} = \lim_{h\to 0} \frac{f(p + hv) - f(p)}{h}$$

the ***directional derivative*** of f at p in direction v. If $(x_0, \dots, x_n)$ is a basis of V, we call the directional derivatives

$$\partial_i f(p) := \frac{\partial}{\partial x_i} f(p)$$

the ***partial derivatives of f at p***. If all partial derivatives exist, we call f ***partially differentiable***. If all maps $\partial_i f : p \mapsto \partial_i f(p)$ are continuous, we call f ***continuously partially differentiable***.

Partial derivatives aren't actually any more complicated than standard one-dimensional derivatives - you can calculate partial derivatives by simply fixing all coordinates except one and then taking a regular derivative in the remaining coordinate. For example, if we have a function

$$\begin{aligned} f : \mathbb{R}^3 &\to \mathbb{R} \\ (x, y, z) &\mapsto f(x, y, z), \end{aligned}$$

we can define a function $f|_1 : \mathbb{R} \to \mathbb{R}$ by restriction to the first coordinate:

$$f|_1(x) := f(x, y, z).$$

Using this function, we can simplify the partial derivative:

$$\begin{aligned} \partial_1 f(x, y, z) &= \lim_{h\to 0} \frac{f((x, y, z) + he_1) - f(x, y, z)}{h} \\ &= \lim_{h\to 0} \frac{f(x + h, y, z) - f(x, y, z)}{h} \\ &= \lim_{h\to 0} \frac{f|_1(x + h) - f(x)}{h} \\ &= \frac{d}{dx} f|_1(x) \end{aligned}$$

This tells us that, for example, the function $f(x, y, z) = x \cdot y \cdot z$ has partial derivatives

$$\begin{aligned} \partial_1 f(x, y, z) &= \frac{d}{dx} xyz = yz \\ \partial_2 f(x, y, z) &= \frac{d}{dy} xyz = xz \\ \partial_3 f(x, y, z) &= \frac{d}{dz} xyz = xy. \end{aligned}$$

Exercise 1.5. 1. *Show that if f is differentiable, then all directional derivatives exist and fulfill $\frac{\partial}{\partial v} f(p) = Df(p)(v)$.*

2. *Conclude that $Df(p)$ is represented in basis $(x_0, \dots, x_n)$ by the* **Jacobian matrix**

$$Jf(p) = \left(\frac{\partial}{\partial x_1} f(p), \dots, \frac{\partial}{\partial x_n} f(p)\right) = \begin{pmatrix} \frac{\partial}{\partial x_1} f_1(p) & \dots & \frac{\partial}{\partial x_n} f_1(p) \\ \vdots & & \vdots \\ \frac{\partial}{\partial x_1} f_n(p) & \dots & \frac{\partial}{\partial x_n} f_n(p) \end{pmatrix}$$

If the image of f is one-dimensional, the Jacobian matrix is simply a row vector. We call the transpose of this vector the ***gradient of f at p*** and denote it

$$\nabla f(p) := Jf(p)^\top = \begin{pmatrix} \frac{\partial}{\partial x_1} f(p) \\ \vdots \frac{\partial}{\partial x_n} f(p) \end{pmatrix}$$

The transposition comes from the fact that we would like to be able to get the directional derivatives of f at p by taking a scalar product with the gradient, which gives us

$$\begin{aligned} \langle \nabla f(p), v \rangle &= \nabla f(p)^\top v \\ &= Jf(p) \cdot v \\ &= Df(p)(v) \\ &= \frac{\partial}{\partial v} f(p) \end{aligned}$$

There is one very important pitfall to watch out for when it comes to differentiability: We have just shown that if f is differentiable, we can get the differential by construction the Jacobian matrix from the partial derivatives. This makes it look like every partially differentiable function is differentiable, which turns out to not be true in general.

For example, there exist functions which are partially differentiable, but not continuous - we showed earlier that all differentiable functions are continuous, so these functions are quick counterexamples.

One example is

$$\begin{aligned} f : \mathbb{R}^2 &\to \mathbb{R} \\ (x, y) &\mapsto \begin{cases} \frac{xy}{(x^2+y^2)^2} & (x, y) \neq 0 \\ 0 & (x, y) = 0 \end{cases}. \end{aligned}$$

This function has $\partial_1 f(0) = \partial_2 f(0) = 0$, since $f(h, 0) = f(0, h) = 0$ for all $h \in \mathbb{R}$, but isn't continuous, since $\lim_{n\to\infty} f\left(\frac{1}{n}, \frac{1}{n}\right) = \lim_{n\to\infty} \frac{n^2}{4} = \infty \neq f(0)$.

For a more involved counterexample, consider the function

$$\begin{aligned} f : \mathbb{R}^2 &\to \mathbb{R} \\ (x, y) &\mapsto \begin{cases} \frac{y^3}{x^2+y^2} & (x, y) \neq 0 \\ 0 & (x, y) = 0 \end{cases} \end{aligned}$$

This function is continuous - for $(x, y) \neq 0$, it is a composition of continuous functions, and thus continuous, and since $x^2 + y^2 \geq y^2$, we additionally have

$$\begin{aligned} |f(x,y)| &= \frac{\left|y^3\right|}{|x^2+y^2|} \\ &= \frac{\left|y^3\right|}{x^2+y^2} \\ &\leq \frac{\left|y^3\right|}{y^2} \\ &= |y|, \end{aligned}$$

so $f(x,y)$ converges to $f(0,0) = 0$ as (x,y) converges to 0, implying that f is also continuous at the origin.

f is also directionally differentiable in all directions:

$$\begin{aligned} \partial_v f(0) &= \lim_{h\to 0} \frac{f(h\cdot v) - f(0)}{h} \\ &= \lim_{h\to 0} \frac{f(h\cdot v)}{h} \\ &= \lim_{h\to 0} \frac{\left(\frac{h^3 v_2^3}{h^2 v_1^2 + h^2 v_2^2}\right)}{h} \\ &= \lim_{h\to 0} \frac{h^3 v_2^3}{h^3 v_1^2 + h^3 v_2^2} \\ &= \lim_{h\to 0} \frac{v_2^3}{v_1^2 + v_2^2} \\ &= \frac{v_2^3}{v_1^2 + v_2^2} \\ &= f(v) \end{aligned}$$

In particular, $\partial_1 f(0) = 0$ and $\partial_2 f(0) = 1$. However, f is not differentiable at 0, since otherwise our representation $Df(0) = Jf(0)$ would give us:

$$\lim_{\vec{h}\to 0} \frac{\left|f(h) - f(0) - Df(0)(\vec{h})\right|}{\left\|\vec{h}\right\|_2} = \lim_{\vec{h}\to 0} \left|\frac{f(h) - Df(0)(h)}{\left\|\vec{h}\right\|_2}\right|$$

$$= \lim_{\vec{h}\to 0} \left|\frac{f(h) - \partial_1 f(0)\cdot h_1 + \partial_2 f(0)\cdot h_2}{\sqrt{h_1^2 + h_2^2}}\right| = \lim_{\vec{h}\to 0} \left|\frac{f(h) - h_2}{\sqrt{h_1^2 + h_2^2}}\right|$$

$$= \lim_{\vec{h}\to 0} \left|\frac{1}{\sqrt{h_1^2 + h_2^2}} \cdot \left(\frac{h_2^3}{h_1^2 + h_2^2} - h_2\right)\right| = \lim_{\vec{h}\to 0} \left|\frac{1}{\sqrt{h_1^2 + h_2^2}} \cdot \left(\frac{h_2^3 - h_2(h_1^2 + h_2^2)}{h_1^2 + h_2^2}\right)\right|$$

$$= \lim_{\vec{h}\to 0} \left|\frac{1}{\sqrt{h_1^2 + h_2^2}} \cdot \left(\frac{h_2^3 - h_2 h_1^2 - h_2^3}{h_1^2 + h_2^2}\right)\right| = \lim_{\vec{h}\to 0} \left|\frac{1}{\sqrt{h_1^2 + h_2^2}} \cdot \left(\frac{h_2 h_1^2}{h_1^2 + h_2^2}\right)\right| = \lim_{\vec{h}\to 0} \left|\frac{h_2 h_1^2}{\sqrt{h_1^2 + h_2^2}^3}\right|$$

Since we are only interested in whether this expression converges to 0, we can square it to simplify slightly:

$$\left| \frac{h_2 h_1^2}{\sqrt{h_1^2 + h_2^2}^3} \right|^2 = \left| \frac{h_2^2 h_1^4}{(h_1^2 + h_2^2)^3} \right|$$

If we now set $h := \left(\frac{1}{n}, \frac{1}{n}\right)$, we get:

$$\left| \frac{h_2^2 h_1^4}{(h_1^2 + h_2^2)^3} \right| = \left| \frac{\frac{1}{n^6}}{\left(\frac{2}{n^2}\right)^3} \right| = \left| \frac{\frac{1}{n^6}}{\frac{8}{n^6}} \right| = \frac{1}{8} \neq 0$$

The important takeaway here is that directional differentiability only guarantees "differentiability along straight paths", while full differentiability also requires "differentiability along curved paths".

Theorem 1.6

$f : V \to W$ is continuously differentiable if and only if it is continuously partially differentiable.

Proof. $\Rightarrow$: We have $\mathrm{Hom}(\mathbb{R}^n, \mathbb{R}^m) \cong \mathbb{R}^{mn}$. If the differential $Jf : \mathbb{R}^n \mapsto \mathbb{R}^{mn}$ exists, it is continuous if and only if every component $(Jf)_i : \mathbb{R}^n \mapsto \mathbb{R}$ is continuous. By the definition of Jf, these are the partial derivatives.

$\Leftarrow$: Since every normed complex vector space is isomorphic to a normed real vector space, we can assume without loss of generality that W is real. Since all norms are equivalent, and since a sequence in W converges iff. all components converge, we can additionally assume that W is one-dimensional, i.e. $W \cong \mathbb{R}$.

We need to show that

$$\lim_{h \to \vec{0}} \frac{f(p+h) - f(p) - Jf(p)h}{\|h\|_2} = \vec{0}$$

Let $h := \sum_{i=1}^n h_i e_i$. We decompose the path from p to $p + h$ into its partial steps, taking one coordinate direction at a time:

$$\forall k \in [0..n] :$$

$$p_k = p + \sum_{i=1}^{k} h_i e_i$$

Now, by the mean value theorem, there exists an $s_k \in [0, 1]$ such that

$$\begin{aligned} f(p_k) - f(p_{k-1}) &= f(p_{k-1} + h_k e_k) - f(p_{k-1}) \\ &= \partial_k f(p_{k-1} - s_k h_n e_n) h_k \end{aligned}$$

Now, since

$$Jf(p)(h) = \sum_{i=1}^{n} \partial_i f(p) h_i,$$

we get:

$$\begin{aligned}
&\frac{|f(p+h)-f(p)-Jf(p)(h)|}{\|h\|_2} \\
=&\frac{1}{\|h\|_2}\left|f(p_n)-f(p_0)-\sum_{i=1}^{n}\partial_i f(p)h_i\right| \\
=&\frac{1}{\|h\|_2}\left|\sum_{i=1}^{n} f(p_i)-f(p_{i-1})-\partial_i f(p)h_i\right| \\
=&\frac{1}{\|h\|_2}\left|\sum_{i=1}^{n} \partial_i f(p_{i-1}+s_i h_i e_i)h_i-\partial_i f(p)h_i\right| \\
=&\frac{1}{\|h\|_2}\left|\sum_{i=1}^{n} (\partial_i f(p_{i-1}+s_i h_i e_i)-\partial_i f(p))h_i\right| \\
=&\frac{1}{\|h\|_2}\left|\sum_{i=1}^{n}\left(\partial_i f\left(p+\sum_{j=1}^{i-1}h_j e_j+s_i h_i e_i\right)-\partial_i f(p)\right)h_i\right| \\
\leq&\frac{1}{\|h\|_2}\sum_{i=1}^{n}\left|\left(\partial_i f\left(p+\sum_{j=1}^{i-1}h_j e_j+s_i h_i e_i\right)-\partial_i f(p)\right)h_i\right| \\
=&\sum_{i=1}^{n}\left|\partial_i f\left(p+\sum_{j=1}^{i-1}h_j e_j+s_i h_i e_i\right)-\partial_i f(p)\right|\cdot\frac{|h_i|}{\|h\|_2} \\
\leq&\sum_{i=1}^{n}\left|\partial_i f\left(p+\sum_{j=1}^{i-1}h_j e_j+s_i h_i e_i\right)-\partial_i f(p)\right|
\end{aligned}$$

Since $s_i \in [0,1]$, we have

$$\lim_{h\to 0} p+\sum_{j=1}^{i-1}h_j e_j+s_i h_i e_i = p$$

And thus, since the partial derivatives $\partial_i f$ are continuous, the right side converges to zero, so f is differentiable.

□

Exercise 1.7. *Let V be an inner product space over a field $\mathbb{F}=\mathbb{R}$ or $\mathbb{F}=\mathbb{C}$. Let $v \in V$. Then the function*

$$\begin{aligned}
\langle\cdot,v\rangle : V &\longrightarrow \mathbb{F} \\
w &\longmapsto \langle w,v\rangle
\end{aligned}$$

is differentiable, with $D\langle\cdot,v\rangle(w)=v^\top$ for all $w\in V$.

Exercise 1.8. *Let b be a symmetric bilinear form. Let*

$$q(v) := b(v,v)$$

be the corresponding quadratic form. Then q is differentiable, with

$$Dq(v)(w) = 2b(v,w)$$

1.2 Rules of Differentiation

Exercise 1.9. *Show that the differential operator D is linear, i.e. that we have*

$$D(\alpha f + \beta g)(p) = \alpha D f(p) + \beta D g(p)$$

for f, g differentiable at p.

Theorem 1.10: Chain Rule in $\mathbb{R}^n$

Let $f : U \to \mathbb{R}^m$ and $g : V \to \mathbb{R}^r$ with $U \subset \mathbb{R}^n$, $V \subset \mathbb{R}^m$ open and $f(U) \subseteq V$. Let f be differentiable at a point $p \in U$ and let g be differentiable at $f(p)$. Then

$$D(g \circ f)(p) = Dg(f(p)) \circ Df(p)$$

Exercise 1.11. *Verify that this version of the chain rule agrees with the familiar chain rule for $f, g : \mathbb{R} \to \mathbb{R}$.*

Proof. By definition of the differential, we have:

$$\begin{aligned} f(x) = f(p) & \\ &+ Df(p)(x-p) \\ &+ \varepsilon_f(x)\|x-p\| \\ g(f(x)) = g(f(p)) & \\ &+ Dg(f(p))(f(x)-f(p)) \\ &+ \varepsilon_g(f(x))\|f(x)-f(p)\| \end{aligned}$$

For some functions $\varepsilon_f : U \to \mathbb{R}^m$, $\varepsilon_g : U \to \mathbb{R}^r$ which are continuous in p and $f(p)$ and fulfill $\varepsilon_f(p) = 0$ and $\varepsilon_g(f(p)) = 0$.
The second equation tells us:

$$\begin{aligned} g(f(x)) = g(f(p)) & \\ &+ Dg(f(p))(f(x)-f(p)) \\ &+ \varepsilon_g(f(x))\|f(x)-f(p)\| \end{aligned}$$

Plugging in our equation for $f(x)$ now lets us regroup our terms into an equation of the same form as the definition of the differential:

$$\begin{aligned}
g(f(x)) = g(f(p))& \\
&+ Dg(f(p))\left[\overbrace{f(p) + Df(p)(x-p) + \varepsilon_f(x)\|x-p\|}^{f(x)} - f(p)\right] \\
&+ \varepsilon_g(f(x))\left\|\overbrace{f(p) + Df(p)(x-p) + \varepsilon_f(x)\|x-p\|}^{f(x)} - f(p)\right\|
\end{aligned}$$

$$\begin{aligned}
&= g(f(p)) \\
&\quad + Dg(f(p))\left[Df(p)(x-p) + \varepsilon_f(x)\|x-p\|\right] \\
&\quad + \varepsilon_g(f(x))\left\|Df(p)(x-p) + \varepsilon_f(x)\|x-p\|\right\|
\end{aligned}$$

$$\begin{aligned}
&= g(f(p)) \\
&\quad + Dg(f(p))Df(p)(x-p) \\
&\quad + Dg(f(p))\varepsilon_f(x)\|x-p\| \\
&\quad + \varepsilon_g(f(x))\left\|Df(p)(x-p) + \varepsilon_f(x)\|x-p\|\right\|
\end{aligned}$$

$$\begin{aligned}
&= g(f(p)) \\
&\quad + Dg(f(p))Df(p)(x-p) \\
&\quad + \underbrace{\left(\varepsilon_f(x)Dg(f(p)) + \varepsilon_g(f(x))\left\|Df(p)\left(\frac{x-p}{\|x-p\|}\right) + \varepsilon_f(x)\right\|\right)}_{:=\varepsilon_{g\circ f}(x)}\|x-p\| \\
&= g(f(p)) \\
&\quad + (Dg(f(p)) \circ Df(p))(x-p) \\
&\quad + \varepsilon_{g\circ f}(x)\|x-p\|
\end{aligned}$$

All that remains is to show that $\varepsilon_{g\circ f}(x)$ is continuous if we set

$$\varepsilon_{g\circ f}(p) = 0.$$

Since we have $\varepsilon_f(p) = 0$ and $\varepsilon_g(f(p)) = 0$, we have our desired result if we can show that

$$\left\|Df(p)\left(\frac{x-p}{\|x-p\|}\right)\right\| = \frac{\|Df(p)(x-p)\|}{\|x-p\|}.$$

is bounded. Since $\mathbb{R}^n$ is finite-dimensional, we can find a constant c such that

$$\|Df(p)(x-p)\| \leq c \cdot \|x-p\|,$$

which means that our quotient is indeed bounded, completing the proof.

□

Theorem 1.12: Product Rule in $\mathbb{R}^n$

Let $U \subseteq \mathbb{R}^n$. Let $f, g : U \to \mathbb{R}$ at a point $p \in U$. Then

$$D(f \cdot g)(p) = Df(p) \cdot g(p) + f(p) \cdot Dg(p).$$

Exercise 1.13. *Verify that this version of the product rule agrees with the familiar product rule for $f, g : \mathbb{R} \to \mathbb{R}$.*

Proof. We define the multiplication function $m(x, y) := x \cdot y$. The partial derivatives are $\partial_1 m(x, y) = y$ and $\partial_2 m(x, y) = x$, i.e.

$$Dm(x, y) = \begin{pmatrix} y & x \end{pmatrix}$$

Applying the chain rule now gives

$$\begin{aligned} D(f \cdot g)(p) &= D(m \circ (f, g))(p) \\ &= Dm((f, g)(p)) \circ D(f, g)(p) \\ &= Dm(f(p), g(p)) \circ (Df(p), Dg(p)) \\ &= Dm(f(p), g(p)) \cdot (Df(p), Dg(p)) \\ &= \begin{pmatrix} g(p) & f(p) \end{pmatrix} \cdot \begin{pmatrix} Df(p) \\ Dg(p) \end{pmatrix} \\ &= Df(p) \cdot g(p) + f(p) \cdot Dg(p). \end{aligned}$$

□

Exercise 1.14. *Let $f : V \to W$ be differentiable. Let $\gamma : [0, 1] \to V$ be differentiable. Then $f \circ \gamma$ is differentiable with*

$$(f \circ \gamma)'(t) = Df(\gamma'(t)) \cdot \gamma(t)$$

Theorem 1.15. Let $U \subseteq V$ be open. Let $f : U \to W$ be differentiable with $Df = 0$. Then f is locally constant.

Corollary 1.16. Let $U \subseteq V$ be open and connected. Let $f : U \to W$ be differentiable with $Df = 0$. Then f is constant.

1.3 Higher Partial Derivatives

Theorem 1.17: Schwartz's Theorem

Let $f : U \subseteq V \to W$ be twice continuously partially differentiable. Then the order of partial derivation does not matter, i.e.

$$\partial_i \partial_j f = \partial_j \partial_i f$$

for all $i, j \in [1..n]$.

Theorem 1.18: First-Order Taylor Expansion in multiple Variables

Let $f : U \subseteq V \to W$. Let f be continuously differentiable. Let $x \in U$, $h \in V$ fulfill $[x, x+h] \subseteq V$. Then

$$f(x+h) = f(x) + Df(x)(h) + r(h),$$

with $r \in o(\|h\|)$.

1.4 Differential Operators

Chapter 2

The Two Queens of Multivariable Calculus

2.1 The Inverse Function Theorem

Theorem 2.1: Inverse Function Theorem

Let $U \subseteq \mathbb{R}^n$ be open. Let $f : U \to \mathbb{R}^n$ be continuously differentiable. Let $x \in U$ be a point such that $Df(x) : \mathbb{R}^n \to \mathbb{R}^n$ is invertible. Then there exists an open neighbourhood V of x such that:

1. $f(V)$ is an open neighbourhood of $f(x)$.
2. $f|_V$ is a C^1-diffeomorphism, i.e. invertible with continuously differentiable inverse.

In short: if a continuously differentiable function has an invertible differential at a given point, there exists an open neighbourhood around that point in which the function is invertible:

Corollary 2.2. Let $U \subseteq \mathbb{R}^n$ be open. Let $f : U \to \mathbb{R}^n$ be continuously differentiable. Let $Df(x)$ be invertible for all $x \in \mathbb{R}^n$. Then $f(U)$ is open.

2.2 The Implicit Function Theorem

Theorem 2.3: Implicit Function Theorem

Let $U \subseteq \mathbb{R}^m \times \mathbb{R}^k$ be open. Let $f : U \to \mathbb{R}^k$ be continuously differentiable. Let $(x, y) \in U$ with $f(x, y) = z$. Let $Df|_{\mathbb{R}^k}(x, y) : \mathbb{R}^k \to \mathbb{R}^k$ be invertible. Then there exist

1. open neighbourhoods V of x and W of y,
2. and a continuously differentiable function $g : V \to W$,

such that the set of values in $V \times W$ for which f takes the value z is the graph of g, i.e.

$$\{(v, w) \in V \times W \mid f(v, w) = z\} = \{(v, g(v)) \mid v \in V\}$$

In short: under mild assumptions, the level sets of a continuously differentiable function are graphs of continuously differentiable functions:

Chapter 3

Multiple Integrals

Lemma 3.1. Let $K \subseteq \mathbb{R}^n$ be compact. Let $f : K \times [a, b] \to \mathbb{R}$ be continuous. Then the function

$$x \mapsto \int_a^b f(x, t)\, dt$$

is uniformly continuous.

> *Proof.* $K \times [a, b]$ is a finite product of compact sets, i.e. compact by Theorem 2.32. Thus, f is uniformly continuous by Theorem 2.5, which means that for every $\varepsilon > 0$, there exists a $\delta > 0$ such that for any $x, y \in \mathbb{R}, t \in \mathbb{R}^n$
>
> $$d((x, t), (y, t)) < \delta \implies d(f(x, t), f(y, t)) < \varepsilon.$$
>
> This tells us that for any $x, y \in \mathbb{R}$,
>
> $$\begin{aligned} d(x, y) < \delta &\implies d((x, t), (y, t)) < \varepsilon \\ &\implies |f(x, t) - f(y, t)| = d(f(x, t), f(y, t)) < \varepsilon \\ &\implies d\left(\int_a^b f(x, t)\, dt, \int_a^b f(y, t)\, dt\right) \\ &\qquad = \left|\int_a^b f(x, t)\, dt - \int_a^b f(y, t)\, dt\right| \\ &\qquad \leq \int_a^b |f(x, t) - f(y, t)|\, dt \\ &\qquad \leq (b - a)\varepsilon, \end{aligned}$$
>
> since ε was arbitrary, our function is uniformly continuous as desired. $\square$

Definition 3.2. Given a compact cuboid $Q = \prod_{i=1}^n [a_i, b_i] \subseteq \mathbb{R}^n$ and a continuous function $f : Q \to \mathbb{R}$, we declare the ***integral of f over Q*** to be the nested integral

$$\int_Q f(x)\, d^n x := \int_{a_n}^{b_n} \dots \int_{a_1}^{b_1} f(x_1, \dots, x_n) dx_1 \dots dx_n$$

By Lemma 3.1, all involved functions are continuous, and thus their integrals are indeed well-defined. This definition will become a fully justified theorem once we begin our study of the Lebesgue integral later on.

Exercise 3.3. *Verify that linearity and monotonicity carry over from the single-variable case, i.e show that for any $f, g : Q \to \mathbb{R}$ and $\lambda \in \mathbb{R}$, we have:*

1. $\int_Q f(x) + g(x)\, d^n x = \int_Q f(x)\, d^n x + \int_Q g(x)\, d^n x$
2. $\int_Q \lambda f(x)\, d^n x = \lambda \int_Q f(x)\, d^n x$
3. $f \leq g \implies \int_Q f(x)\, d^n x \leq \int_Q g(x)\, d^n x$
4. $\left|\int_Q f(x)\, d^n x\right| \leq \int_Q |f(x)|\, d^n x$

Definition 3.4. We define

$$\operatorname{vol} Q = \int_Q 1 = \prod_{i=1}^{n} |b_i - a_i|$$

Definition 3.5. By partitioning the intervals $[a, b] \times [c, d] := Q \subseteq \mathbb{R}^2$ into r equidistant chunks with boundaries

$$a = a_1 \leq a_2 \leq \ldots \leq a_r = b$$
$$c = c_1 \leq c_2 \leq \ldots \leq c_r = d,$$

we partition Q into r^2 equally sized smaller rectangles

$$Q_{ij} := [a_i, c_{i+1}] \times [a_j, c_{j+1}]$$

which must each have area $\operatorname{vol} Q/r^2$. We can thus define ***two-dimensional Riemann sums*** analogously to how we did it in one dimension:

$$S_r^2(f) := \sum_{i=0}^{r-1} \sum_{j=0}^{r-1} f(a_i, c_j) \frac{\operatorname{vol} Q}{r^2}$$

Lemma 3.6: Multivariable Integration via Riemann Sums

Given a compact cuboid $Q \subseteq \mathbb{R}^2$ and a continuous function $f : Q \to \mathbb{R}$, we have

$$\int_Q f(x)\, d^2 x = \lim_{r\to\infty} S_r^2(f)$$

Proof. Applying the single-variable version of this theorem twice, we get:

$$\begin{aligned}\int_Q f(x)\, d^2 x &= \int_c^d \int_a^b f(x_1, x_2)\, dx_1\, dx_2 \\ &= \lim_{r\to\infty} S_r\left(\int_a^b f(x_1, x_2)\, dx_1\right) \\ &= \lim_{r\to\infty} S_r^2(f(x_1, x_2))\end{aligned}$$

□

Chapter 4

Curves

Chapter 5

Surfaces

Chapter 6

Measure Theory

6.1 The Measure Problem

The most basic goal of measure theory is to establish a generalized notion of a "measure function", which assigns a "volume" to a given set. In particular, we would like to establish a function that assigns volume functions to subsets of $\mathbb{R}^n$ and has the following properties:

1. When given a subset with an easily intuitively definable volume, the volume function should agree with that volume. In particular, the volume of a cuboid should be the product of the lengths of its sides, and the length of a real interval (a, b) should be $b - a$.

2. The volume of a countable disjoint union of sets should be the sum of the individual volumes. This property is generally referred to as *σ-additivity*.

3. The volume should be invariant under isometries, i.e. functions like rotations, translations, and reflections should not change the volume of a set.

4. We want our volume to be a positive real number.

We call a σ-additive function a *measure* - we want to eventually find a meausure on $\mathbb{R}^n$ that also fulfills the other properties, but we want measures to be far more generally applicable, and as such we want to be able to define measures much more general spaces that may not have a defined notion of an interval or an isometry.

It will turn out that there exists exactly one function on $\mathbb{R}^n$, called the *Lebesque measure* λ^n, that fulfills these conditions for a very large family of sets - enough to include every "somewhat reasonable" subset of $\mathbb{R}^n$. However, there are still counterexamples.

Proposition 6.1. Every subset of $\mathbb{R}^n$ being Lebesque-measurable is consistent with ZF (without the axiom of choice).

Theorem 6.2. Assuming the axiom of choice, there exist subsets of $\mathbb{R}^n$ that cannot be assigned a volume without arriving at a contradiction.

It turns out that this result crucially relies on the full axiom of choice, and in particular is not implied by commonly used weaker forms of the axiom of choice such as the axiom of dependent choice.

The following two subsections deal with two different ways of proving this theorem by construction *non-measurable sets*: The *Vitali Sets*, and the decomposition of a sphere given in the *Banach-Tarski-Paradox*.

6.1.1 Vitali Sets

Proposition 6.3. The relation $x \sim y \Leftrightarrow x - y \in \mathbb{Q}$ is an equivalence relation on the real numbers.

Theorem 6.4. There exist sets $V \subset [0,1]$ such that for each $r \in \mathbb{R}$, there exists exactly one number $v \in V$ such that $v - r$ is rational. We call such a set a **Vitali Set**.

> *Proof.* Consider the aforementioned equivalence relation $x \sim y$ on $\mathbb{R}$. Each equivalence class must contain at least one representative also contained in $[0,1]$, since if $x - y = q \in \mathbb{Q}$, we have $x - (y + q) \in \mathbb{Q}$ for all $q \in \mathbb{Q}$, letting us pick a q such that $x - (y + q) \in [0,1]$. Being equivalence classes, they must also be disjoint.
>
> This means we can use the axiom of choice to pick exactly one element of each equivalence class of $\sim$, giving us our set V. □

Lemma 6.5. Let $q_1, q_2, \ldots$ be an enumeration of $\mathbb{Q} \cap [-1,1]$. Then for $j \neq k$, we have

$$(q_j + V) \cap (q_k + V) = \emptyset$$

> *Proof.* Assume the intersection is non-empty. Then there exist $v_1, v_2 \in V$ such that $q_j + v_1 = q_k + v_2$, meaning $v_1 - v_2 \in \mathbb{Q}$. Therefore, v_1 must be in the same equivalence class as v_2. Since V contains exactly one element of each equivalence class, we have $v_1 = v_2$, and therefore we also have $q_j = q_k$, i.e. $j = k$. □

Lemma 6.6. We have

$$[0,1] \subset \bigcup_{k \in \mathbb{N}} (q_k + V) \subset [-1,2]$$

> *Proof.* 1. $\bigcup_{k \in \mathbb{N}} (q_k + V) \subset [-1,2]$ follows trivially from $q_k \in [-1,1]$ and $V \subset [0,1]$.

2. $[0,1] \subset \bigcup_{k\in\mathbb{N}}(q_k + V)$ follows from the definition of V, since for every $y \in [0,1]$ we have a unique $v \in V$ such that $y - v := q \in \mathbb{Q}$, and since $y \in [0,1]$ and $v \in [0,1]$, we have $q \in [-1,1]$, i.e. q is contained in our enumeration.

□

Corollary 6.7. Vitali sets are not measurable by a translation-invariant normed measure.

Proof. Assume that λ is translation-invariant, countably additive and that $\lambda([a,b]) = b - a$. Then we have:

$$1 = \lambda([0,1]) \leq \lambda\left(\bigcup_{k\in\mathbb{N}}(q_k + V)\right) \leq \lambda([-1,2]) = 3$$

Since our measure is countably additive and invariant under isometries, we can translate each individual set and preserve the measure:

$$\begin{aligned}\lambda\left(\bigcup_{k\in\mathbb{N}}(q_k + V)\right) &= \sum_{k\in\mathbb{N}} \lambda(q_k + V)\\ &= \sum_{k\in\mathbb{N}} \lambda(V)\end{aligned}$$

Now, if $\lambda(V) \leq 0$, we wouldn't have $1 \leq \sum_{k\in\mathbb{N}} \lambda(V)$, and if $\lambda(V) > 0$, we wouldn't have $\sum_{k\in\mathbb{N}} \lambda(V) \leq 3$. Therefore every possible measure we could assign to V leads to a contradiction. □

6.1.2 The Banach-Tarski Paradox

Theorem 6.8. Given any two sets $A, B \subset \mathbb{R}^n$, with $n \geq 3$, such that both A and B have a nonempty interior, there exist disjoint decompositions $A_1 \sqcup \ldots \sqcup A_k = A$ and $B_1 \sqcup \ldots \sqcup B_k = B$ such that for each i, A_i and B_i can be transformed into each other by an isometry.

Corollary 6.9. The unit sphere can be transformed into two copies of the unit sphere by a finite disjoint decomposition followed by an isometry. The subsets of the decomposition therefore violate the countable union condition we expect from measure functions, and can therefore not be a assigned a meaningful volume.

6.2 Lattices and Boolean Algebras

We have just seen that we cannot define our desired measure function on the full power set of $\mathbb{R}^n$. This means we will have to work on smaller systems of subsets of a given set. Naturally, we want to find the largest such systems that are still well-behaved enough to allow us to define a sensible notion of measure.

We will arrive at different algebraic structures on subsets of power sets, which will serve as the domains of our measure functions.

In order to gain a full birds-eye view of these definitions, we will first introduce the more general notion of *boolean algebras*:

6.2.1 Boolean Algebras

Definition 6.10. A **boolean algebra** is a set X, equipped with two binary operations $\wedge$ and $\vee$, a unary operation $\neg$, and two elements 0 and 1, such that:

1. $\wedge$ and $\vee$ are commutative,
2. 1 is a neutral element of $\wedge$, and 0 is a neutral element of $\vee$,
3. $\wedge$ distributes over $\vee$ and $\vee$ distributes over $\wedge$,
4. $x \wedge \neg x = 0$, and $x \vee \neg x = 1$.

Corollary 6.11. Any boolean algebra also has the following properties:

1. $\wedge$ and $\vee$ are associative,
2. $\wedge$ and $\vee$ have the following **absorption property**:

$$a \wedge (a \vee b) = a$$
$$a \vee (a \wedge b) = a,$$

3. $a = b \wedge a$ if and only if $a \vee b = b$.

There are three "central" boolean algebras, from which most of the terminology describing them is descended:

Theorem 6.12. The set of **propositional formulas** forms a boolean algebra, where 0 is the logical falsum ($\perp$, an unfulfillable formula), 1 is the logical verum ($\top$, a tautological formula), and $\wedge$, $\vee$ and $\neg$ are logical "and", "or" and "not".

In computer science and circuit engineering, one often considers the subalgebra of this boolean algebra where every formula is directly evaluated to "0" or "1".

Theorem 6.13. (Power set algebra): The **power set** $\mathcal{P}(X)$ of any set X forms a boolean algebra, where $0 = \emptyset$, $1 = X$, $\wedge$ is the set intersection operation $\cap$, $\vee$ is the set union operation $\cup$, and $\neg$ is the set complement operation $M \to X \setminus M$.

Theorem 6.14. (Restrictions of Boolean algebras): Let $\mathcal{B}$ be a Boolean algebra on a set X, and let Y be a subset of X. Then the **restriction of $\mathcal{B}$ to Y**, defined as

$$\mathcal{B}|_Y := \{E \cap Y \mid E \in \mathcal{B}\},$$

is a boolean Algebra on Y.

Theorem 6.15. If $Y \in \mathcal{B}$, then

$$\mathcal{B}|_Y = \mathcal{B} \cap \mathcal{P}(Y) = \{E \subset Y \mid E \in \mathcal{B}\}$$

Theorem 6.16. (Atomic algebra): Let X be partioned into a union

$$X = \bigcup_{\alpha \in I} A_\alpha$$

of disjoint sets A_α, which we refer to as *atoms*. Then this partition forms a Boolean algebra

$$\mathcal{A}((A_\alpha)_{\alpha \in I}) := \left\{ E \;\middle|\; E = \bigcup_{\alpha \in J} A_\alpha, J \subset I \right\}$$

of all the sets that can be represented as a union of atoms.

The power set Algebra on X is exactly the atomic algebra where X is partitioned into singleton atoms.

Theorem 6.17. Atomic algebras are uniquely determined by their atoms, up to relabeling. More precisely: Let $(A_\alpha)_{\alpha \in I}$ and $(B_\beta)_{\beta \in J}$ be two partitions of a set X. Then

$$\mathcal{A}((A_\alpha)_{\alpha \in I}) = \mathcal{A}((B_\beta)_{\beta \in J})$$

if and only if there exists a bijection $\varphi : I \longrightarrow J$ such that $B_{\varphi(\alpha)} = A_\alpha$ for all $\alpha \in I$.

Theorem 6.18. Every finite Boolean algebra is an atomic algebra.

Corollary 6.19. Every finite Boolean algebra has cardinality 2^n, where $n \in \mathbb{N}$.

Corollary 6.20. There is a one-to-one correspondence, up to relabeling, between finite Boolean algebras on a set X and finite partitions of X into non-empty sets.

Theorem 6.21. (Dyadic algebras): Let $n, i_1, \ldots, i_d \in \mathbb{Z}$. The **dyadic algeba** $\mathcal{D}_n(\mathbb{R}^d)$ at scale 2^{-n} in $\mathbb{R}^d$ is the atomic algebra generated by the products of the half-open dyadic intervals

$$I_j := \left[\frac{i_j}{2^n}, \frac{i_j + 1}{2^n} \right)$$

of length 2^{-n}.

This algebra consists exactly of the "grid figures" made up of a finite number of "pixels" of length 2^{-n}.

Theorem 6.22. (Intersection of Boolean Algebras): The intersection of a family $(\mathcal{B}_\alpha)_{\alpha \in I}$ of Boolean algebras on a set X is again a Boolean algebra, assuming the convention that, if I is empty, the intersection is the full power set. Furthermore, this Intersection is the finest Boolean algebra that is coarser than every $\mathcal{B}_\alpha$.

Definition 6.23. Let $\mathcal{F}$ be any family of subsets of a set X. Then we define $\langle \mathcal{F} \rangle_{\mathcal{B}}$ to be the intersection of all Boolean algebras that contain $\mathcal{F}$. We call this the **Boolean algebra generated by** $\mathcal{F}$.

Equivalently, $\langle \mathcal{F} \rangle_{\mathcal{B}}$ is the smallest Boolean algebra containing $\mathcal{F}$.

Theorem 6.24. $\mathcal{F}$ is a Boolean algebra if and only if $\langle \mathcal{F} \rangle_{\mathcal{B}} = \mathcal{F}$.

6.2.2 Lattices

Boolean algebras themselves turn out to be specific instances of *lattices*, which play an important role in order theory and universal algebra.

Definition 6.25. A **lattice** is an algebraic structure $(L, \vee, \wedge)$, consisting of a set L, an operation $\vee$, called **join**, and an operation $\wedge$, called **meet**, such that the absorbtion laws $a \vee (a \wedge b) = a$ and $a \wedge (a \vee b) = a$.

Proposition 6.26. Equivalently, a partially ordered set $(L, \leq)$ is a lattice if every pair of elements has a least upper bound $\sup(a, b) := a \vee b \in L$ and a greatest lower bound $\inf(a, b) := a \wedge b \in L$.

Definition 6.27. We call a lattice **bounded** if there exists a **least element** 0, i.e. 0 fulfills $a \vee 0 = a$, and a **greatest element** 1, which fulfills $a \wedge 1 = a$.

Corollary 6.28. A boolean algebra is a bounded lattice such that meet and join are distributive over each other and such that complements exist.

6.3 Set Algebras

Definition 6.29. Let X be a set and $\mathcal{A} \subset \mathcal{P}(X)$. Then we call $\mathcal{A}$ a **set algebra** if it has the following properties:

1. $\emptyset \in \mathcal{A}$
2. For any $A \in \mathcal{A}$, we have $X \setminus A \in \mathcal{A}$ ($\mathcal{A}$ is closed under the operation of taking complements).
3. For any $F, G \in \mathcal{A}$, we have $F \cup G$ in $\mathcal{A}$ ($\mathcal{A}$ is closed under binary unions).

Corollary 6.30. If $\mathcal{A}$ is a set algebra on X, it also fulfills the following:

1. $X \in \mathcal{A}$,
2. For any $F, G \in \mathcal{A}$, we have $F \cap G$ in $\mathcal{A}$,
3. For any $A_1, \ldots, A_n \in \mathcal{A}$, we have $\bigcup_{i=1}^{n} A_i \in \mathcal{A}$,
4. For any $A_1, \ldots, A_n \in \mathcal{A}$, we have $\bigcap_{i=1}^{n} A_i \in \mathcal{A}$.

Thus, we obtain the following more concise (but less readable) definition of set algebras:

Corollary 6.31. A set algebra is a subalgebra of the power set boolean algebra on X.

Corollary 6.32. A topology on X is "simply" a set algebra on X that is closed under arbitrary unions.

Theorem 6.33. (Stone's Representation Theorem for Boolean Algebras): Every boolean algebra is isomorphic to a set algebra.

6.3.1 Set Rings

A very important weakening of the concept of set algebras is given by *set rings*, which contain the empty set and are closed under intersection and union, but don't have to contain the full set X or be closed under complements.

Theorem 6.34. Let $\mathcal{A}$ be a set ring. Then it is closed under finite symmetric difference, and forms a ring in the algebraic sense, with symmetric difference as addition and intersection as multiplication. If it contains the full set X, it forms a ring with identity.

Definition 6.35. Let $I \subset \mathbb{R}$. We call I an **interval** if there exist $a, b \in \mathbb{R}$ such that $(a, b) \subset I \subset [a, b]$.

Theorem 6.36. The set of subsets of the real numbers which can be written as a finite union of intervals forms a set ring.

Theorem 6.37. Let $\mathcal{R}$ and $\mathcal{S}$ be set rings. Then the set of finite unions of cartesian products of elements of R_i and S_i, i.e. of elements of the form,

$$\bigcup_{i \in \mathbb{N}} R_i \times S_i,$$

is also a set ring, which we will denote by $\mathcal{R} \boxtimes \mathcal{S}$.

Corollary 6.38. The set of finite unions of cuboids in $\mathbb{R}^n$, where we define a cuboid to be a product of arbitrary intervals, i.e. we don't care if the boundary on any particular side is open or closed, forms a set ring. We call sets of this form **Elementary Sets**.

6.3.2 Set Semirings

Theorem 6.39. Let $\mathcal{S} \subset \mathcal{P}(X)$. We call $\mathcal{S}$ a **set semiring**, or **semiring of sets**, if:

1. $\emptyset \in \mathcal{S}$,
2. $\mathcal{S}$ is closed under finite intersections,
3. For $A, B \in \mathcal{S}$, there exist disjoint sets $S_1, \ldots, S_n \in \mathcal{S}$ such that $A \setminus B = \bigcup_{i=i}^{n} S_i$.

This means that a set semiring is a weakened form of a set ring where complements are not necessarily contained in the semiring, but can still be "constructed" from elements of the ring. Any set ring is therefore immediately also a set semiring.

Sadly, unlike with rings of sets, there is absolutely no connection between set semirings and the algebraic notion of a semiring - a semiring of sets is exclusively a (semi)(ring of sets), and *not* a (semiring)(of sets). This makes it tempting for me to use an alternative name which makes this distinction more clear, but since I haven't encountered any good alternative names anywhere else (and because I already know I will forget to stick with this convention moving forwards) I will stick with the less than perfect established name.

Set semirings are of fundamental importance to measure theory because the set of cuboids in $\mathbb{R}^n$ forms a set semiring, and we will end up defining our lebesque measure by approximating sets through coverings of the set with cuboids. Of course, we first have to establish a basic theory of set semirings and prove this claim.

Theorem 6.40. The set $\mathcal{I}$ of real intervals forms a set semiring.

Theorem 6.41. The product of two set semirings is again a set semiring.

Corollary 6.42. The set Q of cuboids in $\mathbb{R}^n$ (once again with both open and closed sides allowed) forms a set semiring.

6.4 σ-Algebras

The most important type of set algebra for the purposes of measure theory is the *σ-Algebra*, on which we will eventually define the notion of a "measure" in our desired final form.

Definition 6.43: σ-Algebra

Let X be an arbitrary set and $\mathcal{A} \subset \mathcal{P}(X)$. We call $\mathcal{A}$ a **σ-Algebra on X** if:

1. $X \in \mathcal{A}$
2. For all $A \in \mathcal{A}$, we have $X \setminus A \in \mathcal{A}$ ($\mathcal{A}$ is closed under the operation of taking a complement).
3. For all $A_i \in \mathcal{A}$, we have $\bigcup_{i \in \mathbb{N}} \in \mathcal{A}$ ($\mathcal{A}$ is closed under the operationg of taking countable unions).

If $\mathcal{A}$ is a σ-algebra on X, we call $(X, \mathcal{A})$ a **measure space**, and any set $A \in \mathcal{A}$ **$\mathcal{A}$-measurable**.

The "σ" here once again stands for "countable sum", as it also did for σ-additivity and F_σ sets.

Corollary 6.44. Any σ-algebra contains the empty set and is closed under countable intersection.

Corollary 6.45. A σ-algebra can be more concisely defined as a set algebra that is closed under *countable* union and intersection, not just finite ones.

This also trivially makes every σ-algebra a Boolean algebra.

Theorem 6.46. Every atomic algebra is a σ-algebra.

Theorem 6.47: Induced σ-algebra

Just like for Boolean algebras, the restriction $\mathcal{A}|_Y$ of a σ-algebra $\mathcal{A}$ on X to a subset $Y \subset X$, defined by

$$\mathcal{A}|_Y = \{A \cap Y \mid A \in \mathcal{A}\},$$

is again a σ-algebra on Y.

Theorem 6.48. We have $\mathcal{A}|_Y \subset \mathcal{A}$ if and only if $Y \in \mathcal{A}$.

Proof.

1. Assume $\mathcal{A}|_Y \subset \mathcal{A}$. We have $X \cap Y = Y \in \mathcal{A}|Y$, and thus we get $Y \in \mathcal{A}$.
2. Assume $Y \in \mathcal{A}$. Then since σ-algebras are closed under intersection, we have $A \cap Y \in \mathcal{A}$ for all $A \in \mathcal{A}$, i.e. we have $\mathcal{A}|_Y \subset \mathcal{A}$.

$\square$

Theorem 6.49. Let $X \neq \emptyset$, let $(Y, \mathcal{B})$ be a measurable space and let $f : X \to Y$ be an arbitrary map. Then

$$f^{-1}(\mathcal{B}) = \left\{ f^{-1}(B) \,\middle|\, B \in \mathcal{B} \right\}$$

is a σ-algebra.

Note however that we don't necessarily have $f^{-1}(\mathcal{B}) = \mathcal{A}$, or even $f^{-1}(\mathcal{B}) \subset \mathcal{A}$! We call maps that fulfill the latter property *measurable*. Measurable maps play a key role in measure theory and especially in integration theory, entirely analogous to the role continuous functions play in topology. We will study them in more detail in a later chapter.

Theorem 6.50. The intersection of arbitrarily many σ-algebras is on the same set X is once again a σ-algebra.

Corollary 6.51: Generated σ-algebra

Let $\mathcal{E} \subset \mathcal{P}(X)$. Then $\langle \mathcal{E} \rangle_\sigma$ denotes the intersection of all σ-algebras containing $\mathcal{E}$.

In particular, note that if $\mathcal{A}$ is a σ-Algebra containing $\mathcal{E}$, we have

$$\langle \mathcal{E} \rangle_\sigma \subseteq \mathcal{A}.$$

Theorem 6.52. We have $\langle \mathcal{F} \rangle_{\mathcal{B}} = \langle \mathcal{F} \rangle_\sigma$ if and only if $\langle \mathcal{F} \rangle_{\mathcal{B}}$ is a σ-algebra.

Theorem 6.53. Let X be any set. Let $\mathcal{E}_i \subset \mathcal{P}(X)$ be a family of sets of subsets of X indexed by $i \in I$. Then we have:

$$\left\langle \bigcup_{i \in I} \langle \mathcal{E}_i \rangle_\sigma \right\rangle_\sigma = \left\langle \bigcup_{i \in I} \mathcal{E}_i \right\rangle_\sigma$$

6.4.1 The Borel σ-Algebra

Definition 6.54. Let X be a topological space. The **Borel σ-algebra $\mathcal{B}[X]$ of X** is the σ-Algebra generated by the open subsets of X.

Theorem 6.55. The Borel σ-Algebra $\mathcal{B}[\mathbb{R}^d]$ is equivalently generated by any of the following:

1. The closed subsets of $\mathbb{R}^d$,
2. The compact subsets of $\mathbb{R}^d$,
3. The open balls of $\mathbb{R}^d$,
4. The boxes in $\mathbb{R}^d$,
5. The elementary sets in $\mathbb{R}^d$.

6.5 Dynkin Systems

Definition 6.56: Dynkin System

Let $\mathcal{D} \subset \mathcal{P}(X)$. Then we call $\mathcal{D}$ a **Dynkin system** iff:

1. $X \in \mathcal{D}$
2. $\mathcal{D}$ is closed under complements $B \setminus A$ as long as $A \subset B$
3. $\mathcal{D}$ is closed under coutable pairwise disjoint union

Theorem 6.57

A Dynkin system is a σ-algebra iff. it is closed under intersection.

Theorem 6.58. Let $\mathcal{E} \subset \mathcal{P}(X)$ be closed under intersection. Then $\langle \mathcal{E} \rangle_{\mathcal{D}} = \langle \mathcal{E} \rangle_{\sigma}$.

6.6 Monotone Classes

Definition 6.59: Monotone classes

Let $\mathcal{M} \subset \mathcal{P}(X)$. Then we call $\mathcal{M}$ a **monotone class** if:

1. For any family $(A_i)_{i\in\mathbb{N}} \subset \mathcal{M}$ such that $A_i \subset A_{i+1}$ holds for all $i \in \mathbb{N}$, we have
$$\bigcup_{i=1}^{\infty} A_i \in \mathcal{M}$$
2. For any family $(A_i)_{i\in\mathbb{N}} \subset \mathcal{M}$ such that $A_i \supset A_{i+1}$ holds for all $i \in \mathbb{N}$, we have
$$\bigcap_{i=1}^{\infty} A_i \in \mathcal{M}$$

We denote the smallest monotone class containing a set $\mathcal{S} \subset \mathcal{P}(X)$ as $\langle \mathcal{S} \rangle_{\mathcal{M}}$.

Corollary 6.60. Every σ-algebra is a monotone class.

Corollary 6.61. Given $\mathcal{F} \subset \mathcal{P}(X)$, we have $\langle \mathcal{F} \rangle_{\mathcal{M}} \subseteq \langle \mathcal{F} \rangle_{\sigma}$

Recall that a *set algebra* is a set $\mathcal{A} \subset \mathcal{P}(X)$ that contains the empty set and is closed under complements and binary unions.

Theorem 6.62. Let $\mathcal{F} \subset \mathcal{P}(X)$ be a set algebra and a monotone class. Then $\mathcal{F}$ is a σ-algebra.

Proof. We need to show that $\mathcal{F}$ is closed under countable unions. Let $(F_j)_{j\in\mathbb{N}} \subset \mathcal{F}$ be a family of elements of $\mathcal{F}$. Since $\mathcal{F}$ is closed under finite union, the increasing sequence of sets

$$A_i := \bigcup_{j=1}^{i} F_j$$

must be contained in $\mathcal{F}$. Therefore, since $\mathcal{F}$ is a monotone class, the union

$$\bigcup_{i\in\mathbb{N}} A_i = \bigcup_{i\in\mathbb{N}} \left(\bigcup_{j=1}^{i} F_j \right) = \bigcup_{j\in\mathbb{N}} F_j$$

must also be contained in $\mathcal{F}$. Therefore, $\mathcal{F}$ is closed under countable union, making it a σ-algebra. □

The relevance of monotone classes is given by the following theorem, which gives us an additional way of verifying whether a given subset of a power set forms a σ-algebra:

Theorem 6.63: Monotone class theorem

Let $\mathcal{A} \subset \mathcal{P}(X)$ be a set algebra. Then

$$\langle\mathcal{A}\rangle_{\mathcal{M}} = \langle\mathcal{A}\rangle_{\sigma}$$

Proof. We have already observed that, since any σ-algebra is a monotone class, we have $\langle\mathcal{A}\rangle_{\mathcal{M}} \subset \langle\mathcal{A}\rangle_{\sigma}$. Therefore, it suffices to show that we also have $\langle\mathcal{A}\rangle_{\mathcal{M}} \supset \langle\mathcal{A}\rangle_{\sigma}$. This holds automatically if $\langle\mathcal{A}\rangle_{\mathcal{M}}$ is a σ-algebra, and since we already know that any monotone class is a σ-algebra if it is a set algebra, it suffices to show that $\langle\mathcal{A}\rangle_{\mathcal{M}}$ is a set algebra.

For any $E \in \langle\mathcal{A}\rangle_{\mathcal{M}}$, let $\mathcal{M}_E$ be the system of "good subsets" of X which can be combined with E as desired, i.e.

$$\mathcal{M}_E := \left\{ F \subset X \;\middle|\; \begin{array}{c} F \setminus E \in \langle\mathcal{A}\rangle_{\mathcal{M}} \\ E \setminus F \in \langle\mathcal{A}\rangle_{\mathcal{M}} \\ E \cup F \in \langle\mathcal{A}\rangle_{\mathcal{M}} \end{array} \right\}$$

1. First, we want to show that $\mathcal{M}_E$ is a monotone class.
2. Next, we want to show that for every $E \in \mathcal{A}$, we have $\langle\mathcal{A}\rangle_{\mathcal{M}} \subset \mathcal{M}_E$. Since $\mathcal{A}$ is a set algebra, every element of $\mathcal{A}$ combines with E as desired, giving us $\mathcal{A} \subset \mathcal{M}_E$. Therefore, since $\mathcal{M}_E$ is a monotone class, we have
$$\langle\mathcal{A}\rangle_{\mathcal{M}} \subset \langle\mathcal{M}_E\rangle_{\mathcal{M}} = \mathcal{M}_E.$$
3. Next, we want to show that for every $F \in \langle\mathcal{A}\rangle_{\mathcal{M}}$, we have $\langle\mathcal{A}\rangle_{\mathcal{M}} \subset \mathcal{M}_F$. We just showed that $\langle\mathcal{A}\rangle_{\mathcal{M}} \subset \mathcal{M}_F$, which means we have $F \in \mathcal{M}_E$ for any $E \in \mathcal{A}$. Since the definition of $\mathcal{M}_E$ is entirely symmetric in E and F, we also have $E \in \mathcal{M}_F$. Therefore, since all $E \in \mathcal{A}$ are contained in $\mathcal{M}_F$, we have $\mathcal{A} \subset \mathcal{M}_F$, implying
$$\langle\mathcal{A}\rangle_{\mathcal{M}} \subset \langle\mathcal{M}_F\rangle_{\mathcal{M}} = \mathcal{M}_F.$$
4. We have just shown that, for any $E, F \in \langle\mathcal{A}\rangle_{\mathcal{M}}$, we have $E \in \mathcal{M}_F$, implying that $\langle\mathcal{A}\rangle_{\mathcal{M}}$ is closed under complementation and binary union. Since $X \in \mathcal{A}$, we also have $X \in \langle\mathcal{A}\rangle_{\mathcal{M}}$. Therefore, $\langle\mathcal{A}\rangle_{\mathcal{M}}$ is a set algebra.

Therefore, $\langle\mathcal{A}\rangle_{\mathcal{M}}$ is indeed both a set algebra and a monotone class, making it a σ algebra. □

6.7 Measures

With our different subset systems in place, we can finally give a general formal definition of a measure. Along the way, we will encounter outer measures, contents, and premeasures, which are "weakened" measures that we can use to generate proper ones.

Definition 6.64: Outer measure

Let X be a set and μ be a function $\mathcal{P}(X) \to [0, \infty]$. We call μ an **outer measure on X** if it is σ-subadditive.

Theorem 6.65. Let X be any set. Then the constant function $\mu = 0$ is an outer measure on X, which is known as the **trivial measure**.

Theorem 6.66. Let X be any set. Let $A \subset X$ and $x \in X$. Then the function

$$\delta_x(A) := \begin{cases} 1 & x \in A \\ 0 & x \in X \setminus A \end{cases}$$

is an outer measure on X, which is known as the **Dirac measure**.

Theorem 6.67: Counting measure

Let X be any set. Let $A \subset X$. Then the function

$$\mathrm{card}(A) := \begin{cases} |A| & |A| \text{ is finite} \\ \infty & |A| \text{ is infinite} \end{cases}$$

is an outer measure on X, which is known as the **counting measure**.

Definition 6.68: Measurable set

Let μ be an outer measure on a set X. Then we call a subset $A \subset X$ **μ-measurable**, or just **measurable**, if for all $S \subseteq X$ we have

$$\mu(S) = \mu(S \cap A) + \mu(S \setminus A)$$

The system of all μ-measurable sets is sometimes denoted $\mathcal{M}(\mu)$.

By this definition, μ-measurable sets are exactly the sets for which μ is σ-additive. This means that if we restrict μ to its measurable sets, we get a proper measure.

Note that by the subadditivity of outer measures, we already get that the left side is at most as large as the right side, meaning that this condition can equivalently be weakened to

$$\mu(S) \geq \mu(S \cap A) + \mu(S \setminus A).$$

Theorem 6.69: Importance of σ-algebras

Let μ be an outer measure. Then the set of μ-measurable sets forms a σ-Algebra.

Since the restriction of an outer measure to its measurable sets forms a proper measure, this theorem tells us that σ-algebras give us the most sensible definition for which properties the domain of a measure should fulfill. Therefore, we finally get a proper definition of a measure:

Definition 6.70: Measure

Let $\mathcal{A}$ be a σ-Algebra. Then we call a σ-additive function $\mu : \mathcal{A} \to [0, \infty]$ a **measure**.

Definition 6.71. If $\mathcal{A}$ is a σ-algebra on X, and μ is a measure on $\mathcal{A}$, we call the triple $(X, \mathcal{A}, \mu)$ a **measure space**.

Definition 6.72. Let $(X, \mathcal{A}, \mu)$ be a meausure space. Then we call this space a **probability space**, and μ a **probability measure**, iff $\mu(X) = 1$.

Definition 6.73: Content, Premeasure

Let $\mathcal{S}$ be a set semiring. Then we call a finitely additive function $\mathcal{S} \to [0, \infty]$ a **content**, and a σ-additive function $\mathcal{S} \to [0, \infty]$ a **premeasure**.

In effect, a premeasure is a measure whose domain might not be as big as it could be. Every measure is trivially also a content and a premeasure, and every measure defined on $\mathcal{P}(X)$ is an outer measure.

Theorem 6.74: Restriction of a measure

Let $(X, \mathcal{A}, \mu)$ be a measure space. Let $Y \in \mathcal{A}$. Then if we define

$$\mu|_Y(A \cap M) := \mu(A \cap M),$$

$\mu|_Y$ defines a measure on $\mathcal{A}|_Y$.¸

Theorem 6.75: Image measure

Let X, Y be sets. Let $f : X \to Y$ be a function. Let μ be an outer measure on X. then the function

$$\begin{aligned} f(\mu) : \mathcal{P}(X) &\to [0, \infty] \\ f(\mu)(B) &= \mu\big(f^{-1}(B)\big) \end{aligned}$$

is a measure on Y, which is known as the **image measure of μ under f**.

Proof. 1. First, note that we have:

$$\begin{aligned} f(\mu)(\emptyset) &= \mu\big(f^{-1}(\emptyset)\big) \\ &= \mu(\emptyset) \\ &= 0 \end{aligned}$$

2. Furthermore, for $B \subset \bigcup_{i \in \mathbb{N}} B_i$, we have $f^{-1}(B) \subset \bigcup_{i \in \mathbb{N}} B_i$. By the

subadditivity of μ, we get:

$$\begin{aligned} f(\mu)(B) &= \mu(f^{-1}(B)) \\ &\leq \sum_{i\in\mathbb{N}} \mu(f^{-1}(B_i)) \\ &= \sum_{i\in\mathbb{N}} f(\mu)(B_i). \end{aligned}$$

Therefore, $f(\mu)$ is also subadditive and thus an outer measure.

□

Lemma 6.76. Let X, Y be sets. Let $f : X \to Y$ be a function. Let μ be an outer measure on X and let $f(\mu)$ be the corresponding image measure. Then $B \subset Y$ is $f(\mu)$ measurable if $f^{-1}(B)$ is μ-measurable.

Proof. By the definition of measurability:

1. $B \subset Y$ is $f(\mu)$-measurable if and only if for all $T \subset Y$, we have:

   $$f(\mu)(T) \geq f(\mu)(T \cap B) + f(\mu)(T \setminus B)$$

   i.e.

   $$\mu\big(f^{-1}(T)\big) \geq \mu\big(f^{-1}(T \cap B)\big) + \mu(f^{-1}(T \setminus B))$$

   Since $f^{-1}(T \cap B) = f^{-1}(T) \cap f^{-1}(B)$ and $f^{-1}(T \setminus B) = f^{-1}(T) \setminus f^{-1}(B)$, this is equivalent to:

   $$\mu\big(f^{-1}(T)\big) \geq \mu(f^{-1}(T) \cap f^{-1}(B)) + \mu(f^{-1}(T) \setminus f^{-1}(B))$$

2. $f^{-1}(B)$ is μ-measurable if and only if for all $S \subset X$, we have:

   $$\mu(S) \geq \mu(S \cap f^{-1}(B)) + \mu(S \setminus f^{-1}(B)).$$

   In particular, if $f^{-1}(B)$ is μ-measurable, this equation must hold for $S = f^{-1}(T)$, implying that B is $f(\mu)$-measurable.

□

Definition 6.77: (σ)-finite measure

Let $(X, \mathcal{A}, \mu)$ be a measure space. Then:

1. We call μ **finite** if $\mu(X) < \infty$.
2. We call μ **σ-finite** if X can be written as a countable union of sets of finite measure.

Theorem 6.78: Measures are σ-subadditive

Let $(X, \mathcal{A}, \mu)$ be a measure space. Let $(A_i)_{i\in\mathbb{N}} \subset \mathcal{A}$ be a family of (not necessarily disjoint!) measurable subsets of X. Then we have:

$$\mu\left(\bigcup_{i\in\mathbb{N}} A_i\right) \le \sum_{i\in\mathbb{N}} \mu(A_i)$$

Theorem 6.79: Continuity from above and below

Let $(X, \mathcal{A}, \mu)$ be a measure space. Let $(A_i)_{i\in\mathbb{N}} \subset \mathcal{A}$. Then we have:

1. **Continuity from below:** If $A_i \subset A_{i+1}$ for every $i \in \mathbb{N}$, we have

$$\lim_{i\to\infty} \mu(A_i) = \mu\left(\bigcup_{i\in\mathbb{N}} A_i\right)$$

2. **Continuity from above:** If $\mu(A_k) < \infty$ for any $k \in \mathbb{N}$ and $A_i \supset A_{i+1}$ for every $i \in \mathbb{N}$, we have

$$\lim_{i\to\infty} \mu(A_i) = \mu\left(\bigcap_{i\in\mathbb{N}} A_i\right)$$

Definition 6.80. Let $(X, \mathcal{A}, \mu)$. We call a set $A \in \mathcal{A}$ a ***μ*-zero set**, or simply zero set, iff $\mu(A) = 0$. We denote the set of all μ-zero sets by $\mathcal{N}(\mu)$.

Definition 6.81: μ-almost everywhere

Let $(X, \mathcal{A}, \mu)$ be a measure space. Let P be some property. We say that P holds **μ-almost everywhere on M**, or that P holds for **μ-almost all $x \in M$**, iff the set of $x \in M$ such that $P(x)$ is false is a zero set.

Definition 6.82: Complete measure

We call a measure μ **complete** if any subset of a μ-zero set is also a μ-zero set.

Note that, in particular, any subset of a μ-zero set needs to be μ-measurable in the first place.

Lemma 6.83. Let $(X, \mathcal{A}, \mu)$ be a measure space. Let $\mathcal{Z}_\mu$ be the system of all subsets of X that are also subsets of a μ-zero set. Then μ is complete if and only if $\mathcal{Z}_\mu \subset \mathcal{A}$.

Corollary 6.84. Let μ be an outer measure. Then $\mu|_{\mathcal{M}(\mu)}$ is complete.

Theorem 6.85: Completion of a Measure

Let $(X, \mathcal{A}, \mu)$ be a measure space. Let

$$\overline{\mathcal{A}} = \left\{ A \cup N \,\middle|\, A \in \mathcal{A}, N \in \mathcal{Z}_\mu \right\}.$$

and

$$\overline{\mu}(A \cup N) := \mu(A).$$

Then $\overline{\mathcal{A}}_\mu$ is a σ-algebra and $\overline{\mu}$ is a complete measure on $\overline{\mathcal{A}}_\mu$.

Theorem 6.86. (Uniqueness of Completion): Let $(X, \mathcal{A}, \mu)$ be a measure space. Let $(X, \mathcal{B}, \nu)$ be a complete measure space such that $\mathcal{A} \subset \mathcal{B}$ and $\mu(A) = \nu(A)$ for all $A \in \mathcal{A}$. Then we have $\overline{\mathcal{A}}_\mu \subset \mathcal{B}$ and $\overline{\mu} = \nu$ on $\overline{\mathcal{A}}_\mu$.

6.8 Carathéodory Extension

We can construct outer measures from a very large class of functions by the following construction:

Theorem 6.87: Carathéodory Extension

Let $\mathcal{S}$ be a system of subsets of a set X containing the empty set. Let $\lambda : \mathcal{S} \to [0, \infty]$ be a function such that $\lambda(\emptyset) = 0$. Then the function

$$\mu(E) := \inf \left\{ \sum_{i=1}^{\infty} \lambda(P_i) \,\middle|\, P_i \in \mathcal{S}, E \subset \bigcup_{i=1}^{\infty} P_i \right\}$$

is an outer measure on X.

Definition 6.88: Regular Measure

Let μ be an outer measure on X. Then we call μ **regular** iff. for every $M \subset X$, we can find a μ-measurable set $D \supset M$ such that $\mu(M) = \mu(D)$.

Lemma 6.89. The Carathéodory extension of a premeasure is regular.

Theorem 6.90: Characterization of measurable sets

Let $\lambda : \mathbb{R} \to [0, \infty]$ be a $\underline{\sigma\text{-finite}}$ premeasure on a ring $\mathcal{R} \subset \mathcal{P}(X)$. Let $\overline{\lambda}$ be its Carathéodory extension. Then a set $D \subset X$ is μ measurable if and only if either of the following hold:

1. There exists an $E \supset D$ such that $E \in \langle \mathcal{R} \rangle_\sigma$ and $\overline{\lambda}(E \setminus D) = 0$.
2. There exists a $C \subset D$ such that $C \in \langle \mathcal{R} \rangle_\sigma$ and $\overline{\lambda}(D \setminus C) = 0$.

6.9 Measures on $\mathbb{R}^n$

6.9.1 Jordan Content

Definition 6.91. Let $E \subset \mathbb{R}^n$ be an elementary set. Then we can assign to it the **elementary volume** $\mathrm{vol}(E)$, where the volume of a cuboid is the product of its side lengths and the volume of a finite union of cuboids is the sum of the volumes of the individual cuboids making up E.

Definition 6.92. (Inner and outer Jordan content): Let $E \subset \mathbb{R}^n$.

1. The **inner Jordan measure** $J_*(E)$ is

$$J_*(E) := \sup_{\substack{Q_i \in \mathcal{Q}, \\ \bigcup_{i=1}^n Q_i \subset E}} \mathrm{vol}(Q)$$

2. The **outer Jordan measure** $J_*(E)$ is

$$J^*(E) := \inf_{\substack{Q_i \in \mathcal{Q}, \\ \bigcup_{i=1}^n Q_i \supset E}} \mathrm{vol}(Q)$$

Definition 6.93. (Jordan measurable set, Jordan content): We call E **Jordan-measurable** if $J_*(E) = J^*(E)$. Then we call $J(E) = J_*(E) = J^*(E)$ the **Jordan content**.

Theorem 6.94. The Jordan content is σ-additive.

Theorem 6.95. The following are equivalent:

1. E is Jordan measurable,
2. For every $\varepsilon > 0$, there exist elementary sets $A \subset E \subset B$ such that $\mathrm{vol}(B \setminus A) \leq \varepsilon$,
3. For every $\varepsilon > 0$, there exists an elementary set A such that $J^*(A \triangle E) \leq \varepsilon$.

Theorem 6.96. The collection of subsets of $\mathbb{R}^n$ that are either Jordan measurable or have a Jordan-measurable complement form a Boolean algebra, known as the Jordan algebra.

Theorem 6.97. The Jordan algebra is non-atomic.

Theorem 6.98. (Regions under continuous Graphs are Jordan measurable): Let B be a closed box in $\mathbb{R}^n$, and let $f : B \to \mathbb{R}$ be a continuous function. Then the set

$$\{(x, t) \mid x \in B, 0 \leq t \leq f(x)\} \subset \mathbb{R}^{n+1}$$

is Jordan measurable.

Theorem 6.99. Triangles are Jordan measurable.

Theorem 6.100. Convex polytopes in $\mathbb{R}^n$ are Jordan measurable.

Theorem 6.101. Open and closed Euclidean balls are Jordan measurable.

Theorem 6.102. Every subset of a Jordan null set is Jordan measurable and also a Jordan null set.

Theorem 6.103. The sets $[0,1]^2 \setminus \mathbb{Q}^2$ and $[0,1]^2 \cap \mathbb{Q}^2$ are both not Jordan measurable.

Informally, sets with a lot of "holes" or with very messy, fractal-like boundaries are generally not Jordan-measurable.

Theorem 6.104. There exist countable unions, and countable intersections, of Jordan measurable sets which are not Jordan measurable.

6.9.2 Lebesque Measure

One can extend the Jordan measure to a significantly larger number of subsets of $\mathbb{R}^n$ by simply allowing countable unions of cuboids (instead of finite unions of cuboids). The Lebesque measure is simply this generalization of the Jordan measure. Formally:

Definition 6.105. Let $E \subset \mathbb{R}^n$. The **Lebesque outer measure** of E is given by

$$\lambda^*(E) := \inf_{\substack{E \subset \bigcup_{i=1}^{\infty} Q_n, \\ Q_n \in \mathcal{Q}}} \left\{ \sum_{n=1}^{\infty} \mathrm{vol}^n Q_n \right\},$$

i.e. the Lebesque outer measure of E is the greatest lower bound of the measures of all coverings of E by cuboids.

Theorem 6.106. We have $\lambda^0 = \mathrm{card}$ - the zero-dimensional lebesque measure is the counting measure.

Proof. $\mathbb{R}^0$ is by definition cartesian product of $\mathbb{R}$ with a set of cardinality 0 i.e. the empty set, as its index set:

$$\mathbb{R}^0 = \prod_{\emptyset} \mathbb{R}$$

By definition of cartesian products, its members must be functions $\emptyset \to \mathbb{R}$. There is exactly one such "empty function", which we will denote $f_\emptyset$. Thus, the only subsets of $\mathbb{R}^0$ are the empty set and $\{f_\emptyset\}$.

1. Since λ^0 and card are measures, we have

$$\lambda^0(\emptyset) = 0 = \mathrm{card}(\emptyset)$$

2. By definition, a 0-dimensional cuboid must be a function $\emptyset \to I$ to an interval $I \subset \mathbb{R}$. Since all functions from the empty set to any set are identical, $f_\emptyset$ qualifies as a cuboid. We thus have:

$$\begin{aligned} \lambda^0(\{f_\emptyset\}) &= \mathrm{vol}^0(\{f_\emptyset\}) \\ &= \prod_{\emptyset} \mathrm{vol}(I) \\ &= 1 \\ &= \mathrm{card}(\{f_\emptyset\}) \end{aligned}$$

Where the last step follows since empty products evaluate to the multiplicative identity, in this case the number 1.

□

Theorem 6.107. Let $U \subset \mathbb{R}^n$ be open. Then U is a countable union of cuboids.

Therefore, we can define Lebesque measurability similarly to Jordan measurability - a set is Lebesque measurable if it is "almost" an open set.

Definition 6.108. A set $E \subset \mathbb{R}^n$ is **Lebesque measurable** if for every $\varepsilon > 0$, there exists an open set $U \subset \mathbb{R}^n$ such that $U \subset E$ and $\lambda^*(U \setminus E) \leq \varepsilon$. If E is Lebesque measurable, we refer to $\lambda(E) := \lambda^*(E)$ as the **Lebesque measure** of E. If the dimension n should be emphasized, we sometimes write $\lambda(E)$ as $\lambda^n(E)$.

Theorem 6.109. The Lebesque measure fulfills:

$$E \subset \bigcup_{i \in N} E_i \implies \lambda(E) \leq \sum_{i \in \mathbb{N}} \lambda(E_i)$$

This is a weakened form of σ-additivity, which is known as $\boldsymbol{\sigma}$-subadditivity.

Theorem 6.110. The Lebesque measure defines a σ-additive function on the Lebesque measurable sets.

Theorem 6.111. The collection of subsets of $\mathbb{R}^n$ that are either Lebesque measurable or have a Lebesque-measurable complement form a Boolean algebra.

Theorem 6.112. There exist Lebesque measurable sets which are not Borel.

You may have heard in linear algebra that the determinant of a matrix S tells us how the matrix scales the volume of the unit cube. Since the Lebesque measure is defined using volumes of unit cubes, this intuition also holds for the image of any set E under S:

Theorem 6.113: Linear Transformation Equation

Let $S \in \mathbb{R}^{n \times n}$. Then for all $E \subset \mathbb{R}^n$, we have:

$$\lambda^n(S(E)) = |\det(S)| \cdot \lambda^n(E)$$

Definition 6.114: Smith-Volterra-Cantor Set

The Smith-Volterra-Cantor set is the set obtained by removing the middle $1/8$ of the interval $[0, 1]$, and then iteratively removing the middle $1/8$ of any remaining intervals.

Theorem 6.115: Compact does not imply Jordan measurable

The Smith-Volterra-Cantor set is compact, but not Jordan measurable.

Chapter 7

Measurable Functions

Definition 7.1: Measurable Functions

Let $(X, \mathcal{A})$ and $(Y, \mathcal{B})$ be measurable spaces. Then we call a map $f : X \to Y$ **$\mathcal{A}$-$\mathcal{B}$-measurable**, or just **measurable**, if the preimage of every measurable set is again measurable, i.e.

$$B \in \mathcal{B} \implies f^{-1}(B) \in \mathcal{A}$$

Once again, note the similarities between this definition and the topological definition of a continuous function.

If the σ-algebra on one of the sets is supposed to be clear from context, many authors only specify one of the the two sigma algebras. For example, for a function $f : X \to \overline{\mathbb{R}}$, many authors talk about $\mathcal{A}$-measurability when they implicitly mean $\mathcal{A} - \mathcal{B}(\overline{\mathbb{R}})$-measurability.

Corollary 7.2

The composition of two measurable functions is again measurable.

Lemma 7.3. Let $(X, \mathcal{A})$, $(Y, \mathcal{B})$ be measurable spaces and $f : X \to Y$. Let $\mathcal{E} \subset \mathcal{B}$. Then

$$f^{-1}(\langle \mathcal{E} \rangle_\sigma) = \left\langle f^{-1}(\mathcal{E}) \right\rangle_\sigma$$

Corollary 7.4

Let $\mathcal{E}$ be a base of $\mathcal{B}$, i.e. $\langle \mathcal{E} \rangle_\sigma = \mathcal{B}$. Then $f : X \to Y$ is $\mathcal{A}$-$\mathcal{B}$ measurable if and only if

$$E \in \mathcal{E} \implies f^{-1}(E) \in \mathcal{A},$$

This means we don't need to check the preimage of every single set in $\mathcal{B}$ to show that f is measurable - it suffices to check a base.

Corollary 7.5. Every continuous function between topological spaces is measurable in the corresponding Borel-σ-algebras.

Proof. Let $f : X \to Y$ be continuous. Then the preimage of every open set of Y is open in X, i.e. contained in the Borel σ-algebra on X, and the open sets of Y form a base of the Borel σ-algebra on Y. □

Theorem 7.6: Simple criteria for measurability

Let $(X, \mathcal{A})$ be a measureable space. Let $f : X \to \overline{\mathbb{R}}$. Then the following are equivalent:

1. f is $\mathcal{A}$-$\mathcal{B}(\overline{\mathbb{R}})$ measurable,
2. $\forall c \in \mathbb{R} : \{f > c\} \in \mathcal{A}$,
3. $\forall c \in \mathbb{R} : \{f \geq c\} \in \mathcal{A}$,
4. $\forall c \in \mathbb{R} : \{f < c\} \in \mathcal{A}$,
5. $\forall c \in \mathbb{R} : \{f \leq c\} \in \mathcal{A}$,

Proof.
1. The latter four conditions are equivalent to each other, since:
   (a) $\{f \geq c\} = \bigcap_{k\in\mathbb{N}} \left\{f > c - \frac{1}{k}\right\}$,
   (b) $\{f > c\} = \bigcup_{k\in\mathbb{N}} \left\{f \geq c + \frac{1}{k}\right\}$,
   (c) $\{f < c\} = X \setminus \{f \geq c\}$,
   (d) $\{f \leq c\} = X \setminus \{f > c\}$.
2. The intervals $[c, \infty]$ form a base of $\mathcal{B}(\overline{\mathbb{R}})$, since:
   (a) $\{\infty\} = \bigcap_{k\in\mathbb{N}} [k, \infty]$
   (b) $\{-\infty\} = \bigcap_{k\in\mathbb{N}} [-\infty, -k]$
   (c) $(a, b) = [a, \infty] \setminus ([b, \infty] \cap [-\infty, a])$

□

Theorem 7.7

Let $f, g : X \to \overline{\mathbb{R}}$ be $\mathcal{A}$-$\mathcal{B}(\overline{\mathbb{R}})$ measurable. Then the following sets are contained in $\mathcal{A}$:

1. $\{f > g\}$,
2. $\{f \geq g\}$,
3. $\{f = g\}$,
4. $\{f \neq g\}$.

Theorem 7.8. Let $f, g : X \to \overline{\mathbb{R}}$ be $\mathcal{A}$-$\mathcal{B}[\overline{\mathbb{R}}]$ measurable. Then the following functions are also $\mathcal{A}$-measurable:

1. cf, for all $c \in \mathbb{R}$,

2. $|f|^p$, for all $p \in \mathbb{R}_{>0}$,

3. $f + g$, assuming the sum is defined everywhere on X, i.e. there exists no $x \in X$ such that $f(x) = \infty$ and $g(x) = -\infty$ or vice versa,

4. $f \cdot g$.

Theorem 7.9. Let $(X, \mathcal{A})$ be a measurable space and $\mathbb{1}_E : X \to \mathbb{R}$ be the indicator function of a set $E \subset X$. Then $\mathbb{1}_E$ is $\mathcal{A}$-$\mathcal{B}[\mathbb{R}]$ measurable if and only if $E \in \mathcal{A}$.

Proof. We have $\{1\} = (-\infty, 1) \cup (1, \infty) \in \mathcal{B}[\mathbb{R}]$ and $\mathbb{1}_E^{-1}(\{1\}) = E$, so E has to be in $\mathcal{A}$ for $\mathbb{1}_E$ to be measurable.

$E \in \mathcal{A}$ is also a sufficient condition for $\mathbb{1}_E$ to be measurable, since the only other possible preimages are $\emptyset \in \mathcal{A}$, $X \in \mathcal{A}$, $X \setminus E \in \mathcal{A}$ □

Theorem 7.10. Let $(X, \mathcal{A})$ be a measurable space, let $D \in \mathcal{A}$, and let $f_k : D \to \mathbb{R}$ be $\mathcal{A}$ measurable. Then the following functions are $\mathcal{A}$-measurable:

1. $\inf_{n\in\mathbb{N}} f_n$

2. $\sup_{n\in\mathbb{N}} f_n$

3. $\liminf_{n\to\infty} f_n$

4. $\limsup_{n\to\infty} f_n$

Proof. For $s \in \mathbb{R}$, we have:

1. $\left\{(\inf\limits_{n\in\mathbb{N}} f_n) \geq s\right\} = \bigcap\limits_{k=1}^{\infty} \{f_k \geq s\} \in \mathcal{A}$

2. $\left\{(\sup\limits_{n\in\mathbb{N}} f_n) \leq s\right\} = \bigcap\limits_{k=1}^{\infty} \{f_k \leq s\} \in \mathcal{A}$

Therefore $\inf_{n\in\mathbb{N}} f_n$ and $\sup_{n\in\mathbb{N}} f_n$ are measurable. The same argument holds for $\sup$ and $\inf$ over subsets of $\mathbb{N}$. Therefore, the compositions

$$\liminf_{n\to\infty} f_n = \sup_{n\in\mathbb{N}} \left(\inf_{k\geq n} f_k\right)$$

and

$$\limsup_{n\to\infty} f_n = \inf_{n\in\mathbb{N}} \left(\sup_{k\geq n} f_k\right)$$

are also measurable. □

Corollary 7.11

Let f_n be a sequence of $\mathcal{A}$-measurable functions with a pointwise limit f. Then f is $\mathcal{A}$-measurable.

Definition 7.12: μ-measurable

Let $(X, \mathcal{A}, \mu)$ be a measure space. Let $D \in \mathcal{A}$. Then we call a function $f : D \to \overline{\mathbb{R}}$ μ-measurable iff:

1. $X \setminus D$ is a μ-zero set, and
2. f is $\mathcal{A}|_D$-measurable.

Lemma 7.13. Let $f : D \to \overline{\mathbb{R}}$ be μ-measurable. Then the function

$$g(x) = \begin{cases} f(x) & x \in D \\ 0 & x \notin D \end{cases}$$

is $\mathcal{A}$-measurable.

Lemma 7.14

Let $(X, \mathcal{A}, \mu)$ be a **complete** measure space. Let f be μ-measurable. Then every function g such that $g = f$ μ-almost everywhere is μ-measurable.

Theorem 7.15: μ-measurable pointwise limits

Let $(X, \mathcal{A}, \mu)$ be a **complete** measure space. Let $(f_k)_{k \in \mathbb{N}}$ be a family of μ-measurable functions. Then if f_k converges pointwise to a function f μ-almost everywhere, f is μ-measurable.

Chapter 8

Lebesque Integration

8.1 Step Functions

Definition 8.1. Let $(X, \mathcal{A})$ be a measurable space. Then we call a function $f : Y \to \mathbb{R}$ a **step function** if it can be represented as a finite linear combination of indicator functions of sets in $\mathcal{A}$, i.e there exist $\alpha_i \in \mathbb{R}$ and $A_i \in \mathcal{A}$ such that:

$$f = \sum_{i \leq k} \alpha_i \cdot \mathbb{1}_{A_i}$$

Proposition 8.2. Step functions form a vector space over $\mathbb{R}$.

Proposition 8.3. Step functions can only take finitely different values.

Proposition 8.4. Every step function is (extensionally) equal to a step function over pairwise disjoint sets.

> *Proof.* Let
>
> $$s = \sum_{i \leq k} \alpha_i \cdot \mathbb{1}_{A_i}$$
>
> Then if $\beta_1, \ldots, \beta_m$ are all of the finitely many different values that s can take, s is by definition equal to
>
> $$t := \sum_{i \leq m} \beta_i \cdot \mathbb{1}_{\{s=\beta_i\}}$$
>
> And since a function cannot take two values at once, the sets $\{s = \beta_i\}$ are disjoint. □

Lemma 8.5. (Sum of Step Functions): Let s_1 and s_2 be step functions

$$s_1 = \sum_{i=0}^{m} \alpha_i \mathbb{1}_{A_i}, \qquad s_2 = \sum_{j=0}^{n} \beta_j \mathbb{1}_{B_j}$$

defined on pairwise disjoint sets. Then we have:

$$s_1 + s_2 = \sum_{i=0}^{m} \sum_{j=0}^{n} (\alpha_i + \beta_j) \mathbb{1}_{A_i \cap B_j}$$

Proof. 1. Assume s_1 and s_2 are nonnegative step functions

$$s_1 = \sum_{i=0}^{m} \alpha_i \cdot \mathbb{1}_{A_i}, \qquad s_2 = \sum_{j=0}^{n} \beta_j \cdot \mathbb{1}_{B_j}$$

defined on pairwise disjoint sets.

Then:

$$s_1(x) + s_2(x) = \sum_{i=0}^{m} \alpha_i \cdot \mathbb{1}_{A_i}(x) + \sum_{j=0}^{n} \beta_j \cdot \mathbb{1}_{B_j}(x)$$

Since the A_i and B_j are disjoint coverings of X, for every $x \in X$, there exists exactly one i such that $x \in A_i$ and exactly one j such that $x \in B_j$. Therefore, every x contributes exactly one α_i for $x \in A_i$ and one β_j for $x \in B_j$, i.e. $x \in A_i \cap B_j$ and $s_1(x) + s_2(x) = \alpha_i + \beta_j$. Since this is the only intersection which contains x, we can put everything together to get our desired formula:

$$s_1(x) + s_2(x) = \sum_{i=0}^{m} \sum_{j=0}^{n} (\alpha_i + \beta_j) \cdot \mathbb{1}_{A_i \cap B_j}(x)$$

$\square$

The integral of an indicator function is already clear from our intuition: The points contained in the area under an indicator function $\mathbb{1}_A$ are exactly the cartesian product of A with the interval $[0, 1]$, forming a "rectangle with gaps" whose side lengths are 1 and $\mu(A)$. Therefore the integral should be:

Definition 8.6. (Lebesque integral of an indicator function):

$$\int_X \mathbb{1}_A \, d\mu = \mu(A)$$

The definition for the integral of a step function should follow naturally - a step function is just a finite sum of scaled characteristic functions, therefore the integral of a step function is the sum of the scaled integrals of the step functions.

Definition 8.7. (Lebesque integral of a step function): Let $D \subset X$, $A_i \subset X$ pairwise disjoint, and $\alpha_i \geq 0$. Then:

$$\begin{aligned}
\int_D s \, d\mu &= \int_D \left(\sum_{i \leq k} \alpha_i \cdot \mathbb{1}_{A_i} \right) d\mu \\
&:= \sum_{i \leq k} \left(\alpha_i \cdot \int_D \mathbb{1}_{A_i} \, d\mu \right) \\
&= \sum_{i \leq k} (\alpha_i \cdot \mu(D \cap A_i))
\end{aligned}$$

Theorem 8.8. (Linearity of the step function integral):

Theorem 8.9. (Monotonicity of the step function integral):

8.2 Defining the Lebesgue Integral

We can now use the step function integral to define the integral of more general functions:

Theorem 8.10: Approximation via step functions

Let $(X, \mathcal{A})$ be a measurable space. Let $f : X \to \overline{\mathbb{R}}$ be a nonnegative $\mathcal{A}$-measurable function. Then there exists a monotonically increasing sequence s_n of nonnegative step functions whose pointwise limit is f.

Construction 1. For $n \in \mathbb{N}$ and $k \in \{0, \ldots, n \cdot 2^n\}$, we set:

$$F_{n,k} := \left\{ x \in X \;\middle|\; \frac{k}{2^n} \leq f(x) < \frac{k+1}{2^n} \right\}.$$

Then

$$s_n(x) = \begin{cases} \frac{k}{2^n} & x \in F_{n,k} \\ n & \text{otherwise} \end{cases}$$

are step functions converging to f. □

Construction 2. Define $s_0 = 0$. Then we can inductively define

$$E_n := \left\{ s_{n-1} + \frac{1}{n} \leq f \right\}$$

and

$$s_n := s_{n-1} + \frac{1}{n} \cdot \mathbb{1}_{E_n},$$

meaning we have

$$s_n = \sum_{k=1}^{n} \frac{1}{k} \cdot \mathbb{1}_{E_i}.$$

We now need to show that this series converges pointwise to f. Let $n \in \mathbb{N}$.

1. Assume $x \in E_n$. Then, by definition,

$$f_n(x) = f_{n-1} + \frac{1}{n} \leq f(x)$$

2. Assume $x \notin E_n$. Then, by induction, we have

$$f_n(x) = f_0(x) = 0 \leq f(x).$$

Therefore, we have $f_0 \leq f_1 \leq \ldots$ and $f_n \leq f$ for all n, meaning that

$$\lim_{n \to \infty} f_n(x) \leq f(x).$$

If $f(x) = \infty$, then $x \in E_n$ for every n, and we have

$$\lim_{n \to \infty} f_n(x) = \sum_{k=1}^{\infty} \frac{1}{k} = \infty$$

If $f(x) < \infty$, then there must be an infinite number of $n \in \mathbb{N}$ such that $f_{n-1}(x) > f_n(x) - \frac{1}{n}$, implying

$$\lim_{n\to\infty} f_n(x) \geq f(x)$$

□

Definition 8.11: Lebesque integral of a positive function

Let $f : D \to [0, \infty]$ be $\mathcal{A}$-measurable. Then:

$$\int_D f \, d\mu = \sup_{\substack{s \text{ is a step function,} \\ 0 \leq s \leq f}} \left\{ \int_D s \, d\mu \right\}$$

Definition 8.12. (Lebesque integral of an arbitrary measurable function): Let f be $\mathcal{A}$-mesurable and

$$f^+ = f \cdot \mathbb{1}_{f \geq 0}, \qquad f^- = -f \cdot \mathbb{1}_{f<0}$$

Note that $f^+ \geq 0$ and $f^- > 0$. Then, as long as f^+ or f^- have a finite integral, we can define:

$$\int_D f \, d\mu \int_D f^+ \, d\mu - \int_D f^- \, d\mu$$

Corollary 8.13. (Integrating over subsets):

$$\int_M f \, d\mu = \int_X f \cdot \mathbb{1}_M \, d\mu = \int_M f \, d\mu|_M$$

It is common to derive from this corollary a slight abuse of notation: Assume that f is not $\mathcal{A}$-measurable, but it is μ-measurable, i.e. $f : D \to \overline{\mathbb{R}}$ such that $\mu(X \setminus D) = 0$ and f is $\mathcal{A}|_D$-measurable. Then it is common to implicitly expand the domain of D to X by setting $f(x) = 0$ on $X \setminus D$, and to therefore write:

$$\int_X f \, d\mu := \int_D f \, d\mu$$

Corollary 8.14. (Integrating over zero sets): Let N be a set such that $\mu(N) = 0$. Then

$$\int_N f \, d\mu = 0.$$

Definition 8.15: Integrable function

We call a function $f : X \to \overline{\mathbb{R}}$ **integrable** with regards to a measure μ if it is μ-measurable and

$$\int_X f \, d\mu \in \mathbb{R}$$

Proposition 8.16. (Integrating with the counting measure): Let X be an arbitrary set. Let card be the counting measure on $\mathcal{P}(X)$. Let $f : X \to \overline{\mathbb{R}}$. Then f is integrable with respect to card if and only if $\sum_{x \in X} f(x)$ is absolutely convergent, and we have

$$\int_X f \, d\text{card} = \sum_{x \in X} f(x)$$

Theorem 8.17: Monotonicity of the Lebesque integral

Let μ be an outer measure on X such that $f, g : X \to \overline{\mathbb{R}}$ are μ-measurable. Then if $f \leq g$ μ-almost everywhere and $\int_X f \, d\mu > -\infty$, then the integral of g exists and we have

$$\int_X f \, d\mu \leq \int_X g \, d\mu$$

Proof.

1. Assume f, g are nonnegative. Then if s is a step function such that $s \leq f$. Then if we define $t := \mathbb{1}_{f \leq g} \cdot s$, we have $t \leq g$. Furthermore, for all $c \geq 0$, we have:

$$\mu(\{s = c\}) = \mu(\{s = c\} \cap)$$

2.

□

Theorem 8.18

Let $f, g : X \to \overline{\mathbb{R}}$ and let f be μ-measurable. Then if $g = f$ μ-almost everywhere, then g is μ-measurable, and

$$\int_X g \, d\mu = \int_X f \, d\mu$$

as long as the right integral exists.

Lemma 8.19. (Chebyshev Inequality): Let $f : X \to [0, \infty]$ be μ-measurable with $\int_X f \, d\mu < \infty$. Let $s \in (0, \infty]$. Then:

$$\mu(\{x : f(x) \geq s\}) \leq \begin{cases} \frac{1}{s} \int_X f \, d\mu & s \in (0, \infty) \\ 0 & s = \infty \end{cases}$$

Corollary 8.20. Let $f : X \to \overline{\mathbb{R}}$ be μ-measurable. Then:

1. If $\int_X f \, d\mu < \infty$, $\{x : f(x) = \infty\}$ is a μ-zero set.
2. If $f \geq 0$ and $\int_X f \, d\mu = 0$, $\{x : f(x) > 0\}$ is a μ-zero set.

Theorem 8.21: Monotone Convergence Theorem

Let $f_n : X \to [0, \infty]$ be μ-measurable such that $f_i \leq f_{i+1}$. Let $\lim_{n\to\infty} f_n = f$. Then

$$\int_X \lim_{n\to\infty} f_n \, d\mu = \lim_{n\to\infty} \int_X f_n \, d\mu = \sup_{n\in\mathbb{N}} \int_X f_n \, d\mu$$

Theorem 8.22: Linearity of the Lebesque integral

Let μ be a measure on X. Let $f, g : X \to \overline{\mathbb{R}}$ and $\alpha, \beta \in \mathbb{R}$. Then we have:

$$\int_X (\alpha f + \beta g) \, d\mu = \alpha \int_X f \, d\mu + \beta \int_X g \, d\mu$$

Corollary 8.23. Let $f : X \to \overline{\mathbb{R}}$ be μ-measurable. Then f is integrable if and only if $|f|$ is integrable.

Corollary 8.24. Let $f : X \to \overline{\mathbb{R}}$ be μ-measurable. Then, assuming the integral of f exists, we have

$$\left| \int_X f \, d\mu \right| \leq \int_X |f| \, d\mu$$

Corollary 8.25: Dominated Convergence Theorem

Let $g : X \to [0, \infty]$ be μ-measurable such that $|f| \leq g$ μ-almost everywhere. Then if

$$\int_X g \, d\mu < \infty,$$

f is integrable.

8.3 Comparing Riemann Integration and Lebesque Integration

Theorem 8.26. The Dirichlet function $\mathbb{1}_{\mathbb{Q}}$ is Lebesque integrable, but not Riemann integrable.

Theorem 8.27

A Lebesque-integrable function is not necessarily "almost Riemann-integrable". More formally, there exists a function $f : \mathbb{R} \to \mathbb{R}$ with the following properties:

1. f is Lebesque-integrable.
2. There exists no Riemann-integrable function g such that $f = g$ almost everywhere.

Proof. Consider the indicator function $\mathbb{1}_{SVC}$ of the Smith-Volterra-Cantor set. □

Theorem 8.28

Let $f : [a, b] \to \mathbb{R}$. Then f is Riemann integrable if and only if it is the limit of a uniformly convergent sequence of step functions over Jordan measurable sets.

8.4 Convergence Theorems

Theorem 8.29: Levi's Stronger Monotone Convergence Theorem

The monotone convergence theorem continues to hold if $f_n \leq f_{n+1}$ and $f_n \nearrow f$ only hold μ-almost everywhere.

Corollary 8.30: Exchange of Integrals and Sums

Let f_n be nonnegative and μ-measurable. Then

$$\int_X \left(\sum_{n=1}^{\infty} f_n \right) d\mu = \sum_{n=1}^{\infty} \left(\int_x f_n \, d\mu \right)$$

Theorem 8.31: Fatou's Lemma

Let (f_n) be a sequence of μ-measurable functions $X \to \overline{\mathbb{R}}$. Let $g \in \mathcal{L}^1$. Then if we have $f_n \geq g$ μ-almost everywhere for all $n \in \mathbb{N}$, we have

$$\int_X \left(\liminf_{n\to\infty} f_n \right) d\mu \leq \liminf_{n\to\infty} \int_X f_n \, d\mu$$

Theorem 8.32: Lebesque's Dominated Convergence Theorem

Let f, f_k be μ-measurable functions such that $f_k \to f$ μ-almost everywhere. Assume there exists a nonnegative Function $g \in \mathcal{L}^1$ such that $\sup_k |f_k| \leq g$ μ-almost everywhere. Then we have $f \in \mathcal{L}^1$ and

$$\int_X f \, d\mu = \lim_{k\to\infty} \int_X f_k \, d\mu$$

Lemma 8.33. (Continuity of Parametrized Integrals): Let $(X, \mathcal{A}, \mu)$ be a measure space. Let $U \subset \mathbb{R}^n$ be open. Let $f : U \times X \to \mathbb{R}$. Then

$$\begin{aligned} \varphi &: U \to \mathbb{R} \\ \varphi(y) &= \int_X f(y, x) \, d\mu(x) \end{aligned}$$

is continuous if the following hold:

1. $f(\cdot, x)$ is continuous in U for μ-almost all $x \in X$,
2. $f(y, \cdot)$ is μ-measurable for all $y \in U$,
3. there exists a function $g \in \mathcal{L}^1(\mu)$ such that for all $y \in U$ and μ-almost all $x \in X$, we have

$$\|f(y, x)\| \leq g(x)$$

Theorem 8.34: Differentiation under the Integral Sign

Let $(X, \mathcal{A}, \mu)$ be a measure space, $U \subset \mathbb{R}^n$ open, and $f : U \times X \to \mathbb{R}$. Then

$$\varphi(y) := \int_X f(y, x)\, d\mu(x)$$

is continuously differentiable with

$$\frac{\partial}{\partial y_i} \int_X f(y, x)\, d\mu(x) = \int_x \frac{\partial}{\partial y_i} f(y, x)\, d\mu(x)$$

if the following hold:

1. $f(\cdot, x)$ is continuously differentiable in U for almost all x,
2. $f(y, \cdot)$ is μ-measurable for almost all $y \in U$,
3. There exists a function $g \in \mathcal{L}^1(\mu)$ such that

$$\left| \frac{\partial}{\partial y_i} f(y, x) \right| \leq g(x)$$

for all $y \in U$, μ-almost all $x \in X$ and $i \in \{1, \ldots, n\}$.

Theorem 8.35. (Absolute continuity of the Lebesque integral): Let $f \in \mathcal{L}^1(\mu)$. Let $\varepsilon > 0$. Then there exists a $\delta > 0$ such that, for all $A \in \mathcal{A}$:

$$\mu(A) < \delta \implies \int_A |f|\, d\mu < \varepsilon$$

8.5 L^p-Spaces

Definition 8.36: $\mathcal{L}^p$

Let $1 \leq p < \infty$. We denote by $\mathcal{L}^p = \mathcal{L}^p(X, \mathcal{A}, \mu)$ the set of all μ-measurable functions f such that

$$\int_X |f|^p \, d\mu < \infty$$

Definition 8.37: L^p **Norm**

Let $1 \leq p < \infty$ and $f \in \mathcal{L}^p(X, \mathcal{A}, \mu)$. We call

$$\|f\|_p = \sqrt[p]{\int_X |f|^p \, d\mu}$$

The L^p-Norm, or just p-Norm of f.

Note that, if we view integration as the natural "continuous analogue of summation"[1] and f as a vector with uncountably many components, this is a direct generalization of the familiar p-norm of vectors, such as the euclidean norm

$$\|\vec{x}\|_2 = \sqrt{\sum_{i=1}^{n} x_i^2}$$

Theorem 8.38: L^∞**-Norm**

We have

$$\|f\|_\infty := \lim_{p \to \infty} \|f\|_p = \inf_{s>0} \mu\{|f| > s\}$$

Lemma 8.39. (Young Inequality): Let $p, q \in (1, \infty)$ such that $\frac{1}{p} + \frac{1}{q} = 1$. Then, for nonnegative $a, b \geq 0$, we have:

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q}$$

Proof. From the concavity of the logarithm, we get:

$$\ln\left(\frac{a^p}{p} + \frac{b^q}{q}\right) \geq \frac{1}{p}\ln(a^p) + \frac{1}{q}\ln(b^q)$$

Which we can immediately simplify to get:

$$\frac{1}{p}\ln(a^p) + \frac{1}{q}\ln(b^q) = \ln(a) + \ln(b) = \ln(ab)$$

Exponentiating both sides gives us our desired result. □

[1]We can actually formalize this by going the other way - summation is just integration over a discrete space, using the counting measure as the canonical measure.

Comment 8.40. Keep in mind that:

$$\begin{aligned} \frac{1}{p} + \frac{1}{q} &= 1 \\ \iff \frac{1}{q} &= 1 - \frac{1}{p} \\ &= \frac{p-1}{p} \\ \iff q &= \frac{p}{p-1} \end{aligned}$$

Now, letting $\frac{1}{\infty} = 0$, we get:

Corollary 8.41: Hölder Inequality

Let $p, q \in [1, \infty]$ such that $\frac{1}{p} + \frac{1}{q} = 1$. Let $f \in \mathcal{L}^p$ and $g \in \mathcal{L}^q$. Then we have $fg \in \mathcal{L}^1$ and, in particular,

$$\left| \int_X f(x) \cdot g(x) \, d\mu(x) \right| \\ \leq \|fg\|_1 \leq \|f\|_p \cdot \|g\|_q$$

Proof.

$p = 1$: In this case, we have $q = \infty$. We have

$$g(x) < \inf_{s>0} \{\mu(|g| > s) = 0\} = \|g\|_\infty$$

μ-almost everywhere, and thus

$$\begin{aligned} \int_X |f(x) \cdot g(x)| \, d\mu(x) &\leq \int_X |f(x)| \cdot \|g\|_\infty \, d\mu(x) \\ &= \int_X |f(x)| \, d\mu(x) \cdot \|g\|_\infty \\ &= \|f\|_1 \cdot \|g\|_\infty \end{aligned}$$

$p > 1$: Assume that $\|f\|_p \cdot \|g\|_q > 0$. Then, applying the Young inequality with $a = \dfrac{|f(x)|}{\|f\|_p}$ and $b = \dfrac{|g(x)|}{\|g\|_q}$, we get

$$\begin{aligned} \frac{f(x)g(x)}{\|f\|_p \|g\|_q} &\leq \frac{|f(x)||g(x)|}{\|f\|_p \|g\|_q} \\ &\leq \frac{|f(x)|^p}{p\|f\|_p^p} + \frac{|g(x)|^q}{q\|g\|_q^q} \end{aligned}$$

Integrating both sides gives:

$$\begin{aligned}
\frac{\|fg\|_1}{\|f\|_p\|g\|_q} &= \frac{1}{\|f\|_p\|g\|_q} \cdot \int_X f(x)g(x)\, d\mu(x) \\
&\leq \frac{1}{\|f\|_p\|g\|_q} \cdot \int_X |f(x)||g(x)|\, d\mu(x) \\
&\leq \frac{1}{p\|f\|_p^p} \cdot \int_X |f(x)|^p\, d\mu(x) + \frac{1}{q\|g\|_q^q} \cdot \int_X |g(x)|^q\, d\mu(x) \\
&= \frac{1}{p\|f\|_p^p} \cdot \|f\|_p^p + \frac{1}{q\|g\|_q^q} \cdot \|g\|_q^q \\
&= \frac{1}{p} + \frac{1}{q} \\
&= 1
\end{aligned}$$

Which gives us our desired inequality.

□

Lemma 8.42. $\mathcal{L}^p$ is closed under addition.

Proof. 1. $p = 1$: Follows from the triangle inequality on $\mathbb{R}$:

$$\begin{aligned}
\|f+g\|_1 &= \int_X |f(x) + g(x)|\, d\mu(x) \\
&\leq \int_X |f(x)| + |g(x)|\, d\mu(x) \\
&= \int_X |f(x)|\, d\mu(x) + \int_X |g(x)|\, d\mu(x) \\
&= \|f\|_1 + \|g\|_1
\end{aligned}$$

2. $p \in (1, \infty)$: Since the function $t \mapsto t^p$ is convex on $[0, \infty)$, we have:

$$\begin{aligned}
|f+g|^p &= 2^p \left| \frac{f+g}{2} \right|^p \\
&\leq 2^{p-1}(|f|^p + |g|^p)
\end{aligned}$$

3. $p = \infty$: We have $|f(x)| \leq \|f\|_\infty$ and $|g(x)| \leq \|g\|_\infty$ for μ-almost all x, which immediately gives us

$$|f(x) + g(x)| \leq |f(x)| + |g(x)| \leq \|f\|_\infty + \|g\|_\infty$$

μ-almost everywhere and thus

$$\|f+g\|_\infty = \inf_{s>0} \left\{ |f+g| \leq s \ \mu\text{-almost everywhere} \right\} \leq \|f\|_\infty + \|g\|_\infty$$

□

We now want to show that the L^p norm defines a seminorm, which we will then easily be able to turn into a proper norm.

Theorem 8.43: Minkowski Inequality

Let $f, g \in \mathcal{L}^p$. Then we have

$$\|f+g\|_p \le \|f\|_p + \|g\|_p$$

Proof. We already showed the cases $p = 1$ and $p = \infty$ while proving $\mathcal{L}^p$ is closed under addition. Thus, let $p \in (1, \infty)$. Then the triangle inequality on $\mathbb{R}$ gives:

$$\begin{aligned}
|f(x)+g(x)|^p &= |f(x)+g(x)| \cdot |f(x)+g(x)|^{p-1} \\
&\le (|f(x)| + |g(x)|) \cdot |f(x)+g(x)|^{p-1} \\
&= |f(x)||f(x)+g(x)|^{p-1} + |g(x)||f(x)+g(x)|^{p-1}
\end{aligned}$$

Applying the Hölder inequality with $q = \frac{p}{p-1}$ gives:

$$\begin{aligned}
&\int_X |f(x)||f(x)+g(x)|^{p-1}\, d\mu(x) \\
&\le \|f\|_p \cdot \left\| |f+g|^{p-1} \right\|_q \\
&= \|f\|_p \cdot \left(\int_X |f(x)+g(x)|^{(p-1)q}\, d\mu(x) \right)^{\frac{1}{q}} \\
&= \|f\|_p \cdot \left(\int_X |f(x)+g(x)|^{p}\, d\mu(x) \right)^{\frac{p-1}{p}} \\
&= \|f\|_p \cdot \|f+g\|_p^{p-1}
\end{aligned}$$

Switching the roles of f and g similarly gives:

$$\int_X |g(x)||f(x)+g(x)|^{p-1}\, d\mu(x) \le \|g\|_p \cdot \|f+g\|_p^{p-1}$$

And putting everything together, we get:

$$\begin{aligned}
\|f+g\|_p^p &= \int_X |f(x)+g(x)|^p\, d\mu(x) \\
&\le \int_X |f(x)||f(x)+g(x)|^{p-1}\, d\mu(x) + \int_X |g(x)||f(x)+g(x)|^{p-1}\, d\mu(x) \\
&\le \|f\|_p \cdot \|f+g\|_p^{p-1} + \|g\|_p \cdot \|f+g\|_p^{p-1} \\
&= \|f+g\|_p \cdot \|f+g\|_p^{p-1}
\end{aligned}$$

From which we get our desired result by dividing out the factor of $\|f+g\|_p^{p-1}$. □

Corollary 8.44

The $\mathcal{L}^p$-norm is a seminorm on $\mathcal{L}^p$. In particular, we have:

1. $\|f\|_p \geq 0$, with $\|f\|_p = 0$ iff. $f = 0$ μ-almost everywhere
2. $\|\lambda f\|_p = |\lambda| \cdot \|f\|_p$
3. $\|f + g\|_p \leq \|f\|_p + \|g\|_p$

Corollary 8.45: L^p-space

Define L^p-space to be the quotient space of $\mathcal{L}^p$ under μ-almost-everywhere equality. Then $\|\cdot\|_p$ forms a proper norm on L^p.

Lemma 8.46. Let $1 \leq p < \infty$. Let $u_j \in L^p$ and

$$f_k = \sum_{j=1}^{k} u_j$$

Then, if $\sum_{j=1}^{\infty} \|u_j\|p < \infty$, we have:

1. The pointwise limit $f(x) = \lim_{k\to\infty} f_k(x)$ exists for μ-almost all $x \in X$,
2. $f \in L^p$,
3. $\|f - f_k\|_p \to 0$

Lemma 8.47: Convergence in L^p

If we have $f_k \to f$ in L^p Norm for $1 \leq p \leq \infty$, then there exists a subsequence f_{k_i} such that $f_{k_i} \to f$ pointwise μ-almost everywhere.

Theorem 8.48: Riesz-Fischer

For $1 \leq p \leq \infty$, $(L^p, \|\cdot\|_p)$ is a Banach space.

Definition 8.49. Let $1 \leq p < \infty$. Let the sequence $(f_n) \subset L^p$ converge pointwise μ-almost everywhere against the μ-measurable function f. Let

$$\nu(A) := \limsup_{n\to\infty} \int_A |f_n|^p \, d\mu$$

Then we call f_k **uniformly integrable** iff, for any $A \in \mathcal{A}$, we have:

1. $\forall \varepsilon > 0 : \exists \delta > 0 : \forall A \in \mathcal{A} : \mu(A) < \delta \implies \nu(A) < \varepsilon$
2. $\forall \varepsilon > 0 : \exists E \in \mathcal{A} : \mu(E) < \infty, \nu(X \setminus E) < \varepsilon$

Theorem 8.50: Vitali Convergence Theorem

Let $1 \leq p < \infty$. Let the sequence $(f_n) \subset L^p$ converge pointwise μ-almost everywhere against the μ-measurable function f. Then the following are equivalent:

1. $f \in L^p$ and $\|f_n - f\|_p \to 0$,
2. f_k is uniformly integrable.

Definition 8.51. Let $\Omega \subset \mathbb{R}^n$ be open. Then the **support of a function** $f : \Omega \to \mathbb{R}$ is the set

$$\operatorname{spt} f := \overline{\{x \in \Omega : f(x) \neq 0\}}$$

The space of k times continuously differentiable functions with compact support in Ω is denoted $C_c^k(\Omega)$.

Theorem 8.52: $C_c^0(\Omega)$ is dense in $L^p(\Omega)$

Let $\Omega \subset \mathbb{R}^n$ be open. Let $1 \leq p < \infty$. Then, for every $f \in L^p(\Omega)$, there exists a sequence f_k in $C_c^0(\Omega)$ such that $\|f - f_k\|_p \to 0$.

8.6 Density Functions

Theorem 8.53

Let $(X, \mathcal{A}, \mu)$ be a measure space. Let $\theta : X \to \overline{\mathbb{R}}$ be nonnegative and μ-measurable. Then the map

$$\begin{aligned} \nu : \mathcal{A} &\to \overline{\mathbb{R}} \\ A &\mapsto \int_A \theta \, d\mu \end{aligned}$$

is a measure, which we denote μ_θ. We call θ the **density of ν with respect to μ**.

Corollary 8.54. The following hold for μ_θ:

1. $\mu(A) = 0$ implies $\mu_\theta(A) = 0$.
2. For every nonnegative μ-measurable function f, we have

$$\int_X f \, d\mu_\theta = \int_X f \cdot \theta \, d\mu.$$

3. θ is unique up to equality μ-almost everywhere.

Definition 8.55

Let μ and ν be measures on $(X, \mathcal{A})$. Then we call ν **absolutely continuous with respect to μ**, which we denote $\nu \ll \mu$, if $\mu(A) = 0$ implies $\nu(A) = 0$.

Lemma 8.56. Let σ and ν be finite measures on $(X, \mathcal{A})$ such that $\nu(A) \leq \sigma(A)$ for every $A \in \mathcal{A}$. Then there exists a density function θ such that $\nu = \sigma_\theta$.

This lemma is actually a special case of the significantly more general Riesz representation theorem for Hilbert spaces, which I sadly do not have the time to go into at this point (but look it up, it's really neat!).

Lemma 8.57. Let μ, ν be measures on $(X, \mathcal{A})$. Let $\sigma := \mu + \nu$. Then if $f \in \mathcal{L}^*(\sigma)$, then $f \in \mathcal{L}^*(\mu)$ and $f \in \mathcal{L}^*(\nu)$ and we have

$$\int_X f \, d\sigma = \int_X f \, d\mu + \int_X f \, d\nu$$

Theorem 8.58. (Mini-Radon-Nikodym): Let μ, ν be finite measures on $(X, \mathcal{A})$ such that $\nu \ll \mu$. Then there exists a density function $\theta \in L^1(\mu)$ such that $\nu = \mu_\theta$.

In this case, θ is sometimes also called the **Radon-Nikodym derivative of ν with respect to μ** and denoted $\frac{d\nu}{d\mu}$.

Theorem 8.59. Let μ and ν be finite measures on $(X, \mathcal{A})$. Then the following are equivalent:

1. $\nu \ll \mu$,
2. There exists a density function $\theta \in L^1(\mu)$ such that $\nu = \mu_\theta$,
3. For all $\varepsilon > 0$, there exists a $\delta > 0$ such that $\mu(A) < \delta$ implies $\nu(A) < \varepsilon$.

Proof. (i) $\implies$ (ii) is our mini-Radon-Nikodym theorem. (ii) $\implies$ (iii) follows from absolute continuity of the Lebesque integral. (iii) $\implies$ (i) follows immediately from the definition of $\ll$. □

Theorem 8.60: Radon-Nikodym

Let μ, ν be σ-finite measures on $(X, \mathcal{A})$ such that $\nu \ll \mu$. Then there exists a density function $\theta \in L^1(\mu)$ such that $\nu = \mu_\theta$.

Definition 8.61

Let μ and ν be measures on $(X, \mathcal{A})$. Then we call μ and ν **singular with respect to each other**, which we denote $\mu \perp \nu$, if there exists a set $M \in \mathcal{A}$ such that

$$\mu(M) = \nu(X \setminus M) = 0.$$

Theorem 8.62: Lebesque's decomposition theorem

Let μ and ν be measures on $(X, \mathcal{A})$, and let ν be σ-finite. Then there exists a unique decomposition $\nu = \nu_a + \nu_s$ such that $\nu_a \ll \mu$ and $\nu_a \perp \mu$.

Chapter 9

Integration in $\mathbb{R}^n$

9.1 Products of Measure Spaces

In this section, we will be interested in finding a canonical σ-Algebra on cartesian products of measurable spaces. Category theory tells us the correct solution directly: The product σ-algebra of an indexed family of σ-algebras as the sigma algebra fulfilling the universal property of product objects. We are working in the category of measurable spaces, where morphisms are measurable functions.

Therefore, our definition simply amounts to defining product sigma algebras as the σ-algebras induced by the projection functions, which in turn is the smallest σ-algebra such that all projections are measurable.

Definition 9.1: Product σ-Algebra

Let $(X_i, \mathcal{A}_i)_{i\in I}$ be a family of measurable spaces. Then the product σ-algebra on the cartestian product $\prod_{i\in I} X_i$ is the smallest σ-Algebra $\bigotimes_{i\in I} \mathcal{A}_i$ such that every projection π_j is $\bigotimes_{i\in I} \mathcal{A}_i - \mathcal{A}_j$-measurable.

Explicitly, this means we have:

$$\bigotimes_{i\in I} \mathcal{A}_i = \left\langle \bigcup_{i\in I} \pi_i^{-1}(\mathcal{A}_i) \right\rangle_\sigma$$

Note that this is also entirely analogous to the definition of the product topology, which is the coarsest (i.e. "smallest") topology such that projections are continuous.

If the cardinality of I is a small finite number, we may sometimes write out the product σ-algebra as:

$$\bigotimes_{i\in I} \mathcal{A}_i = \mathcal{A}_1 \otimes \mathcal{A}_2 \otimes \dots$$

Theorem 9.2

Let $(X, \mathcal{A})$ and $(Y_i, \mathcal{B}_i)$ be measurable spaces. Let $g : X \to \prod_{i\in I} Y_i$. Then g is $\mathcal{A} - \bigotimes_{i\in I} \mathcal{B}_i$-measurable if and only if all maps $\pi_i \circ g$ are $\mathcal{A} - \mathcal{B}_i$-measurable.

Proof. 1. Assume g is measurable. We know that π_i are measurable by definition, and we know that compositions of measurable functions are measurable. Therefore, $\pi_i \circ g$ are measurable.

2. Assume $\pi_i \circ g$ are measurable. Then we have:

$$\begin{aligned} g^{-1}\left(\bigcup_{i\in I} \pi_i^{-1}(\mathcal{A}_i)\right) &= \bigcup_{i\in I} g^{-1}(\pi_i^{-1}(\mathcal{A}_i)) \\ &= \bigcup_{i\in I} (\pi_i \circ g)^{-1}(\mathcal{A}_i)) \subset \mathcal{A} \end{aligned}$$

Where the last step follows from the measurability of $\pi_i \circ g$. Therefore, since $\bigcup_{i\in I} \pi_i^{-1}(\mathcal{A}_i)$ forms a basis of $\bigotimes_{i\in I} \mathcal{A}_i$, we have that g is measurable.

□

As a corollary, we get that our product space actually fulfills the universal property of product objects:

Corollary 9.3. Let $(X, \mathcal{A})$ and $(Y_i, \mathcal{B}_i)_{i\in I}$ be measurable spaces. Let $g_i : X \to Y_i$ be a family of $\mathcal{A} - \mathcal{B}_i$-measurable functions. Then there exists a $\mathcal{A} - \mathcal{B}_i$-measurable function $g : X \to \prod_{i\in I} Y_i$ such that, for each $i \in I$, $g_i = \pi_i \circ g$.

Proof. Define g as the cartesian product of all g_i:

$$\begin{aligned} g : I &\to \left(X \to \prod_{i\in I} Y_i\right) \\ i &\mapsto g_i \end{aligned}$$

Then our previous lemma tells us that, since all $g_i = \pi_i \circ g$ are measurable, g is measurable as well. □

Corollary 9.4. Let $(X_i, \mathcal{A}_i)_{i\in I}$ be measurable spaces, and let $J \subset I$ be nonempty. Then the projection π_J is $\bigotimes_{i\in I} \mathcal{A}_i - \bigotimes_{j\in J} \mathcal{A}_j$-measurable.

Definition 9.5. (Embedding into a larger product space): Let $p_j \in X_j$. Then we define the **embedding of $\prod_{i\in I\setminus j} X_i$ into $\prod_{i\in I} X_i$ by x_j** to be the map e_{x_j} which takes a product $(x_i)_{i\in I\setminus j}$ and "adds in x_j as a new component", i.e. it maps $(x_i)_{i\in I\setminus j}$ to the product $(x'_i)_{i\in I}$ with $x'_j = x_j$ and $(x'_i)_{i\in I\setminus j} = (x_i)_{i\in I\setminus j}$.

Definition 9.6: Cuts of subsets

Let $M \subseteq \prod_{i\in I} X_i$, and let $x_j \in X_j$. Then we define the **cut of M through x_j** to be the set $M^{x_j} = e_{x_j}^{-1}(M)$.

More explicitly, M^{x_j} is the set such that for for every $y \in M$, we have $y|_{I\setminus\{j\}} \in M^{x_j}$ if and only if $y_j = x_j$.

Corollary 9.7. Let $(X_i, \mathcal{A}_i)_{i\in I}$ be measurable spaces, let $M \in \bigotimes_{i\in I} \mathcal{A}_i$, and let

$x_j \in X_j$ for $j \in I$. Then we have

$$M^{x_j} \in \bigotimes_{i \in I \setminus j} \mathcal{A}_i$$

Proof. Let $x_j \in X_j, j \in I$. We want to show that e_{x_j} is measurable, which would follow immediately if we could show that all projections $\pi_i^I \circ e_{x_j}$ are measurable.

1. Let $i \in I \setminus \neq j$. Then $\pi_i^I \circ e_{x_j} = \pi_i^{I \setminus j}$, which is measurable.
2. Let $i = j$. Then $\pi_i^I \circ e_{x_j} = \pi_j^I \circ e_{x_j}$, which is just the constant map onto x_j, which is measurable.

Since all projections are measurable, e_{x_j} itself must also be measurable. Therefore, since M is measurable, the preimage $M^{x_j} = e_{x_j}^{-1}(M)$ must also be measurable. □

Note that, in particular, this tells us that if $(X_1, \mathcal{A}_1)$ and $(X_2, \mathcal{A}_2)$ are measurable spaces, then for any $M \in \mathcal{A}_1 \otimes \mathcal{A}_2$, $x_1 \in X_1$, and $x_2 \in X_2$, we have

$$M^{x_2} = \{x \in X_1 \mid (x, x_2) \in M\} \in \mathcal{A}_1$$

and

$$M^{x_1} = \{x \in X_2 \mid (x_1, x) \in M\} \in \mathcal{A}_2$$

Theorem 9.8: A basis of the product σ-Algebra

Let $(X_i, \mathcal{A}_i)_{i \in I}$ be a family of measurable spaces. For each $i \in I$, let $\mathcal{E}_i$ be a basis of $\mathcal{A}_i$. Then we have

$$\bigotimes_{i \in I} \mathcal{A}_i = \left\langle \bigcup_{i \in I} \pi_i^{-1}(\mathcal{E}_i) \right\rangle_\sigma .$$

Proof. Recall that, for any map f and any system of subsets $\mathcal{S} \subset \mathcal{A}$ of a σ-algebra $\mathcal{A}$, we have

$$f^{-1}(\langle \mathcal{S}_i \rangle_\sigma) = \left\langle f^{-1}(\mathcal{S}_i) \right\rangle_\sigma$$

and

$$\left\langle \bigcup_{i \in I} \mathcal{S}_i \right\rangle_\sigma = \left\langle \bigcup_{i \in I} \langle \mathcal{S}_i \rangle_\sigma \right\rangle_\sigma ,$$

which gives us:

$$
\begin{aligned}
\bigotimes_{i\in I} \mathcal{A}_i &= \left\langle \bigcup_{i\in I} \pi_i^{-1}(\mathcal{A}_i) \right\rangle_\sigma \\
&= \left\langle \bigcup_{i\in I} \pi_i^{-1}(\langle \mathcal{E}_i \rangle_\sigma) \right\rangle_\sigma \\
&= \left\langle \bigcup_{i\in I} \left\langle \pi_i^{-1}(\mathcal{E}_i) \right\rangle_\sigma \right\rangle_\sigma \\
&= \left\langle \bigcup_{i\in I} \pi_i^{-1}(\mathcal{E}_i) \right\rangle_\sigma
\end{aligned}
$$

□

If I is finite, we get the following representation of the product σ-algebra:

Corollary 9.9. For each $i \in \{1, \ldots, n\}$, let $(X_i, \mathcal{A}_i)$ be a measurable space such that $\mathcal{E}_i$ is a basis of $\mathcal{A}_i$ and such that X_i can be represented as a countable union of basis elements $E_i^k \in \mathcal{E}_i$. Then

$$
Q_0 := \left\{ \prod_{i=1}^n E_i \;\middle|\; E_i \in \mathcal{E}_i \right\}
$$

is a basis of $\bigotimes_{i=1}^n \mathcal{A}_i$.

Proof. We need to show $\langle Q_0 \rangle_\sigma = \bigotimes_{i=1}^n \mathcal{A}_i$.

1. Let $E_i \in \mathcal{E}_i$. We know that

$$
\pi_i^{-1}(E_i) = E_i \times \prod_{j\neq i} X_j,
$$

which gives us the following representation for the basis Q we just established in the last theory:

$$
Q = \bigcup_{i=1}^n \pi_i^{-1}(\mathcal{E}_i) = \bigcup_{i=1}^n \left\{ E_i \times \prod_{j\neq i} X_j \;\middle|\; E_i \in \mathcal{E}_i \right\}
$$

Furthermore, $\cap$-stability of σ-algebras gives us that any element

$$
\prod_{i=1}^n E_i = \bigcap_{i=1}^n \pi_i^{-1}(E_i)
$$

of Q_0 must be contained in $\langle Q \rangle_\sigma$.

Therefore, we have

$$
\text{ť}\, \langle Q_0 \rangle_\sigma \subset \langle Q \rangle_\sigma = \bigotimes_{i=1}^n \mathcal{A}_i
$$

2. Now, let $E_i \in \mathcal{E}_i$ and $k \in \mathbb{N}$. By definition of Q_0, we have

$$
E_i \times \prod_{j\neq i} E_j^k \in Q_0,
$$

which gives us:

$$\pi_i^{-1}(E_i) = \bigcup_{k=1}^{\infty}\left(E_i \times \prod_{j\neq i} E_j^k\right) \in \langle \mathcal{Q}_0\rangle_\sigma\,,$$

and by extension:

$$\mathcal{Q} = \bigcup_{i=1}^{n} \pi_i^{-1}(E_i) \subset \langle \mathcal{Q}_0\rangle_\sigma$$

which means we also have

$$\bigotimes_{i=1}^{n} \mathcal{A}_i = \langle \mathcal{Q}\rangle_\sigma \subset \sigma(\mathcal{Q}_0).$$

□

Corollary 9.10. The Borel σ-algebra $\mathcal{B}(\mathbb{R}^n)$ is the product of the Borel σ-algebra $\mathcal{B}(\mathbb{R})$, i.e. we have:

$$\mathcal{B}(\mathbb{R}^n) = \bigotimes_{i=1}^{n} \mathcal{B}(\mathbb{R})$$

Proof. We know that the set $\mathcal{I}$ of real intervals forms a basis of $\mathcal{B}(\mathbb{R})$. Given this basis, the set $\mathcal{Q}_0$ is exactly the set of n-dimensional real cuboids, which forms a basis of $\mathcal{B}(\mathbb{R}^n)$. Therefore, we have:

$$\mathcal{B}(\mathbb{R}^n) = \langle \mathcal{Q}_0\rangle_\sigma = \bigotimes_{i=1}^{n} \mathcal{B}(\mathbb{R})$$

□

For the final part of this chapter, we establish some notation for commonly encountered subsets of product σ-algebras.

Definition 9.11. Let I be a set. Then we denote by $\mathcal{P}_0(I)$ the finite, nonenmpty subsets of I.

Definition 9.12: Measurable Cuboids

Let $(X_i, \mathcal{A}_i)_{i\in I}$ be a family of measurable spaces. Then we define the set of **measurable cuboids in $\prod_{i\in I} X_i$** as:

$$\mathcal{Q} := \bigcup_{J\in\mathcal{P}_0(I)} \left\{ \prod_{j\in J} A_j \times \prod_{I\setminus J} X_i \;\middle|\; A_j \in \mathcal{A}_j \right\},$$

i.e. a finite number of the sides of our cuboid can be arbitrary measurable sets, and the remaining sides must correspond to the remaining domain sets $(X_i)_{i\in I\setminus J}$.

Definition 9.13: Cylindrical Sets

Let $(X_i, \mathcal{A}_i)_{i\in I}$ be a family of measurable spaces. Then we define the set of **cylinder sets in $\prod_{i\in I} X_i$** as:

$$\begin{aligned}\mathcal{Z} &:= \bigcup_{J\in\mathcal{P}_0(I)} \left\{ A_J \times \prod_{i\in I\setminus J} X_i \,\middle|\, A_J \in \bigotimes_{j\in J} \mathcal{A}_j \right\} \\ &= \bigcup_{J\in\mathcal{P}_0(I)} \pi_J^{-1}\left(\bigotimes_{j\in J} \mathcal{A}_j \right),\end{aligned}$$

i.e. the base of the cylinder is a measurable set $A_j \subset \prod_{i\in I} X_i$ and all other sides are once again the remaining domain sets $(X_i)_{i\in I\setminus J}$.

Note that if we view an everyday geometric cylinder $Z \subset \mathbb{R}^3$ as the product of of the circle at its base with a finite real interval, it doesn't actually qualify as a cylindrical set in $\mathbb{R}^3$, since it would have to be a product of the base circle with all of $\mathbb{R}$, i.e. only "infinite" cylinders immediately qualify.

By technicality, normal cylinders are still cylindrical sets though, since we can set $J = I$ and have view the entire cylinder as the base, without adding any additional "sides".

Therefore, this definition is not very useful if I is finite, since every measurable set is a cylindrical with itself as its base and every cylindrical set is a product of measurable sets, i.e. measurable.

Theorem 9.14

Let $(X_i, \mathcal{A}_i)_{i\in I}$ be a family of measurable spaces. Then the measurable cuboids in $\prod_{i\in I} X_i$ and the cylindrical sets in $\prod_{i\in I} X_i$ form bases of $\bigotimes_{i\in I} \mathcal{A}_i$.

Proof. We will show $\mathcal{Q} \subset \mathcal{Z} \subset \bigotimes_{i\in I} \mathcal{A}_i \subset \langle \mathcal{Q} \rangle_\sigma$, which implies $\langle \mathcal{Q} \rangle_\sigma \subset \langle \mathcal{Z} \rangle_\sigma \subset \bigotimes_{i\in I} \mathcal{A}_i \subset \langle \mathcal{Q} \rangle_\sigma$, proving the theorem.

1. $\mathcal{Q} \subset \mathcal{Z}$: Let $A \in \mathcal{Q}$. Then A is given as

$$\begin{aligned} A &= \prod_{j\in J}^{n} A_j \times \prod_{i\notin J} X_i \\ &= \pi_J^{-1}\left(\prod_{j\in J} A_j \right). \end{aligned}$$

   Therefore, since $\prod_{j\in J} A_j \in \bigotimes_{j\in J} \mathcal{A}_j$, A is a cylinder set.

2. $\mathcal{Z} \subset \bigotimes_{i\in I} \mathcal{A}_i$: Recall that π_J is $\bigotimes_{i\in I} \mathcal{A}_i - \mathcal{A}_J$-measurable. Therefore, for any $J \in \mathcal{P}_0(I)$, we have that $\pi_J^{-1}(\mathcal{A}_j) \subset \bigotimes_{i\in I} \mathcal{A}_i$, which means any cylinder set is contained in $\bigotimes_{i\in I} \mathcal{A}_i$.

3. $\bigotimes_{i\in I}\mathcal{A}_i \subset \langle \mathcal{Q}\rangle_\sigma$: For any $i \in I$ and $A_i \in \mathcal{A}_i$, we have:

$$\pi_i^{-1}(A_i) = A_i \times \prod_{j\neq i} X_j \in \mathcal{Q},$$

i.e. $\pi_i^{-1}(\mathcal{A}_i) \subset \mathcal{Q}$ for all $i \in I$, which means we also have:

$$\bigcup_{i\in I}\pi_i^{-1}(\mathcal{A}_i) \subset \mathcal{Q}.$$

By definition of $\bigotimes_{i\in I}\mathcal{A}_i$, we get:

$$\bigotimes_{i\in I}\mathcal{A}_i = \left\langle \bigcup_{i\in I}\pi_i^{-1}(\mathcal{A}_i)\right\rangle_\sigma \subset \langle \mathcal{Q}\rangle_\sigma.$$

□

9.2 Product Measures and Fubini's Theorem

In this section, given two measure spaces $(X, \mathcal{A}, \mu)$ and $(Y, \mathcal{B}, \nu)$, we want to derive a canonical measure $\mu \otimes \nu$ on $\mathcal{A} \otimes \mathcal{B}$. Our intuition already tells us what the measure of a cuboid should be: Given a cuboid

$$Q = A \times B \subset X \times Y,$$

where $A \in \mathcal{A}$ and $B \in \mathcal{B}$, we want the measure of the cuboid to be the product of the measures of its sides:

$$(\mu \otimes \nu)(M) = \mu(A) \cdot \nu(B)$$

Recall that, for any $M \in X \times Y$, the cuts through M by $x \in X$ and $y \in Y$ are simply defined as

$$\begin{aligned} M^x &= \{y \in Y \mid (x, y) \in M\}, \\ M^y &= \{x \in X \mid (x, y) \in M\}. \end{aligned}$$

Theorem 9.15

Let $(X, \mathcal{A}, \mu)$ and $(Y, \mathcal{B}, \nu)$ be σ-finite measure spaces. Let $M \in \mathcal{A} \otimes \mathcal{B}$. Then:

1. The function $x \mapsto \nu(M^x)$ is $\mathcal{A}$-measurable,
2. The function $y \mapsto \mu(M^y)$ is $\mathcal{B}$-measurable,
3. We have

$$\int_X \nu(M^x)\, d\mu(x) = \int_Y \mu(M^y)\, d\nu(y)$$

Proof. Let $\mathcal{S}$ be the system of all sets $M \in \mathcal{A} \otimes \mathcal{B}$ for which this theorem holds.

Let $\mathcal{Q}$ be the system of all measurable cuboids in $X \times Y$, which we have just established are simply given by $Q = A \times B$ such that $A \in \mathcal{A}$ and $B \in \mathcal{B}$.

Let $\mathcal{F}$ be the system of finite unions of cuboids (which by our definition of cuboids inherently includes complements of cuboids). Since we count $X \times Y$ as a cuboid, $\mathcal{F}$ forms a set algebra.

We will show that $\mathcal{S}$ is a monotone class containing $\mathcal{F}$, which will give us

$$\mathcal{A} \otimes \mathcal{B} = \langle \mathcal{Q} \rangle_\sigma = \langle \mathcal{F} \rangle_\sigma = \langle \mathcal{F} \rangle_M \subseteq \mathcal{S}$$

since $\mathcal{S} \subseteq \mathcal{A} \otimes \mathcal{B}$ by definition, this implies $\mathcal{A} \otimes \mathcal{B} = \mathcal{S}$.

The proof the $\mathcal{S}$ is a monotone class is however left as an exercise to the reader :) □

Theorem 9.16: Product measure

Let $(X, \mathcal{A}, \mu)$ and $(Y, \mathcal{B}, \nu$ be σ-finite measure spaces. Then there exists exactly one measure $\mu \otimes \nu$ on $\mathcal{A} \otimes \mathcal{B}$ such that for any $A \in \mathcal{A}$ and $B \in \mathcal{B}$, we have

$$(\mu \otimes \nu)(A \times B) = \mu(A)\nu(B)$$

This measure is given by

$$(\mu \otimes \nu)(M) = \int_X \nu(M^x)\, d\mu(x) = \int_Y \mu(M^y)\, d\mu(y)$$

Corollary 9.17: Cavalieri's Principle

Let $(X, \mathcal{A}, \mu)$ and $(Y, \mathcal{B}, \nu)$ be σ-finite measure spaces. Then, for all $M \in \mathcal{A} \otimes \mathcal{B}$, we have

$$\begin{aligned}(\mu \otimes \nu)(M) &= \int_X \nu(M^x)\, d\mu(x) \\ &= \int_X \int_Y \chi_M(x, y)\, d\nu(y)\, d\mu(x) \\ &= \int_Y \mu(M^Y)\, d\nu(x) \\ &= \int_Y \int_X \chi_M(x, y)\, d\mu(x)\, d\nu(y)\end{aligned}$$

Corollary 9.18. Let $(X, \mathcal{A}, \mu)$ and $(Y, \mathcal{B}, \nu)$ be σ-finite measure spaces. Then, for any $M \in \mathcal{A} \otimes \mathcal{B}$, the following are equivalent:

1. $(\mu \otimes \nu)(M) = 0$,
2. $\mu(M^y) = 0$ ν-almost everywhere,
3. $\nu(M^x) = 0$ μ-almost everywhere.

Lemma 9.19. Let $(X, \mathcal{A}, \mu)$ and $(Y, \mathcal{B}, \nu)$ be σ-finite measure spaces. Let $f : X \times Y \to \mathbb{R}$ be $\mathcal{A} \otimes \mathcal{B}$ measurable. Then the function $f(x, \cdot) : Y \to \mathbb{R}$ is $\mathcal{B}$-measurable for all $x \in X$.

Theorem 9.20: Fubini's Theorem

Let $(X, \mathcal{A}, \mu)$ and $(Y, \mathcal{B}, \nu)$ be σ-finite measure spaces. Let $f \in \mathcal{L}^*(\mu \otimes \nu)$. Then:

$$\begin{aligned}\int_{X \times Y} f\, d(\mu \otimes \nu) &= \int_X \int_Y f(x, y)\, d\nu(y)\, d\mu(x) \\ &= \int_Y \int_X f(x, y)\, d\mu(x)\, d\nu(y)\end{aligned}$$

Corollary 9.21. The volume of the open ball

$$B_r^n(0) = \left\{\vec{x} \in \mathbb{R}^n \;\middle|\; \left\|\vec{x}\right\|_2 < r\right\}.$$

is given by:

$$\lambda^n(B_r^n(0)) = \frac{\pi^{n/2}}{\Gamma(\frac{n}{2}+1)} \cdot r^n = \begin{cases} \frac{\pi^k}{k!} \cdot r^n & n = 2k \\ \frac{\pi^k}{\prod_{i=1}^k \frac{1}{2}+k-i} \cdot r^n & n = 2k+1 \end{cases}$$

In particular:

1. The 1-dimensional unit ball is simply the interval $(-1, 1)$ and thus has length 2,
2. The 2-dimensional unit ball is a circle of radius 1 and has area π
3. The 3-dimensional unit ball has volume $\frac{4\pi}{3}$
4. The 4-dimensional unit ball has volume $\frac{\pi^2}{2}$
5. The 5-dimensional unit ball has volume $\frac{8\pi^2}{15}$
6. ...

Proof. By Cavalieri's principle and the linear transformation formula for the Lebesque measure, we get:

$$\begin{aligned}
\lambda^n(B_r^n(0)) &= \int_{\mathbb{R}} \lambda^{n-1}(B_r^n(0)^y)\, dy \\
&= \int_{(-r,r)} \lambda^{n-1}(B_r^n(0)^y)\, dy \\
&= \int_{-1}^{1} = \det\left(\sqrt{r^2-y^2} \cdot I_n\right) \cdot \lambda^{n-1}\left\{x \in \mathbb{R}^{n-1} \;\middle|\; \|x\|_2 < 1\right\} dy \\
&= r^n \cdot \int_{-1}^{1} \left(\sqrt{r^2-y^2}\right)^{n-1} \cdot \lambda^{n-1}\left\{x \in \mathbb{R}^{n-1} \;\middle|\; \|x\|_2 < 1\right\} dy \\
&= r^n \cdot \lambda^{n-1}(B_1^{n-1}(0)) \cdot \int_{-1}^{1} \left(\sqrt{1-y^2}\right)^{n-1} dy
\end{aligned}$$

We can thus get our result by solving the integral

$$I_n := \int_{-1}^{1} \left(\sqrt{1-y^2}\right)^{n-1} dy$$

Substituting $\cos(\theta) = y$ gives:

$$\begin{aligned}
I_n &= \int_{-1}^{1} \left(\sqrt{1-y^2}\right)^{n-1} dy \\
&= \int_{-1}^{1} \left(\sqrt{1-\cos(\theta)^2}\right)^{n-1} \cdot \sin(\theta)\, d\theta \\
&= \int_{0}^{\pi} \sin^n(\theta)\, d\theta
\end{aligned}$$

Now, partial integration gives:

$$\begin{aligned}
I_n &= \int_0^\pi \sin^n(\theta)\, d\theta \\
&= \int_0^\pi \sin(\theta) \cdot \sin^{n-1}(\theta)\, d\theta \\
&= \left[-\cos(\theta) \cdot \sin^{n-1}(\theta)\right]_0^\pi + (n-1) \cdot \int_0^\pi \cos^2(\theta) \cdot \sin^{n-2}(\theta)\, d\theta \\
&= (n-1) \cdot \int_0^\pi \cos^2(\theta) \cdot \sin^{n-2}(\theta)\, d\theta \\
&= (n-1) \cdot \int_0^\pi (1-\sin^2) \cdot \sin^{n-2}(\theta)\, d\theta \\
&= (n-1) \cdot (I_{n-2} - I_n),
\end{aligned}$$

which means:

$$\begin{aligned}
I_n &= (n-1) \cdot I_{n-2} - (n-1) \cdot I_n \\
\implies nI_n &= (n-1) \cdot I_{n-2} \\
\implies I_n &= \frac{n-1}{n} \cdot I_{n-2}
\end{aligned}$$

Unrolling this recursive formula, we get:

$$\begin{aligned}
I_{2k} &= I_0 \cdot \prod_{i=1}^{n} \frac{2i-1}{2i} \\
I_{2k+1} &= I_1 \cdot \prod_{i=1}^{n} \frac{2i}{2i+1}
\end{aligned}$$

We can easily calculate I_0 and I_1 explicitly:

$$\begin{aligned}
I_0 &= \int_0^\pi \sin^0(\theta)\, d\theta \\
&= \int_0^\pi 1\, d\theta \\
&= \pi
\end{aligned}$$

$$\begin{aligned}
I_1 &= \int_0^\pi \sin(\theta)\, d\theta \\
&= (-\cos(\pi)) - (-\cos(0)) \\
&= 2
\end{aligned}$$

Thus, we have:

$$\begin{aligned}
I_{2k} &= \pi \cdot \prod_{i=1}^{k} \frac{2i-1}{2i} \\
I_{2k+1} &= 2 \cdot \prod_{i=1}^{k} \frac{2i}{2i+1}
\end{aligned}$$

This gives us:

$$\begin{aligned}
I_{2k+1}I_{2k} &= 2\pi \cdot \prod_{i=1}^{k} \frac{2i-1}{2i} \cdot \prod_{i=1}^{k} \frac{2i}{2i+1} \\
&= 2\pi \cdot \prod_{i=1}^{k} \frac{2i-1}{2i+1} \\
&= \frac{2\pi}{2k+1} \\
&= \frac{\pi}{k+\frac{1}{2}}
\end{aligned}$$

$$\begin{aligned}
I_{2k}I_{2k-1} &= 2\pi \cdot \prod_{i=1}^{k} \frac{2i-1}{2i} \cdot \prod_{i=1}^{k-1} \frac{2i}{2i+1} \\
&= 2\pi \cdot \frac{2k-1}{2k} \cdot \prod_{i=1}^{k-1} \frac{2i-1}{2i} \cdot \prod_{i=1}^{k-1} \frac{2i}{2i+1} \\
&= 2\pi \cdot \frac{2k-1}{2k} \cdot \frac{1}{2k-1} \\
&= \frac{\pi}{k}
\end{aligned}$$

We know that

$$\lambda(B_1^1(0)) = \lambda((-1,1)) = 2$$

And, recalling the fact that λ^0 is the counting measure, we also know that

$$\lambda^0(B_1^0(0)) = \lambda^0(\{()\}) = 1$$

Thus, at long last, we can calculate the volumes of our unit balls:

$$\begin{aligned}
\lambda^{2k}(B_1^{2k}(0)) &= (I_{2k}I_{2k-1}) \cdot (I_{2k-2}I_{2k-3}) \cdot \ldots \cdot (I_2 I_1) \cdot \lambda^0(B^0(0)) \\
&= \left(\prod_{i=1}^{k} \frac{\pi}{i}\right) \cdot 1 \\
&= \frac{\pi^k}{k!} = \frac{\pi^{n/2}}{\Gamma\left(\frac{n}{2}+1\right)}
\end{aligned}$$

$$
\begin{aligned}
\lambda^{2k+1}(B_1^{2k+1}(0)) &= (I_{2k+1}I_{2k}) \cdot (I_{2k-1}I_{2k-2}) \cdot \ldots \cdot (I_3 I_2) \cdot \lambda^1(B^1(0)) \\
&= \left(\prod_{i=1}^{k} \frac{\pi}{i + \frac{1}{2}} \right) \cdot 2 \\
&= \frac{\pi^k}{\prod_{i=1}^{k} i + \frac{1}{2}} \cdot 2 \\
&= \frac{\pi^k}{\prod_{i=0}^{k} i + \frac{1}{2}} \\
&= \frac{\pi^k \cdot \sqrt{\pi}}{\left(\prod_{i=0}^{k} i + \frac{1}{2}\right) \cdot \sqrt{\pi}} \\
&= \frac{\pi^{\frac{n}{2}}}{\left(\prod_{i=0}^{k} i + \frac{1}{2}\right) \cdot \Gamma\left(\frac{1}{2}\right)} \\
&= \frac{\pi^{\frac{n}{2}}}{\left(\prod_{i=1}^{k+1} i - \frac{1}{2}\right) \cdot \Gamma\left(\frac{1}{2}\right)} \\
&= \frac{\pi^{\frac{n}{2}}}{\Gamma\left(k + \frac{1}{2} + 1\right)} \\
&= \frac{\pi^{n/2}}{\Gamma\left(\frac{n}{2} + 1\right)}
\end{aligned}
$$

Which gives us

$$
\lambda^n(B_r^n(0)) = \frac{\pi^{n/2}}{\Gamma(\frac{n}{2} + 1)} \cdot r^n
$$

as desired. □

Proposition 9.22. The 5-dimensional unit ball has the highest volume out of any unit ball. As the dimension increases, the volume of the unit ball converges to zero.

Proof Sketch. Plugging small natural numbers into our formula, we see that the formula is monotonically increasing for $n \leq 5$ and monotonically decreasing for $n \geq 5$. Furthermore, we have $\Gamma(\frac{n}{2} + 1) \geq \Gamma(k + 1) = k!$, which grows asymptotically faster than $\pi^{n/2} \leq \sqrt{\pi} \cdot \pi^k$. ”□”

Lemma 9.23

Let $j \in \{1, \ldots, n\}$. Let $f : \mathbb{R}^n \to \mathbb{R}$ be continously differentiable with $f, \partial_j f \in L^1(\mathbb{R}^n)$. Then

$$
\int_{\mathbb{R}^n} \partial_j f(x) \, d\lambda^n(x) = 0
$$

Proof. 1. First, assume there is an $R > 0$ such that $f(x) = 0$ for all $\|x\|_2 \geq R$. Let $x = (y, x_n) \in \mathbb{R}^{n-1} \times \mathbb{R}$. Then by Fubini we get:

$$\begin{aligned}\int_{\mathbb{R}^n} \partial_j f(x)\, d\lambda^n(x) &= \int_{\mathbb{R}^n} \partial_j f(y, x_n)\, d\lambda^n(y, x_n) \\ &= \int_{\mathbb{R}^{n-1}} \int_{\mathbb{R}} \partial_j f(y, x_n)\, d\lambda^1(x_n)\, d\lambda^{n-1}(y).\end{aligned}$$

For $j = n$, we apply the fundamental theorem of calculus and our assumption that $f(x) = 0$ for all $\|x\|_2 \geq R$ and get:

$$\begin{aligned}\int_{\mathbb{R}} \partial_j f(y, x_n)\, d\lambda^1(x_n) &= \lim_{x_n \to \infty} f(y, x_n) - f(y, -x_n) \\ &= 0\end{aligned}$$

By induction, the same holds for all j.

2. Now, let $\varphi : \mathbb{R} \to \mathbb{R}$ be a smooth function fulfilling:

$$\varphi(t) = \begin{cases} 1 & |t| \leq 1, \\ 0 & |t| \geq 2. \end{cases}$$

And let

$$\begin{aligned}\varphi_R &: \mathbb{R}^n \to \mathbb{R} \\ \varphi_R(x) &= \varphi\left(\frac{\|x\|_2}{R}\right)\end{aligned}$$

Then we have

$$(\varphi_R f)(x) = 0$$

for all $\|x\|_2 \geq 2R$, meaning that by the product rule of differentiation and step 1 we get:

$$\begin{aligned}\int_{\mathbb{R}^n} \varphi_R(x) \cdot \partial_j f(x)\, d\lambda^n(x) + \int_{\mathbb{R}^n} \partial_j \varphi_R(x) \cdot f(x)\, d\lambda^n(x) \\ = \int_{\mathbb{R}^n} \partial_j (\varphi_R(x) \cdot f(x))\, d\lambda^n(x) \\ = 0,\end{aligned}$$

i.e.

$$\int_{\mathbb{R}^n} \varphi_R(x) \cdot \partial_j f(x)\, d\lambda^n(x) = -\int_{\mathbb{R}^n} \partial_j \varphi_R(x) \cdot f(x)\, d\lambda^n(x)$$

Now, notice we also have $\varphi_R(x) = 1$ for $\|x\| \leq R$ - in particular, we have $\varphi_R \to 1$ and $\partial_j \varphi_R \to 0$ pointwise on $\mathbb{R}^n$ for $R \to \infty$, which means we have

$$\int_{\mathbb{R}^n} \partial_j f(x)\, d\lambda^n(x) = \int_{\mathbb{R}^n} \lim_{R \to \infty} \varphi_R(x) \cdot \partial_j f(x)\, d\lambda^n(x)$$

and

$$\int_{\mathbb{R}^n} \lim_{R \to \infty} \partial_j \varphi_R(x) \cdot f(x)\, d\lambda^n(x) = 0$$

Finally, we also have:

$$\left|\varphi_R \cdot (\partial_j f)\right| \leq \left|\partial_j f\right| \in L^1(\mathbb{R}^n)$$

and

$$\begin{aligned}\left|(\partial_j \varphi_R) \cdot f\right| &\leq \frac{\max|\varphi'|}{R} \cdot |f| \\ &\leq (\max|\varphi'|) \cdot |f| \in L^1(\mathbb{R}^n)\end{aligned}$$

Thus, we can apply Lebesque's dominated convergence theorem to pull out the limits and connect the two equalities, giving us:

$$\begin{aligned}\int_{\mathbb{R}^n} \partial_j f(x)\, d\lambda^n(x) &= \int_{\mathbb{R}^n} \lim_{R\to\infty} \varphi_R(x) \cdot \partial_j f(x)\, d\lambda^n(x) \\ &= \lim_{R\to\infty} \int_{\mathbb{R}^n} \varphi_R(x) \cdot \partial_j f(x)\, d\lambda^n(x) \\ &= -\lim_{R\to\infty} \int_{\mathbb{R}^n} \partial_j \varphi_R(x) \cdot f(x)\, d\lambda^n(x) \\ &= -\int_{\mathbb{R}^n} \lim_{R\to\infty} \partial_j \varphi_R(x) \cdot f(x)\, d\lambda^n(x) \\ &= 0\end{aligned}$$

□

Corollary 9.24: Partial Integration-ish Formula

Let $f, g : \mathbb{R}^n \to \mathbb{R}$ be continuously differentiable such that $f, \partial_j f \in L^p(\mathbb{R}^n)$ and $g, \partial_j g \in L^q(\mathbb{R}^n)$ with $\frac{1}{p} + \frac{1}{q} = 1$. Then we have

$$\int_{\mathbb{R}^n} (\partial_j f(x)) \cdot g(x)\, d\lambda^n(x) = 0 - \int_{\mathbb{R}^n} f(x) \cdot (\partial_j g(x))\, d\lambda^n(x)$$

Proof. Applying the product rule of differentiation, and then the last lemma, we get:

$$\begin{aligned}\int_{\mathbb{R}^n} (\partial_j f(x)) \cdot g(x)\, d\lambda^n(x) + \int_{\mathbb{R}^n} f(x) \cdot (\partial_j g(x))\, d\lambda^n(x) \\ = \int_{\mathbb{R}^n} \partial_j (f(x) \cdot g(x))\, d\lambda^n(x) \\ = 0\end{aligned}$$

□

9.3 Change of Variables

Lemma 9.25. Let $U \subset \mathbb{R}^n$ and $x_0 \in U$. Let $\varphi : U \to \mathbb{R}^n$ be a function such that $D\varphi(x_0)$ is invertible.

Then for a sequence $Q_j = Q(x_j, r_j) \subset U$ of cuboids of sidelength r_j with center x_j such that $r_j \to 0$ and $x_0 \in Q_j$, we have:

$$\limsup_{j\to\infty} \frac{\lambda^n(\varphi(Q_j))}{\lambda^n(Q_j)} \leq |\det D\varphi(x_0)|$$

Proof. We can assume $x_0 = 0$ and $\varphi(0) = 0$, since otherwise we can translate space as needed before doing any calculations without breaking any of our assumptions.

1. Assume $D(\varphi(0)) = E_n$. Then, by definition of differentiablity and equivalence of norms on finite-dimensional vector spaces, we have

$$\begin{aligned} 0 &= \lim_{x\to 0} \frac{\|\varphi(x) - \varphi(0) - D\varphi(0)x\|\infty}{\|x\|_\infty} \\ &= \lim_{x\to 0} \frac{\|\varphi(x) - x\|_\infty}{\|x\|_\infty}, \end{aligned}$$

Let $\varepsilon > 0$. Then, by the definition of convergence, for every x with a sufficiently small norm, we have:

$$\frac{\|\varphi(x) - x\|_\infty}{\|x\|_\infty} \leq \varepsilon,$$

which means

$$\|\varphi(x) - x\|_\infty \leq \varepsilon \|x\|_\infty.$$

Furthermore, for $x \in Q_j$, we have:

$$\begin{aligned} \|x\|_\infty &= \left\|x - \vec{0}\right\|_\infty \\ &= \|x - x_0\|_\infty \\ &\leq \left\|x - x_j\right\| + \left\|x_j - x_0\right\| \\ &\leq 2r_j \end{aligned}$$

For sufficiently large j, these imply:

$$\begin{aligned} \|\varphi(x) - x\|_\infty &\leq \varepsilon \|x\|_\infty \\ &\leq 2\varepsilon r_j. \end{aligned}$$

Further applying the triangle inequality, we get:

$$\begin{aligned} \|\varphi(x) - \varphi(x)\| &\leq \|\varphi(x) - x\|_\infty \\ &\quad + \left\|x - x_j\right\|_\infty \\ &\quad + \left\|x_j - \varphi(x_j)\right\|_\infty \\ &\leq 2\varepsilon r_j + r_j + 2\varepsilon r_j \\ &\leq (1 + 4\varepsilon) r_j, \end{aligned}$$

which means that φ increases the side length of our cube by a factor of at most $(1+4\varepsilon)$. Therefore, it increases the volume by a factor at most $(1+4\varepsilon)^n$, i.e:

$$\frac{\lambda^n(\varphi(Q_j))}{\lambda^n(Q_j)} \leq (1+4\varepsilon)^n$$

Letting $j \to \infty$ and $\varepsilon \searrow 0$, we have

$$\begin{aligned}\limsup_{j\to\infty} \frac{\lambda^n(\varphi(Q_j))}{\lambda^n(Q_j)} &\leq \lim_{\varepsilon\to 0}(1+4|\varepsilon|)^n \\ &= 1 \\ &= |\det E_n|\end{aligned}$$

2. Now, let $S := D\varphi(0)$ and $\varphi_0 := S^{-1} \circ \varphi$, i.e. $\varphi = S \circ \varphi_0$. Then $D\varphi_0(0) = E_n$. By the linear transformation equation $\lambda^n(S(E)) = |\det(S)|\lambda^n(E)$ (6.9.2), we have:

$$\begin{aligned}\limsup_{j\to\infty} \frac{\lambda^n(\varphi(Q_j))}{\lambda^n(Q_j)} &= \limsup_{j\to\infty} \frac{\lambda^n(S(\varphi_0(Q_j)))}{\lambda^n(Q_j)} \\ &= |\det S| \limsup_{j\to\infty} \frac{\lambda^n(\varphi_0(Q_j))}{\lambda^n(Q_j)} \\ &\leq |\det S| \\ &= |\det D\varphi(0)|\end{aligned}$$

□

Theorem 9.26: Multivariable Substitution Formula

Let $U \subset \mathbb{R}^n$ be open. Let $\varphi : U \to \mathbb{R}^n$ be C^1. Then if $f : V \to \overline{\mathbb{R}}$ is λ^n-measurable, we have:

$$\int_V f(y)\,dy = \int_{\varphi^{-1}(V)} f(\varphi(x)) \cdot |\det D\varphi(x)|\,dx.$$

Corollary 9.27

Let $U, V \subset \mathbb{R}^n$ be open. Let $\varphi : U \to \mathbb{R}^n$ be C^1. Then if $A \subset U$ is λ^n-measurable, so is $\varphi(A)$, and we have

$$\lambda^n(\varphi(A)) = \int_A |\det D\varphi(x)|\,dx.$$

Proof. Apply the previous equation to $f = \mathbb{1}_{\varphi(A)}$. □

Definition 9.28: Gram Matrix of a Diffeomorphism

Let $\varphi : U \to V$ be a C^k-diffeomorphism between open subsets of $\mathbb{R}^n$. Then the **Gram matrix** $g(\varphi) \in C^{k-1}(U, \mathbb{R}^{n\times n})$ is given by:

$$g(\varphi)(x) = D\varphi(x)^\top D\varphi(x),$$

or equivalently:

$$g(\varphi)_{ij}(x) = \left\langle \frac{\partial}{\partial x_i}\varphi(x), \frac{\partial}{\partial x_j}\varphi(x) \right\rangle$$

Since $g(\varphi)(x)$ is the product of a matrix with its own transpose, it is symmetric and strictly positive definite. This lets us alternatively write the multivariable substitution formula as:

$$\int_V f(y)dy = \int_{\varphi^{-1}(V)} f(\varphi(x))\sqrt{\det g(\varphi)(x)}\, dx$$

9.4 Alternative coordinate systems on $\mathbb{R}^2$ and $\mathbb{R}^3$

9.4.1 Polar Coordinates

A commonly used alternative coordinate system on $\mathbb{R}^2$ is given by **polar coordinates**, which describes a point $x \in \mathbb{R}^2$ based on its distance r from the origin and the *polar angle* ω between x and $(0, r)^\top$. These might be familiar from dealing with complex numbers, where any $z \in \mathbb{C}$ can be represented as

$$z = r \cdot e^{\omega i}$$

Definition 9.29: Polar Coordinates on $\mathbb{R}^2$

Let $U = (0, \infty) \times (0, 2\pi)$ and $V = \mathbb{R}^2 \setminus \{(x, 0)^\top : x \geq 0\}$. The **polar coordinate transformation** is the map:

$$\varphi : U \to V$$
$$\begin{pmatrix} r \\ \omega \end{pmatrix} \mapsto \begin{pmatrix} r\cos\omega \\ r\sin\omega \end{pmatrix}$$

which maps a point given in polar coordinates to its corresponding point in the cartesian coordinate system.

The Jacobian of the polar coordinate transform is:

$$\begin{aligned} D\varphi(r, \omega) &= \begin{pmatrix} \frac{\partial}{\partial r} r\cos\omega & \frac{\partial}{\partial \omega} r\cos\omega \\ \frac{\partial}{\partial r} r\sin\omega & \frac{\partial}{\partial \omega} r\sin\omega \end{pmatrix} \\ &= \begin{pmatrix} \cos\omega & -r\sin\omega \\ \sin\omega & r\cos\omega \end{pmatrix} \end{aligned}$$

Which means the determinant is:

$$\begin{aligned} \det D\varphi(r, \omega) &= \det \begin{pmatrix} \cos\omega & -r\sin\omega \\ \sin\omega & r\cos\omega \end{pmatrix} \\ &= r\cos(\omega)^2 + r\sin(\omega)^2 \\ &= r \end{aligned}$$

Corollary 9.30: Polar Coordinate Integral Transform

Let $f : \mathbb{R}^2 \to \mathbb{R}$ be integrable. Then we have:

$$\int_{\mathbb{R}^2} f(x, y)\, d\lambda(x, y) = \int_0^{2\pi} \int_0^\infty r \cdot f(r\cos(\omega), r\sin(\omega))\, dr\, d\omega$$

Proposition 9.31. The inverse of the polar coordinate transform φ is:

$$\varphi^{-1}(x, y) = \begin{cases} \left(r,\ \arccos\left(\frac{x}{r}\right)\right)^\top & y \geq 0 \\ \left(r,\ 2\pi - \arccos\left(\frac{x}{r}\right)\right)^\top & y < 0 \end{cases}$$

where $r = \sqrt{x^2 + y^2}$.

Example 9.32. (The Gaussian Integral): We want to find the area under the Gaussian bell curve pre-normalization, i.e.

$$\int_{\mathbb{R}} e^{-x^2}\ dx$$

To do this, we add an additional dimension and exploit the resulting rotational symmetry. By Fubini's Theorem, we have:

$$\begin{aligned}
\int_{\mathbb{R}^2} e^{-(x^2+y^2)}\ d\lambda^2(x,y) &= \int_{\mathbb{R}} \left(\int_{\mathbb{R}} e^{-(x^2+y^2)}\ d\lambda(y) \right) d\lambda(x) \\
&= \int_{\mathbb{R}} \left(\int_{\mathbb{R}} e^{-x^2} \cdot e^{-y^2}\ d\lambda(y) \right) d\lambda(x) \\
&= \int_{\mathbb{R}} e^{-x^2} \cdot \left(\int_{\mathbb{R}} e^{-y^2}\ d\lambda(y) \right) d\lambda(x) \\
&= \int_{\mathbb{R}} e^{-x^2}\ d\lambda(x) \cdot \int_{\mathbb{R}} e^{-y^2}\ d\lambda(y) \\
&= \left(\int_{\mathbb{R}} e^{-x^2}\ d\lambda(x) \right)^2
\end{aligned}$$

Therefore, we have

$$\int_{\mathbb{R}} e^{-x^2}\ d\lambda(x) = \sqrt{\int_{\mathbb{R}^2} e^{-(x^2+y^2)}\ d\lambda(x,y)^2}$$

We can now calculate the two-dimensional integral using multivariable substitution to transform to polar coordinates:

$$\begin{aligned}
\int_{\mathbb{R}^2} e^{-(x^2+y^2)}\ d\lambda^2(x,y) &= \int_{(0,\infty)\times(0,2\pi)} re^{-r^2}\ d\lambda^2(r,\omega) \\
&= \int_0^\infty \left(\int_0^{2\pi} re^{-r^2}\ d\omega \right) dr \\
&= \int_0^\infty 2\pi re^{-r^2}\ dr \\
&= 2\pi \int_0^\infty re^{-r^2}\ dr \\
&= 2\pi \int_0^\infty \frac{1}{2} e^{-r^2}\ 2r dr \\
&= 2\pi \int_{-\infty}^0 \frac{1}{2} e^s\ ds \\
&= \pi \int_{-\infty}^0 e^s\ ds \\
&= \pi
\end{aligned}$$

Which means that the area under the bell curve is $\sqrt{\pi}$.

Example 9.33. (Beta function): For $p, q \in \mathbb{R}_{>0}$, the beta function B is defined as:

$$B(p,q) = \int_0^1 t^{p-1} \cdot (1-t)^{q-1}\, dt$$

The beta function is of interest primarily because of its close relationship to the gamma function:

Theorem 9.34. We have:

$$B(p,q) = \frac{\Gamma(p)\Gamma(q)}{\Gamma(p+q)}$$

Proof. Substituting $t = g(\omega) := \sin(\omega)^2$ into the definition of the beta function gives us:

$$\begin{aligned} g'(\omega) &= \frac{\mathrm{d}}{\mathrm{d}\omega} \sin(\omega)^2 \\ &= \sin(\omega)\cos(\omega) + \cos(\omega) + \sin(\omega) \\ &= 2\sin(\omega)\cos(\omega) \end{aligned}$$

and:

$$1 = \sin(\omega)^2 \Leftrightarrow \omega = \frac{\pi}{2},$$

which means we have:

$$\begin{aligned} B(p,q) &= \int_0^1 t^{p-1} \cdot (1-t)^{q-1}\, dt \\ &= \int_0^{\pi/2} \sin(\omega)^{2p-2} \cdot (1-\sin(\omega)^2)^{q-1} \cdot 2\sin(\omega)\cos(\omega)\, d\omega \\ &= \int_0^{\pi/2} \sin(\omega)^{2p-2} \cdot \cos(\omega)^{2q-2} \cdot 2\sin(\omega)\cos(\omega)\, d\omega \\ &= 2 \cdot \int_0^{\pi/2} \sin(\omega)^{2p-1} \cdot \cos(\omega)^{2q-1}\, d\omega \end{aligned}$$

Meanwhile, performing a reverse substitution with $t := g(r) = r^2$ in the definition of the gamma function gives:

$$\begin{aligned} \Gamma(m) &= \int_0^\infty t^{m-1} e^{-t}\, dt \\ &= \int_0^\infty (r^2)^{m-1} e^{-r^2} \cdot 2r\, dr \\ &= 2\int_0^\infty r^{2m-1} e^{-r^2}\, dr \end{aligned}$$

Now, we get:

$$\begin{aligned} B(p,q)\Gamma(p+q) &= 2\int_0^\infty B(p,q) \cdot r^{2p+2q-1} \cdot e^{-r^2}\, dr \\ &= 4\int_0^\infty \int_0^{\pi/2} (\sin\omega)^{2p-1} \cdot r^{2p-1} \cdot (\cos\omega)^{2q-1} \cdot r^{2q-1} \cdot r \cdot e^{-r^2}\, d\omega\, dr \\ &= 4\int_0^\infty \int_0^{\pi/2} (r\sin\omega)^{2p-1} \cdot (r\cos\omega)^{2q-1} \cdot r \cdot e^{-r^2}\, d\omega\, dr \end{aligned}$$

By applying the polar coordinate integral transform in reverse, we get:

$$\begin{aligned}
B(p,q)\Gamma(p+q) &= 4\int_0^\infty \int_0^{\pi/2} (r\sin\omega)^{2p-1} \cdot (r\cos\omega)^{2q-1} \cdot r \cdot e^{-r^2}\ d\omega\ dr \\
&= 4\int_0^\infty \int_0^\infty x^{2p-1} \cdot y^{2q-1} \cdot e^{-\sqrt{x^2+y^2}^2}\ dy\ dx \\
&= 4\int_0^\infty \int_0^\infty x^{2p-1} \cdot y^{2q-1} \cdot e^{-x^2-y^2}\ dy\ dx \\
&= \left(2\int_0^\infty x^{2p-1}e^{-x^2}\ dx\right) \cdot \left(2\int_0^\infty y^{2q-1}e^{-y^2}\ dy\right) \\
&= \Gamma(p)\Gamma(q)
\end{aligned}$$

□

9.4.2 Spherical Coordinates

The most natural generalization of polar coordinates to $\mathbb{R}^3$ are *spherical coordinates*, in which $\theta \in (0, \pi)$ describes the *polar angle* between a point P and the z-Axis and $\omega \in (0, 2\pi)$ describes the *azimuthal angle* between the x-axis and the projection P' of P onto the xy-plane. We can derive spherical coordinates from two dimensional coordinates as follows:

1. Take a point $P = (x, y, z) \in \mathbb{R}^3$. Notice that $z = r\cos(\theta)$.
2. Project P to a point $P' = (x, y)$. The distance of P' from $(0, 0)$ is exactly $\rho := r\sin\theta$. Therefore, P' has polar coordinates (ρ, ω), which gives us:

$$P = \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} \rho\cos\omega \\ \rho\sin\omega \\ r\cos\theta \end{pmatrix} = \begin{pmatrix} r\sin\theta\cos\omega \\ r\sin\theta\sin\omega \\ r\cos\theta \end{pmatrix}$$

Definition 9.35: Spherical Coordinates on $\mathbb{R}^3$

Let $U = (0, \infty) \times (0, \pi) \times (0, 2\pi)$ and $V = \mathbb{R}^3 \setminus \{(x, 0, z)^\top : x \geq 0\}$. The **spherical coordinate transformation** is the map:

$$\begin{aligned} \varphi : U &\to V \\ \begin{pmatrix} r \\ \theta \\ \omega \end{pmatrix} &\mapsto \begin{pmatrix} r\sin\theta\cos\omega \\ r\sin\theta\sin\omega \\ r\cos\theta \end{pmatrix} \end{aligned}$$

which maps a point given in spherical coordinates to its corresponding point in the cartesian coordinate system.

The Jacobian of the spherical coordinate transform is:

$$\begin{aligned} D\varphi(r, \theta, \omega) &= \begin{pmatrix} \frac{\partial}{\partial r} r\sin\theta\cos\omega & \frac{\partial}{\partial \theta} r\sin\theta\cos\omega & \frac{\partial}{\partial \omega} r\sin\theta\cos\omega \\ \frac{\partial}{\partial r} r\sin\theta\sin\omega & \frac{\partial}{\partial \theta} r\sin\theta\sin\omega & \frac{\partial}{\partial \omega} r\sin\theta\sin\omega \\ \frac{\partial}{\partial r} r\cos\theta & \frac{\partial}{\partial \theta} r\cos\theta & \frac{\partial}{\partial \omega} r\cos\theta \end{pmatrix} \\ &= \begin{pmatrix} \sin\theta\cos\omega & r\cos\theta\cos\omega & -r\sin\theta\sin\omega \\ \sin\theta\sin\omega & r\cos\theta\sin\omega & r\sin\theta\cos\omega \\ \cos\theta & -r\sin\theta & 0 \end{pmatrix} \end{aligned}$$

And a long, tedious and unenlightening calculation eventually gives the determinant as:

$$\det D\varphi(r, \theta, \omega) = r^2\sin(\theta)$$

Which gives us:

Corollary 9.36: Spherical Coordinate Integral Transform

Let $f : \mathbb{R}^3 \to \mathbb{R}$ be integrable. Then we have:

$$\begin{aligned} &\int_{\mathbb{R}^3} f(x, y, z)\, d\lambda^3(x, y, z) \\ &= \int_0^{2\pi} \int_0^{\pi} \int_0^{\infty} r^2 \sin\theta \cdot f(r\sin\theta\sin\omega, r\sin\theta\cos\omega, r\cos\theta)\, dr\, d\theta\, d\omega \end{aligned}$$

Example 9.37. (Another derivation of the volume of B_R^3):

$$\begin{aligned}
\lambda^3(B_R^3) &= \int_{\mathbb{R}^3} \mathbb{1}_{B_R^3}(x,y,z)\, d\lambda^3(x,y,z) \\
&= \int_{\mathbb{R}^3} \mathbb{1}_{\left\{\sqrt{x^2+y^2+z^2} \,<\, R\right\}}(x,y,z)\, d\lambda^3(x,y,z) \\
&= \int_0^{2\pi} \int_0^{\pi} \int_0^{\infty} \mathbb{1}_{\{r \,<\, R\}}(r) \cdot r^2 \sin\theta \; dr \; d\theta \; d\omega \\
&= \int_0^{2\pi} \int_0^{\pi} \int_0^{\infty} \mathbb{1}_{(0,R)}(r) \cdot r^2 \sin\theta \; dr \; d\theta \; d\omega \\
&= \left(\int_0^{\infty} \mathbb{1}_{(0,R)}(r) \cdot r^2 \; dr\right) \cdot \left(\int_0^{\pi} \sin\theta \; d\theta\right) \cdot \left(\int_0^{2\pi} 1 \; d\omega\right) \\
&= \left(\int_0^{R} r^2 \; dr\right) \cdot \left(\int_0^{\pi} \sin\theta \; d\theta\right) \cdot \left(\int_0^{2\pi} 1 \; d\omega\right) \\
&= \frac{1}{3}R^3 \cdot (-\cos(\pi) + \cos(0)) \cdot 2\pi \\
&= \frac{4}{3}\pi R^3
\end{aligned}$$

Example 9.38. (Gravitational Potential of a Sphere:): The gravitational potential that a body $B \subset \mathbb{R}^3$ with mass distribution $m : B \longrightarrow \mathbb{R}$ exerts on a point $x \in \mathbb{R}^3 \setminus B$ is given by:

$$V(x) = -G \cdot \int_B \frac{m(r)}{\|x-y\|_2} \; d\lambda^3(y)$$

We will derive a closed form expression for the case $B = B_R^3$. If we assume that the mass distribution is rotationally symmetric, we can replace $m : \mathbb{R}^3 \longrightarrow \mathbb{R}$ with a function $\widetilde{m} = m(\|y\|_2) : \mathbb{R} \longrightarrow \mathbb{R}$. We will commit a slight abuse of notation here and let m be this new function $\widetilde{m}$ from now on. Because of this rotational symmetry, we also have $V(x) = V(0, 0, \|x\|)$.

Now, if y has spherical coordinates (r, θ, ω), then assuming y lies outside of our ball we can use the Pythagorean theorem to get:

$$\begin{aligned}
\|x-y\|_2^2 &= (\|x\|_2 - r\cos\theta)^2 + r^2 \sin^2\theta \\
&= \|x\|_2^2 + r^2 - 2r\|x\|_2 \cos\theta
\end{aligned}$$

Thus, we have:

$$\begin{aligned}
V(x) &= -G \cdot \int_{B_R^3} \frac{m(r)}{\|x-y\|_2} \; d\lambda^3(y) \\
&= -G \int_0^{2\pi} \int_0^{\pi} \int_0^{R} \left(\frac{m(r)}{\sqrt{\|x\|_2^2 + r^2 - 2r\|x\|_2 \cos\theta}} \cdot r^2 \sin\theta \right) dr \; d\theta \; d\omega \\
&= -2\pi G \int_0^{R} r^2 \cdot m(r) \cdot \left(\int_0^{\pi} \frac{\sin\theta}{\sqrt{\|x\|_2^2 + r^2 - 2r\|x\|_2 \cos\theta}} \; d\theta \right) dr
\end{aligned}$$

Substituting $t = g(\theta) = -\cos\theta$ into the inner integral, we get:

$$\begin{aligned}
&\int_0^\pi \frac{\sin\theta}{\sqrt{\|x\|_2^2 + r^2 - 2r\|x\|_2\cos\theta}}\,d\theta \\
=&\int_{-1}^1 \frac{1}{\sqrt{\|x\|_2^2 + r^2 + 2r\|x\|_2 t}}\,dt \\
=&\left[\frac{1}{r\cdot\|x\|_2}\cdot\sqrt{\|x\|_2^2 + r^2 + 2r\|x\|_2 t}\right]_{t=-1}^{t=1} \\
&= \frac{1}{r\cdot\|x\|_2}\cdot\left(\sqrt{\|x\|_2^2 + r^2 + 2r\|x\|_2} - \sqrt{\|x\|_2^2 + r^2 - 2r\|x\|_2}\right) \\
&= \frac{1}{r\cdot\|x\|_2}\cdot((\|x\|_2 + r) - \|x\|_2 - r) \\
&= \frac{2}{\|x\|_2}
\end{aligned}$$

Therefore, we have:

$$V(x) = \frac{4\pi}{\|x\|_2}\cdot\int_0^R m(r)r^2\,dr$$

Now, we can once again use the rotational symmetry of m to describe the total mass M of our body in terms of spherical coordinates:

$$\begin{aligned}
M &= \int_{B_R^3(0)} m(y)\,d\lambda^3(y) \\
&= \int_0^{2\pi}\int_0^\pi\int_0^R m(r)r^2\sin\theta\,dr\,d\theta\,d\omega \\
&= 4\pi\int_0^R m(r)r^2\,dr
\end{aligned}$$

Thus, the total mass distribution of our sphere is simply:

$$V(x) = -\frac{GM}{\|x\|}$$

which is exactly the same mass distribution we would get if we concentrated the entire mass of our sphere at its center. This relation is part of a theorem in classical mechanics known as the **shell theorem**.

9.4.3 Cylindrical Coordinates

Alternatively, one can trivially extend polar coordinates into three dimensions by simply leaving the z coordinate unchanged, leading to *cylindrical coordinates*:

Definition 9.39: Cylindrical Coordinates on $\mathbb{R}^3$

Let $U = (0, \infty)\times(0, 2\pi)\times\mathbb{R}$ and $V = \mathbb{R}^3 \setminus \{(x, 0, z)^\top : x \geq 0\}$. The **spherical coordinate transformation** is the map:

$$\varphi : U \to V$$
$$\begin{pmatrix} r \\ \omega \\ z \end{pmatrix} \mapsto \begin{pmatrix} r\cos\omega \\ r\sin\omega \\ z \end{pmatrix}$$

which maps a point given in cylindrical coordinates to its corresponding point in the cartesian coordinate system.

Since the cylindrical coordinate transform leaves the z coordinate unchanged, its Jacobian and its determinant end up being almost identical:

$$\begin{aligned} D\varphi(r,\omega,z) &= \begin{pmatrix} \frac{\partial}{\partial r} r\cos\omega & \frac{\partial}{\partial \omega} r\cos\omega & \frac{\partial}{\partial z} r\cos\omega \\ \frac{\partial}{\partial r} r\sin\omega & \frac{\partial}{\partial \omega} r\sin\omega & \frac{\partial}{\partial z} r\sin\omega \\ \frac{\partial}{\partial r} z & \frac{\partial}{\partial \omega} z & \frac{\partial}{\partial z} z \end{pmatrix} \\ &= \begin{pmatrix} \cos\omega & -r\sin\omega & 0 \\ \sin\omega & r\cos\omega & 0 \\ 0 & 0 & 1 \end{pmatrix} \end{aligned}$$

$$\begin{aligned} \det D\varphi(r,\omega,z) &= \det \begin{pmatrix} \cos\omega & -r\sin\omega & 0 \\ \sin\omega & r\cos\omega & 0 \\ 0 & 0 & 1 \end{pmatrix} \\ &= \det \begin{pmatrix} \cos\omega & -r\sin\omega \\ \sin\omega & r\cos\omega \end{pmatrix} \\ &= r\cos(\omega)^2 + r\sin(\omega)^2 \\ &= r \end{aligned}$$

Corollary 9.40: Cylindrical Coordinate Integral Transform

Let $f : \mathbb{R}^2 \to \mathbb{R}$ be integrable. Then we have:

$$\int_{\mathbb{R}^3} f(x,y,z)\, d\lambda(x,y,z) = \int_{\mathbb{R}} \int_0^{2\pi} \int_0^{\infty} r \cdot f(r\cos(\omega), r\sin(\omega), z)\, dr\, d\omega\, dz$$

Proposition 9.41. The inverse of the cylindrical coordinate transform φ is:

$$\varphi^{-1}(x,y,z) = \begin{cases} \left(r,\ \arccos\left(\frac{x}{r}\right),\ z\right)^\top & y \geq 0 \\ \left(r,\ 2\pi - \arccos\left(\frac{x}{r}\right),\ z\right)^\top & y < 0 \end{cases}$$

where $r = \sqrt{x^2 + y^2}$.

Theorem 9.42: Volume of a Solid of Revolution

Let S be a solid constructed by rotating a function $\rho(z)$ around the z axis, i.e.

$$S = \left\{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 < \rho(z)^2\right\}$$

Then we have:

$$\lambda^3(S) = \pi \int_{\mathbb{R}} \rho(z)^2 \, d\lambda^1(z)$$

Proof.

$$\begin{aligned}
\lambda^3(S) &= \int_{\mathbb{R}^3} \mathbb{1}_{\{x^2+y^2 < \rho(z)^2\}}(x, y, z) \, d\lambda^3(x, y, z) \\
&= \int_{\mathbb{R}} \int_0^{2\pi} \int_0^{\infty} r \cdot \mathbb{1}_{\{r^2 < \rho(z)^2\}}(r, \omega, z) \, dr \, d\omega \, dz \\
&= \int_{\mathbb{R}} \int_0^{2\pi} \int_0^{\infty} r \cdot \mathbb{1}_{\{r < \rho(z)\}}(r, \omega, z) \, dr \, d\omega \, dz \\
&= \int_{\mathbb{R}} \int_0^{2\pi} \int_0^{\rho(z)} r \, dr \, d\omega \, dz \\
&= \int_{\mathbb{R}} \int_0^{2\pi} \frac{1}{2}\rho(z)^2 \, d\omega \, dz \\
&= \int_{\mathbb{R}} \pi\rho(z)^2 \, dz \\
&= \pi \int_{\mathbb{R}} \rho(z)^2 \, dz
\end{aligned}$$

□

9.5 The Surface Area Measure

Definition 9.43. Let $U \supset \mathbb{R}^n$ be open. We call a map $f : U \to \mathbb{R}^{n+k}$ an *immersion* iff f is continuously differentiable and $Df(x)$ is invertible.

Definition 9.44. Let $M \subset \mathbb{R}^{n+k}$ be an n-dimensional C^1 submanifold. Then a *local parametrization of* M is an injective immersion $f : \mathbb{R}^n \supset U \to M \subset \mathbb{R}^{n+k}$

Definition 9.45: Gram Matrix

Let $U \subset \mathbb{R}^n$ and $f : U \to \mathbb{R}^{n+k}$. Then the *Gram matrix* $g_f(x)$ is the matrix

$$g_f(x) = Df(x)^\top Df(x)$$

Our goal now is to find a sensible definition of *surface area*, i.e. of an n-dimensional measure associated with an object embedded into $n+k$-dimensional space.

Corollary 9.46: Area of an Embedded Diffeomorphism of a Set

Let $f = S \circ \varphi$, where $\varphi : U \to V$ is a diffeomorphism with $U, V \subset \mathbb{R}^n$ and S is a linear isometry. Then we have

$$\lambda^n(\varphi(E)) = \int_E \sqrt{\det g_f}$$

This theorem suggests that it is sensible to use the same definition for arbitrary immersions:

Definition 9.47: Area of an Immersion of a Set

Let $f \in C^1(U, \mathbb{R}^{n+k})$ be an n-dimensional immersion. Let $E \subset U$ be λ^n-measurable. Then the n-dimensional area of $f(E)$ is given by:

$$A(f, E) = \int_E \sqrt{\det g_f}$$

$\sqrt{\det g_f}$ is known as the *Jacobian of* f, and often denoted Jf, but I dislike notation that hides the actual calculation being done behind multiple layers of nested notation and will thus be sticking to $\sqrt{\det g_f}$.

Corollary 9.48: Length of a Regular Curve

An Immersion $f : I = (a, b) \to \mathbb{R}^n$ is known as a *regular curve*. The one-dimensional area of the curve, i.e. its length, is given by

$$L(f) := A(f, I) = \int_a^b \|Df(t)\|_2 \, dt$$

Proof. We have:

$$Df(t) = \begin{pmatrix} f_1'(t) \\ \vdots \\ f_n'(t) \end{pmatrix}$$

and thus

$$g_f(t) = Df(t)^\top Df(t) = \|Df(t)\|_2^2$$

□

Note that if some areas of the curve are covered multiple times, then this length formula counts them multiple times as well.

Corollary 9.49: Area of a Regular Surface

An immersion $f : U \supset \mathbb{R}^2 \to \mathbb{R}^3$, with U open, is known as a *regular surface*. Its area is given by

$$\begin{aligned} A(f) := A(f, U) &= \int_U \sqrt{\left\|\frac{\partial}{\partial x} f\right\|_2^2 \cdot \left\|\frac{\partial}{\partial y} f\right\|_2^2 - \left\langle \frac{\partial}{\partial x} f, \frac{\partial}{\partial y} f \right\rangle^2} \, d\lambda^2(x, y) \\ &= \int_U \left\| \frac{\partial}{\partial x} f \times \frac{\partial}{\partial y} f \right\| d\lambda^2(x, y) \end{aligned}$$

Proof. We have

$$Df(x, y) = \begin{pmatrix} \frac{\partial}{\partial x} f(x, y) & \frac{\partial}{\partial y} f(x, y) \end{pmatrix},$$

therefore our Gram matrix is

$$g_f = \begin{pmatrix} \left\|\frac{\partial}{\partial x} f\right\|_2^2 & \left\langle \frac{\partial}{\partial x} f, \frac{\partial}{\partial y} f \right\rangle_2 \\ \left\langle \frac{\partial}{\partial x} f, \frac{\partial}{\partial y} f \right\rangle & \left\|\frac{\partial}{\partial y} f\right\|_2^2 \end{pmatrix},$$

giving our desired result for $\sqrt{\det g_f}$.´ □

Theorem 9.50: Area of a Graph

Let $U \subset \mathbb{R}^n$ be open, $u : U \to \mathbb{R}^k$ and

$$\begin{aligned} f : U &\to \mathbb{R}^{n+k} \\ x &\mapsto (x, u(x)) \end{aligned}$$

Then we have:

$$A(f, U) = \int_U \sqrt{\det(E_n + Du(x)^\top Du(x))}\, d\lambda^n(x)$$

If $k = 1$, i.e. $U \subset \mathbb{R}^n$, $u : U \to \mathbb{R}$, then we can simplify significantly and get:

$$A(f, U) = \int_U \sqrt{1 + \|Du(x)\|^2 \text{ÿ}}\, d\lambda^n(x)$$

Theorem 9.51: Surface Measure

Let M be an n-dimension C^1-submanifold of $\mathbb{R}^{n+k}$ locally parametrized by $f : U \to M$. Let $E \subset M$. Then there exists a measure ω_M such that

$$\omega_M(E) = A(f, E)$$

Theorem 9.52: Surface of Revolution

Let S be a solid of revolution generated by $\rho(z)$. Then the surface area of S is:

$$\omega_S(S) = 2\pi \int_{\mathbb{R}} \rho(z) \sqrt{1 + \left(\frac{\mathrm{d}}{\mathrm{d}z}\rho(z)\right)^2}\, dz$$

Proof. This follows from our formula for the surface area of a regular surface. However, as additional practice, let us derive the formula from scratch. We can parametrize S up to a zero set via:

$$f : \begin{pmatrix} \theta \\ z \end{pmatrix} \mapsto \begin{pmatrix} \rho(z)\cos(\theta) \\ \rho(z)\sin(\theta) \\ z \end{pmatrix}$$

Which gives us:

$$\begin{aligned} Df(\theta, z) &= \begin{pmatrix} \frac{\partial}{\partial\theta}\rho(z)\cos(\theta) & \frac{\partial}{\partial z}\rho(z)\cos(\theta) \\ \frac{\partial}{\partial\theta}\rho(z)\sin(\theta) & \frac{\partial}{\partial z}\rho(z)\sin(\theta) \\ \frac{\partial}{\partial\theta} z & \frac{\partial}{\partial z} z \end{pmatrix} \\ &= \begin{pmatrix} -\rho(z)\sin(\theta) & \cos(\theta)\cdot\frac{\partial}{\partial z}\rho(z) \\ \rho(z)\cos(\theta) & \sin(\theta)\cdot\frac{\partial}{\partial z}\rho(z) \\ 0 & 1 \end{pmatrix} \end{aligned}$$

This gives us the Gram matrix:

$$g_f(\theta,z) = \begin{pmatrix} -\rho(z)\sin(\theta) & \rho(z)\cos(\theta) & 0 \\ \cos(\theta)\cdot\frac{\partial}{\partial z}\rho(z) & \sin(\theta)\cdot\frac{\partial}{\partial z}\rho(z) & 1 \end{pmatrix} \cdot \begin{pmatrix} -\rho(z)\sin(\theta) & \cos(\theta)\cdot\frac{\partial}{\partial z}\rho(z) \\ \rho(z)\cos(\theta) & \sin(\theta)\cdot\frac{\partial}{\partial z}\rho(z) \\ 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} \rho(z)^2\cdot(\sin^2(\theta)+\cos^2(\theta)) & \begin{matrix} -\rho(z)\sin(\theta)\cos(\theta)\frac{\mathrm{d}}{\mathrm{d}z}\rho(z) \\ +\rho(z)\sin(\theta)\cos(\theta)\frac{\mathrm{d}}{\mathrm{d}z}\rho(z) \end{matrix} \\ \begin{matrix} -\rho(z)\sin(\theta)\cos(\theta)\frac{\mathrm{d}}{\mathrm{d}z}\rho(z) \\ +\rho(z)\sin(\theta)\cos(\theta)\frac{\mathrm{d}}{\mathrm{d}z}\rho(z) \end{matrix} & (\cos^2(\theta)+\sin^2(\theta)\left(\frac{\partial}{\partial z}\rho(z)\right)^2+1 \end{pmatrix}$$

$$= \begin{pmatrix} \rho(z)^2 & 0 \\ 0 & \left(\frac{\partial}{\partial z}\rho(z)\right)^2+1 \end{pmatrix}$$

and thus:

$$\begin{aligned} \sqrt{\left|\det g_f(\theta,z)\right|} &= \sqrt{\rho(z)^2\left(\frac{\partial}{\partial z}\rho(z)\right)^2+\rho(z)^2} \\ &= \rho(z)\sqrt{1+\frac{\partial}{\partial z}\rho(z))^2} \end{aligned}$$

Which leaves us with:

$$\begin{aligned} \omega_S(S) &= \int_0^{2\pi}\int_{\mathbb{R}} \rho(z)\sqrt{1+\frac{\partial}{\partial z}\rho(z))^2}\,dz\,d\theta \\ &= 2\pi\int_{\mathbb{R}} \rho(z)\sqrt{1+\frac{\partial}{\partial z}\rho(z))^2}\,dz \end{aligned}$$

□

Example 9.53. (Gabriel's Horn): We want to show that the solid of rotation G generated by $z\in[1,\infty)$ and $\rho(z)=\frac{1}{z}$ has finite volume, but infinite surface area.

1. We already showed in the section on cylindrical coordinates that the volume is simply:

$$\begin{aligned} \lambda^3(G) &= \pi\cdot\int_1^\infty \rho(z)^2\,dz \\ &= \pi\cdot\int_1^\infty \frac{1}{z^2}\,dz \\ &= -\frac{\pi}{2}\cdot\left[\frac{1}{z}\right]_{z=1}^{z=\infty} \\ &= -\frac{\pi}{2}\cdot(0-1) \\ &= \frac{\pi}{2} \end{aligned}$$

2. We have just shown that the surface area is given by:

$$\begin{aligned}
\omega_S(S) &= 2\pi \int_{\mathbb{R}} \rho(z)\sqrt{1 + \frac{\partial}{\partial z}\rho(z))^2}\, dz \\
&= 2\pi \int_1^\infty \frac{1}{z}\sqrt{1 - \frac{1}{z^2}}\, dz \\
&> 2\pi \int_1^\infty \frac{1}{z}\, dz \\
&= 2\pi(\ln(\infty) - \ln(1)) \\
&= \infty
\end{aligned}$$

Theorem 9.54: Surface Integral

Let $M \subset \mathbb{R}^{n+k}$ be an n-dimensional C^1 submanifold of $\mathbb{R}^{n+k}$, and let $M = \bigsqcup_{i\in I} M_i$ be a decomposition of M into countably many pairwise disjoint measure sets such that $M_i \subset V_i$ for a local parametrization $f_i : U_i \to V_i$. Then for any measurable function $u : M \to \overline{\mathbb{R}}$, we have:

$$\int_M u(x)\, d\omega_M(x) = \sum_{i\in I} \int_{f_i^{-1}(M_i)} u(f_i(x)) \cdot \left|\sqrt{\det g_{f_i}(x)}\right| d\lambda^n(x)$$

In particular, if we can find a local parametrization $f : U \to V$ such that $M \setminus V$ is a zero set, we have:

$$\int_M u(x)\, d\omega_M(x) = \int_{f^{-1}(M)} u(f(x)) \cdot |\det Df(x)|\, d\lambda^n(x)$$

Theorem 9.55: Onion Formula

Integrating a function f over $\mathbb{R}^n$ is the same thing as integrating over all surface integrals of the function over all balls B_r^n, i.e. we have:

$$\int_{\mathbb{R}^n} f(x)\, d\lambda^n(x) = \int_0^\infty \left(\int_{\partial B_r^n} f(y)\, d\omega_{\partial B_r^n}(y)\right) dr$$

Note that via multivariable substitution we also get:

$$\int_0^\infty \left(\int_{\partial B_r^n} f(y)\, d\omega_{\partial B_r^n}(y)\right) dr = \int_0^\infty \left(\int_{\partial B_1^n} r^n f(ry)\, d\omega_{\partial B_1^n}(y)\right) dr$$

Corollary 9.56. We have:

$$\omega_{\partial B_1^n}(\partial B_1^n) = (n+1) \cdot \lambda^{n+1}(B_1^{n+1}),$$

i.e. the surface area of the n-dimensional unit ball is $(n+1)$ times the volume of the $n+1$-dimensional unit ball.

Proof. Applying the onion formula and then Fubini twice, we get:

$$\begin{aligned}
\lambda^{n+1}(B_1^{n+1}) &= \int_{\mathbb{R}^n} \mathbb{1}_{B_1^n}(x)\, d\lambda^n(x) \\
&= \int_0^\infty \int_{\partial B_r^n} \mathbb{1}_{B_1^n}(y)\, d\omega_{\partial B_r^n}(y)\, dr \\
&= \int_{\partial B_r^n} \int_0^\infty \mathbb{1}_{B_1^n}(y)\, dr\, d\omega_{\partial B_r^n}(y) \\
&= \int_{\partial B_r^n} \int_0^1 1\, dr\, d\omega_{\partial B_r^n}(y) \\
&= \int_0^1 \int_{\partial B_r^n} 1\, d\omega_{\partial B_r^n}(y)\, dr \\
&= \int_0^1 \omega_{\partial B_r^n}(\partial B_r^n)\, dr \\
&= \int_0^1 r^n \cdot \omega_{\partial B_1^n}(\partial B_1^n)\, dr \\
&= \frac{\omega_{\partial B_1^n}(\partial B_1^n)}{n+1}
\end{aligned}$$

□

9.6 The Divergence Theorem

Definition 9.57. Let $\Omega \in \mathbb{R}^n$. We say Ω has a C^1 boundary if $\partial\Omega$ is an $n-1$ dimensional C^1-manifold such that Ω locally lies on only one side of $\partial\Omega$.

More precisely, Ω has a C^1 boundary iff for every point $p \in \partial\Omega$, there exists an open set $U \subset \mathbb{R}^n$, an open interval $I \subset \mathbb{R}$, an a C^1-function $u : U \to I$ such $p \in U \times I$ and

$$\Omega \cap (U \times I) = \{(x, y) \in U \times I : y < u(x)\}$$

Lemma 9.58. Let $\Omega \in \mathbb{R}^n$ be open with C^1 boundary. Then we have:

1. $\partial\Omega \cap (U \times I) = \{(x, y) \in U \times I : y = u(x)\}$
2. $(\mathbb{R}^n \setminus \overline{\Omega}) \cap (U \times I) = \{(x, y) \in U \times I : y \geq u(x)\}$

In particular, $\partial\Omega$ is an $n-1$ dimensional manifold given via the graph criterion as $(x, u(x))$.

Theorem 9.59: Outer Normal Field

Let $\Omega \subset \mathbb{R}^n$ be open with C^1 boundary. Then there exists exactly one map $\nu_\Omega : \partial\Omega \to \mathbb{R}^n$ such that:

1. $\nu_\Omega(p) \perp T_p(\partial\Omega)$,
2. $\|\nu_\Omega(p)\|_2 = 1$,
3. $\exists \varepsilon > 0 : t \in (0, \varepsilon) \implies p + t\nu_\Omega(p) \notin \Omega$.

This map is continuous and unique and is known as the *outer normal field of* Ω. If u is given as in the definition of C^1 boundary, and $p = (x, u(x))$, then we have:

$$\nu_\Omega(p) = \frac{(-Du(x), 1)^\top}{\sqrt{1 + \|Du(x)\|_2^2}}$$

Lemma 9.60: Partition of Unity

Let $W_i, i \in I$ be an open cover of a compact set $K \subset \mathbb{R}^n$. Then there exists a finite family of smooth functions $\chi_j : \mathbb{R}^n \to \mathbb{R}^n$ with compact support such that

1. For all $x \in K$, we have $\sum_{j \in J} \chi_j(x) = 1$
2. For all $j \in J$, there exists an i such that $\mathrm{spt}\chi_j \subset W_i$.

Theorem 9.61: Divergence Theorem / Gauss's Theorem

Let $\Omega \subset \mathbb{R}^n$ be a bounded open set with C^1 boundary and outer normal $\nu_\Omega : \partial\Omega \to \mathbb{R}^n$. Let $f : \overline{\Omega}\mathbb{R}$ be continuously differentiable. Then we have:

$$\int_\Omega \mathrm{div} f(x)\, d\lambda^n(x) = \int_{\partial\Omega} \langle f(x), \nu(x)\rangle \; d\omega(x).$$

Theorem 9.62: Multidimensional Partial Integration

Let $\Omega \subset \mathbb{R}^n$ be a bounded open set with C^1 boundary and outer normal $\nu_\Omega : \partial\Omega \to \mathbb{R}^n$. Let $f : \mathbb{R}^n \to \mathbb{R}$ and $G : \mathbb{R}^n \to \mathbb{R}^n$ be continuously differentiable. Then we have:

$$\begin{aligned}
&\int_\Omega f \cdot \operatorname{div}(G)\, d\lambda^n(x) \\
&= \int_{\partial\Omega} f \cdot \langle G, \nu_\Omega \rangle \; d\omega(x) - \int_\Omega \left\langle \vec{\nabla} f, G \right\rangle \; d\lambda^n(x)
\end{aligned}$$

Proof. By the product rule for divergence, we have:

$$\begin{aligned}
&f(x) \cdot \operatorname{div}(G(x)) + \left\langle \vec{\nabla} f(x), G(x) \right\rangle \\
&= \operatorname{div}(f(x) \cdot G(x))
\end{aligned}$$

Now, simply apply the divergence theorem, together with basic properties of scalar products, to get:

$$\begin{aligned}
&\int_\Omega f(x) \cdot \operatorname{div}(G(x)) + \left\langle \vec{\nabla} f(x), G(x) \right\rangle \; d\lambda^n(x) \\
&= \int_\Omega \operatorname{div}(f(x) \cdot G(x))\, d\lambda^n(x) \\
&= \int_{\partial\Omega} \langle f(x) \cdot G(x), \nu_\Omega(x) \rangle \; d\omega(x) \\
&= \int_{\partial\Omega} f(x) \cdot \langle G(x), \nu_\Omega(x) \rangle \; d\omega(x),
\end{aligned}$$

which we can rearrange to get our desired identity. □

If we plug $G(x) := \vec{\nabla} g(x)$ into this formula, we get:

Corollary 9.63: Green's First Identity

Let $\Omega \subset \mathbb{R}^n$ be open and bounded with C^1 boundary. Let $f : \overline{\Omega} \to \mathbb{R}$ be continuously differentiable and let $g : \overline{\Omega} \to \mathbb{R}$ be twice continuously differentiable. Then we have:

$$\begin{aligned}
&\int_\Omega f \cdot \Delta g + \left\langle \vec{\nabla} f, \vec{\nabla} g \right\rangle \; d\lambda^n \\
&= \int_{\partial\Omega} f \cdot \left\langle \nu_\Omega, \vec{\nabla} g \right\rangle \; d\omega \\
&\left(= \int_{\partial\Omega} f \cdot \frac{\partial}{\partial \nu_\Omega} g \; d\omega(x) \right)
\end{aligned}$$

Theorem 9.64: Green's Second Identity

Let $\Omega \subset \mathbb{R}^n$ be open and bounded with C^1 boundary. Let $f : \overline{\Omega} \to \mathbb{R}$ and $g : \overline{\Omega} \to \mathbb{R}$ be twice continuously differentiable. Then we have:

$$\begin{aligned}
&\int_{\Omega} f \cdot \Delta g - g \cdot \Delta f \; d\lambda^n \\
&= \int_{\partial\Omega} f \cdot \left\langle \nu_\Omega, \vec{\nabla} g \right\rangle - g(x) \cdot \left\langle \nu_\Omega(x), \vec{\nabla} f(x) \right\rangle \; d\omega(x)
\end{aligned}$$

Proof. This is just two instances of Green's first identity subtracted from each other:

$$\begin{aligned}
&\int_{\Omega} f(x) \cdot \Delta g(x) - g(x) \cdot \Delta f(x) \; d\lambda^n \\
&= \int_{\Omega} f(x) \cdot \Delta g(x) + \left\langle \vec{\nabla} f(x), \vec{\nabla} g(x) \right\rangle \; d\lambda^n \\
&- \int_{\Omega} g(x) \cdot \Delta f(x) + \left\langle \vec{\nabla} f(x), \vec{\nabla} g(x) \right\rangle \; d\lambda^n \\
&= \int_{\partial\Omega} f(x) \cdot \left\langle \nu_\Omega(x), \vec{\nabla} g(x) \right\rangle \; d\omega(x) \\
&- \int_{\partial\Omega} g(x) \cdot \left\langle \nu_\Omega(x), \vec{\nabla} f(x) \right\rangle \; d\omega(x) \\
&= \int_{\partial\Omega} f(x) \cdot \left\langle \nu_\Omega(x), \vec{\nabla} g(x) \right\rangle - g(x) \cdot \left\langle \nu_\Omega(x), \vec{\nabla} f(x) \right\rangle \; d\omega(x)
\end{aligned}$$

□

Chapter 10

Convolutions

Convolution is a way of creating a function by taking a weighted average of two functions. Among other things, it can be used to smooth a given function.

Lemma 10.1. $\tau_h : \mathbb{R}^n \to \mathbb{R}^n$ be the function that translates a vector by the vector h, i.e. $\tau_h(x) = x + h$. Let $f \in L^p(\mathbb{R}^n)$ for $1 \leq p < \infty$. Then the following hold:

1. $f \circ \tau_h \in L^p(\mathbb{R}^n)$
2. $\|f \circ \tau_h\|_p = \|f\|_p$
3. $\lim_{h \to 0} \|f \circ \tau_h - f\|_p = 0$

Definition 10.2: Convolution

Let $1 \leq p < \infty$, $f \in L^p(\mathbb{R}^n)$ and $g \in L^1(\mathbb{R}^n$. The *convolution of f and g* is defined to be:

$$\begin{aligned} f * g &: \mathbb{R}^n \to \overline{\mathbb{R}} \\ (f * g)(x) &= \int_{\mathbb{R}^n} f(x - y) g(y) \, d\lambda^n(y) \end{aligned}$$

Theorem 10.3. Let $1 \leq p < \infty$, $f \in L^p(\mathbb{R}^n)$ and $g \in L^1(\mathbb{R}^n$. Then we have:

1. $f * g \in L^p(\mathbb{R}^n)$
2. $\|f * g\|_p \leq \|f\|_p \|g\|_1$
3. $f * g = g * f$

Theorem 10.4. (Approximation through convolution): Let $\eta \in L^1(\mathbb{R}^n)$ such that

$$\int_{\mathbb{R}^n} \eta(x)\, d\lambda^n(x).$$

Let $\rho > 0$ and

$$\eta_\rho(x) := \rho^{-n} \cdot \eta\left(\frac{x}{\rho}\right).$$

Let $1 \leq p < \infty$ and $f \in L^p(\mathbb{R}^n)$. Then we have:

1. $f * \eta_\rho \in L^p(\mathbb{R}^n)$
2. $\left\|f * \eta_\rho\right\|_p \leq \|f\|_p \cdot \|\eta\|_1$
3. $\left\|(f * \eta_\rho) - f\right\|_p \to 0$

Definition 10.5: Multi-index Notation

Let $\alpha = (\alpha_1, \dots, \alpha_n) \in \mathbb{N}_0^n$. Then we define:

1. $|\alpha| = \displaystyle\sum_{i=1}^n \alpha_i$
2. $\partial^\alpha f(x) = \frac{\partial^{\alpha_1}}{\partial x_1^{\alpha_1}} \dots \frac{\partial^{\alpha_n}}{\partial x_n^{\alpha_n}} f(x)$

Theorem 10.6: Smoothing

Let $\eta \in C^k(\mathbb{R}^n)$ such that for $|\alpha| \leq k$, we have

$$\|\partial^\alpha \eta\|_{C^0(\mathbb{R}^n)} \leq K.$$

Let $f \in L^1(\mathbb{R}^n)$. Then we have:

1. $f * \eta \in C^k(\mathbb{R}^n)$,
2. $\partial^\alpha(f * \eta) = f * (\partial^\alpha \eta)$,
3. $\|\partial^\alpha(f * \eta)\|_{C^0(\mathbb{R}^n)} \leq K\|f\|_1$

Smoothing lets us strengthen our result on the density of continous functions in $L^p(\Omega)$:

Corollary 10.7: Density of C_0^∞ in $L^p(\Omega)$

Let $\Omega \subset \mathbb{R}^n$ be open. Let $1 \leq p < \infty$. Then, for every $f \in L^p(\Omega)$, there exists a sequence $f_k \in C_0^\infty(\Omega)$ such that $\|f - f_k\|_{L^p(\Omega)} \to 0$.

Definition 10.8: Locally integrable Function

Let $\Omega \subset \mathbb{R}^n$ open and $1 \leq p \leq \infty$. Then we call a function $f : \Omega \to \mathbb{R}$ *locally integrable* iff for any compact set $K \subset \Omega$, we have $\mathbb{1}_K \cdot f \in L^p(\Omega)$. We denote the set of all locally integrable functions $f : \Omega \to \mathbb{R}$ by

$$L^p_{\text{loc}}(\Omega)$$

Lemma 10.9: Fundamental Lemma of the Calculus of Variations

Let $\Omega \subset \mathbb{R}^n$ be open. Let $f \in L^1_{\text{loc}}(\Omega)$ such that

$$\begin{aligned} &\forall \varphi \in C_0^\infty(\Omega) : \\ &\varphi \geq 0 \implies \int_\Omega f\varphi \, d\lambda^n(x) \geq 0 \end{aligned}$$

Then we have $f(x) \geq 0$ for λ^n-almost all $x \in \Omega$.

Corollary 10.10: There is no identity of convolution.

There exists no function $\eta \in L^1(\mathbb{R}^n)$ such that for all $f \in C_0^\infty(\mathbb{R}^n)$, we have $f * n = f$.